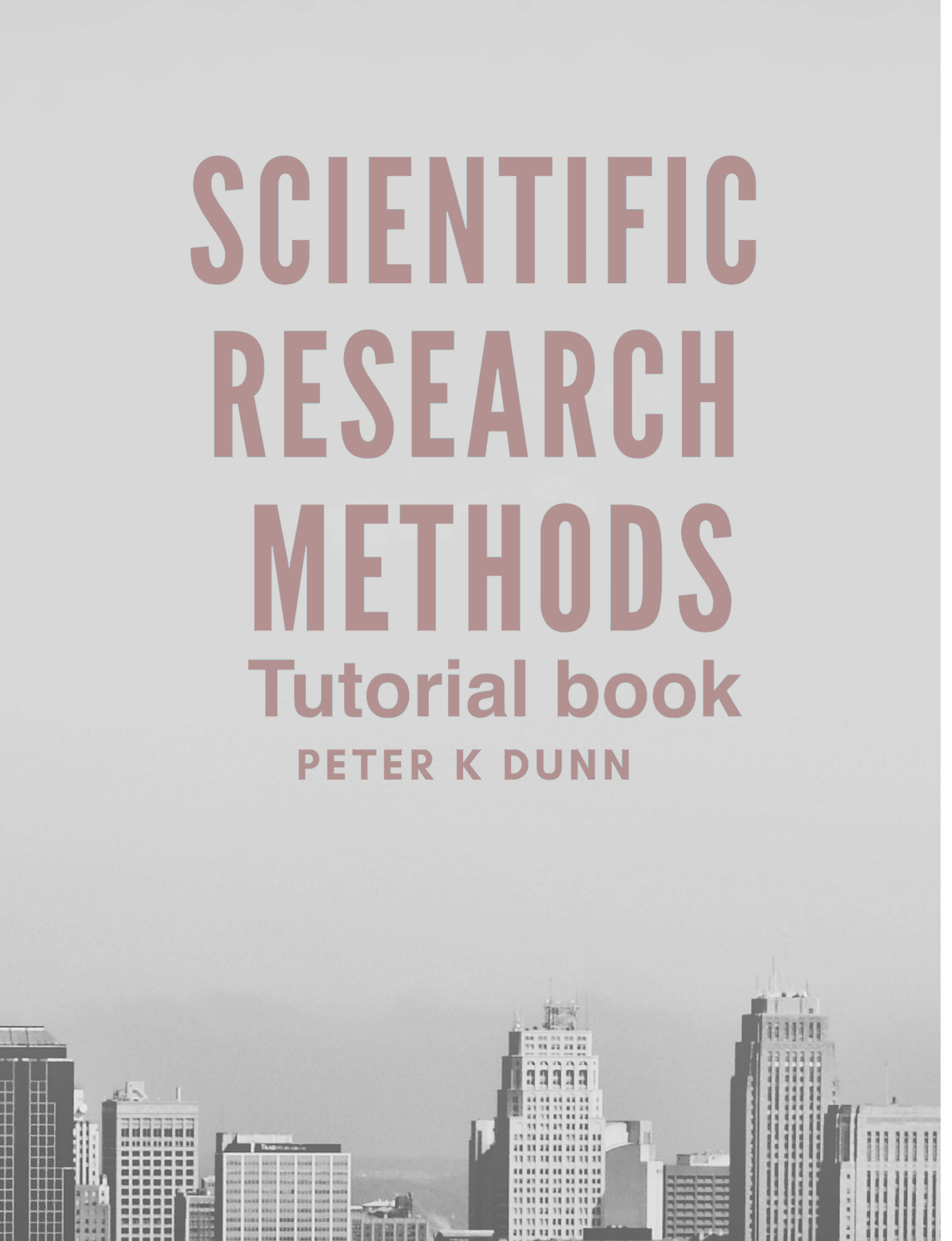
SCIENTIFIC
RESEARCH
METHODS
Tutorial book
PETER K DUNN

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Science Research Methods: Tutorials
INSTRUCTOR EDITION
An introduction to quantitative research and statistics

Contents

Preface

This book has been prepared for use with the book *Scientific Research and Methodology*, to be used in the course *Science Research Methods* at the University of the Sunshine Coast (UniSC).

This course is an introduction to quantitative research methods in the scientific, engineering and health disciplines. It introduces the whole research process, from asking a research question to analysis and reporting of the data. The focus, however, is on the analysis of data.

Answers to questions

The answers to most of the tutorial questions are given in App. A. In the online version, some answers are implied by working through the online exercises.

Statistical software

Most of this book can be read without relying on any specific statistical software. However, some parts explicitly mention and refer to jamovi (The jamovi Project). jamovi is *free* to download and use.

Data sets

Many of the data sets used in this book are available in the **R** package `SRMData`, and are listed and available in the companion textbook.

Call-outs used in this book

The call-outs used in this book have meanings:

Call-outs to *orient* your study

These call-outs introduce the objectives for the sections of the book.

These call-out show how the content links to the course assessment.

These call-outs indicate the relevant chapter of the textbook to read.

Call-outs to indicate *activities*

These call-outs highlight drill questions, for practising some of the mathematical computations.

These call-outs are optional discussion topics that your tutor *may* choose to use in tutorials.

These call-outs highlight **optional** questions, that you might like to complete for personal study.

Call-outs that give *support*

These call-outs highlight common mistakes or warnings, about (for example) a particular concept or about using a formula.

These call-outs offer helpful information.

This icon flags questions that have *video solutions* in the online book: a narrator works through the question, explaining the solution.

Call-outs for *tutors only*

iconmonstr-computer-6-240.png Questions with this icon beside them have MentiMeter availability.

How this book was made

This book was made using **R** (R Core Team 2018), and the **bookdown** package (Xie 2016), based on Markdown syntax, using **knitr** (Xie 2015). Numerous other **R** packages were used, including:

- **diagram** (Soetaert 2017) for making some diagrams.
- **kableExtra** (Zhu 2018) for nicer tables.
- **DT** (Xie et al. 2018) for displaying some data tables in the online version.
- **viridis** (Garnier et al. 2024) for some colour specifications that make colours easier for those with colour-blindness, and for better greyscale printing.
- **webexercises** (Barr and DeBruine 2021) for creating interactive web exercises (i.e., *Quick Revision* questions).

All of this software is *free* and open source. Other resources used include:

- The online quizzes are embedded using H5P iframes.
- Icons are from **iconmonstr**, and are freely available.
- The cover for the book was made using a free image with Canva.

Learning outcomes

In this book, you will learn to:

- Develop quantitative research questions and testable hypotheses.
- Design quantitative studies to answer simple scientific research questions.
- Select and produce appropriate graphical, numerical and statistical analyses.
- Select, apply and interpret the results of the correct statistical technique to analyse data.
- Comprehend, apply and communicate in the language of research and statistics.
- Demonstrate professional integrity in planning, interpreting and reporting the results of quantitative studies.

How to cite this book

Peter K. Dunn (2025). *Scientific Research and Methodology: An introduction to quantitative research and statistics in science, engineering and health*. Chapman & Hall. https://peterkdunn.github.io/SRM-Tutorial-Workbook/index.html

1

Teaching Week 1: tutorial

You will learn to:

- identify types of research studies.
- ask quantitative research questions.
- use and understand the language used in research.

The Week 1 content is essential for:

- **Task 2A**, where you need to *write your own RQ*;
- **Quiz 1**, which includes questions about *writing RQs and RQ components*; and
- the **Exam**, which will contain questions about *RQs and RQ components*.

You will learn and practice the content associated with these chapters of the textbook:

- Chapter 1 (Research: an introduction).
- Chapter 2 (Research questions).
- Chapter 3 (Overview of research design).
- Chapter 4 (Types of research studies).

1.1 Quick revision

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Consider this RQ (based on Jaworowska et al. (2012)):

> For London take-away restaurants, is the average amount of salt in a Chinese meal the same as in an Indian meal?

1. What *type* of RQ is this? Descriptive, relational, repeated-measures or correlational? Explain.

2. What is the *purpose* this RQ? Estimation or decision-making? Explain.

3. Is an *intervention* present? If so, what is it?

4. What is the *outcome* in this RQ? Explain.

5. What is the *response variable*?

6. What is the *unit of analysis*? Explain.

7. What is the *unit of observation*? Explain.

8. The article states (p. 518) that "a takeaway meal was defined as food purchased from out of home food outlets or ordered for home delivery". What *type* of definition is this?

1.2 Class discussion

Discuss: In *this* course, in *this* semester, what percentage of students are female?

1.3 Research process

Organise the steps of the research process (Fig. 1.1) into the usual order in which they are used.

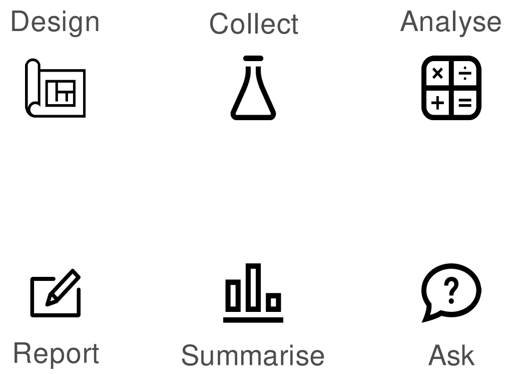

FIGURE 1.1: *The six basic steps in research: what is the usual order in which they are used?*

1.4 Forming RQs

A café owner doesn't believe that his customers can taste the difference between diet and regular colas. As a result, the owner decides to conduct a study of his customers.

1. Two RQs that the café owner is considering are given below. Which do you prefer, and why?
   a. Among my customers, is the *average* customer-taste rating (on a five-point scale) the same for diet and regular cola drinks?
   b. Among my customers, is the *percentage* of customers who prefer diet cola drinks the same as the *percentage* of customers who prefer regular cola drinks?
2. Develop your own RQ to answer the café owner's question, by carefully and precisely describing P, O, C and I (where relevant).
3. Make notes about *how* the café owner could gather data to answer the RQ.

1.5 Terminology

Explain the *fundamental differences* between the following pairs of often-confused terms (Table 1.1), highlighting the important differences.

TABLE 1.1: *Distinguish the similar terms.*

Hypothesis	◀ **Pair 1** ▶	Research question
Relational RQ	◀ **Pair 2** ▶	Repeated-measures RQ
Response variable	◀ **Pair 3** ▶	Explanatory variable
Response variable	◀ **Pair 4** ▶	Outcome
Observational study	◀ **Pair 5** ▶	Experimental study
Quasi-experiment	◀ **Pair 6** ▶	True experiment
Conceptual definition	◀ **Pair 7** ▶	Operational definition
Unit of observation	◀ **Pair 8** ▶	Unit of analysis

1.6 Terminology

The use of mobile phones has changed peoples' behaviour. Barkley and Lepp (2016) examined how the use of mobile phones impacts walking speed. Read the following information from the research article (p. 2, 3):

> The purpose of this study was to compare average walking speed [...] while individuals were holding a cell phone to their ear (*talking*), actively utilizing the phone with their hands and looking at the screen (*texting*), or not using a cell phone (*no use*).
>
> Naturalistic observations were made, unbeknownst to the subjects [...] as individuals transversed a 50 m straight walkway on an American college campus.
>
> We hypothesized that individuals who were talking or texting on a cell phone would walk more slowly than individuals who were not using their cell phone [...] The walkway was straight, had limited ingress and egress, and was simple to navigate.
>
> All observations were made while subjects were walking in one direction (west to east) approaching the overpass where research personnel were positioned. The starting and end points to the 50 m walkway were selected as they had physical landmarks (e.g., a park bench bolted to the walkway) that were easily seen by the observer.
>
> Walking speed was recorded, via a stopwatch, as the time (seconds) it took subjects to walk from the

start to end points on the 50 m walkway.

Walking speed was recorded for every fifth individual, walking independent of others (i.e., alone), to cross the starting point on the 50 m walkway.

Answer the questions that follow, **explaining** the reason for your choices. **Make sure you explain why the other options are incorrect.**

1. Which one of these is most likely the *population* under study?
   a. Phones.
   b. People.
   c. People at an American college.
   d. Students at an American college.

Explain why the other options are *incorrect*.

2. Which one of these is most likely the *outcome* in the study?
   a. The way subjects were using their phone.
   b. The time taken to walk 50 m.
   c. The **average** time taken to walk 50 m.
   d. The difference between the times taken, between those texting and those not texting.

Explain why the other options are *incorrect*.

3. Which one of these is most likely the *comparison* in the study?
   a. Between the ways that subjects were using their phone (talking; texting; not at all).
   b. Between those walking slowly and those walking quickly.
   c. Between those walking and those not walking.
   d. Between those using and not using a mobile phone.

Explain why the other options are *incorrect*.

4. Which one of these is most likely the *intervention* in the study?
   a. The way subjects were using their phones.
   b. The timing of the walk.
   c. There is no intervention.
   d. The phone.

Explain why the other options are *incorrect*.

5. What is the *explanatory variable*?
   a. The phone.
   b. The way that the phone is being used.
   c. The **average** time to walk $50\,\mathrm{m}$.
   d. The walking speed.

Explain why the other options are *incorrect*.

6. What is the *response variable*?
   a. The differences between the times taken.
   b. The way subjects were using the phone.
   c. The **average** time to walk $50\,\mathrm{m}$.
   d. The time taken to walk $50\,\mathrm{m}$.

Explain why the other options are *incorrect*.

7. Which one of these is most likely the *unit of observation* in the study?
   a. The phone.
   b. *Individual* subjects.
   c. *Groups* of subjects.
   d. Minutes.

Explain why the other options are *incorrect*.

8. Which one of these is most likely the *unit of analysis* in the study?
   a. The phone.
   b. *Individual* subjects.
   c. *Groups* of subjects.
   d. Minutes.

Explain why the other options are *incorrect*.

1.7 Creating research questions and studies

Consider the following scenario:

> While at Café C one day enjoying a small, organic, almond-milk, half-strength, decaffeinated latte (sometimes called a *Why-Bother*…), Nim sighed, ‘Exams start next week. I suppose we’ll all start feeling stressed soon.’
>
> ‘Ah,’ Jane said, ‘I had better buy more teabags then.’
>
> ‘Tea bags? Why?’ asked Nim, eyebrows raised.
>
> ‘Everyone knows that Earl Grey tea relaxes students…’ said Jane dreamily.
>
> ‘How do you know that?’ Nim said.
>
> ‘It *does*,’ snapped Jane. ‘Earl Grey relaxes people.’

Nim considers studying Jane’s assertion as the basis for a **feasible** and **practical** SCI110 project.

1. Nim needs to determine a *population* to study. Which of these would be a useful and practical population to use? Or is there a different population that is ‘better’ to use? Explain.
    a. All university students.
    b. Female university students about to do an exam.
    c. UniSC students.
    d. Queensland university students.
2. Identify an *outcome* that Nim can measure to answer the research question. Justify your answer.
3. Which of these *between-individuals comparisons* do you think Nim should use? Why?
    a. Between drinking Earl Grey tea and coffee.
    b. Between drinking Earl Grey tea and black (ordinary) tea.
    c. Between drinking Earl Grey tea and water.
    d. Between exam time and other times of the semester.
    e. Between students and non-students.
4. Explain why ‘Between before and after drinking Earl Grey tea’ is a *within-individuals* comparison.

5. Construct a well-worded *relational* research question *with an intervention* that Nim can ask to assess Jane's assertion, clearly identifying P, O, C and I.

6. *Briefly* describe an *experimental* study for answering your proposed research question.

7. Construct a well-worded *relational* research question *without an intervention* that Nim can ask to assess Jane's assertion, clearly identifying P, O and C.

8. *Briefly* describe an *observational* study for answering your proposed research question.

9. List any necessary word or terms that need operational or conceptual definitions. (You do not need to *give* the definitions.)

10. Using this research question, identify the response and explanatory variables, and hence the data that *needs* to be collected to answer the question.

1.8 Identifying units of analysis and units of observation

Bamboo is a fast-growing, strong grass often used for environmentally-friendly building practices. A small research study explored the hardness of bamboo when used as flooring material.

The *Janka hardness*[1] of bamboo flooring provided by Bamboo Flooring Australia Pty Ltd was measured by the Queensland Department of Primary Industries (Gerber 2004). Five floorboards had two hardness measurements taken on *each* board (units probably kilonewtons; Table 1.2).

TABLE 1.2: *Two Janka hardness measurements from five different bamboo boards..*

Board 1	Board 2	Board 3	Board 4	Board 5
10.5	8.0	11.5	10.3	10.2
7.5	8.0	11.2	9.9	9.3

1. What is the unit of analysis: the test, the board, the Janka hardness, each measurement, kilonewtons, or something else? Explain your answer.
2. *How many* units of analysis are there?
3. *How many* units of observation are there?
4. Suppose the measurements were all taken from 10 *different* places on the *same* board (rather than from five different boards). How many units of analysis are there now? Explain your answer.

[1] The force required to embed an 11.28 mm steel ball into wood to a depth of half the diameter of the ball.

5. Comment on the amount of variation *between* the boards compared to the amount of variation *within* boards.

1.9 Optional questions

These questions are **optional**; e.g., if you need more practice, or you are studying for the exam. (Answers appear in Sect. A.1.)

1.9.1 (Optional) Identifying different types of studies

1. A lecturer wishes to compare the average blood alcohol concentrations (BAC) of students in two of her classes. She measures the BAC of students in two of her Monday morning classes.

   What *type* of study is this: observational, true experimental, quasi-experimental? Explain.

2. A lecturer wishes to compare the average BAC of students in two of her classes, to determine if knowing in advance leads to different average BAC.

   She informs one class on a Friday that she will be measuring BAC in students on a Monday morning, but does not tell another of her Friday classes.

   What type of study is this: observational, true experimental, quasi-experimental? Explain.

1.9.2 (Optional) Terminology

Complete the crossword in Fig. 1.2 to help you understand some of the words used in the content.

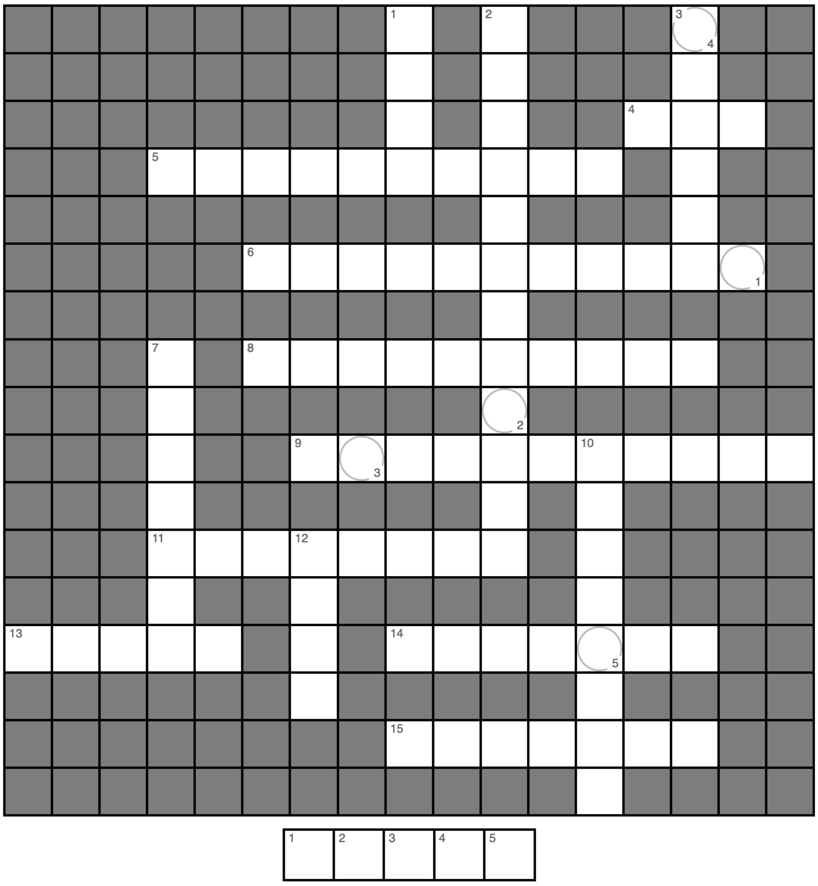

FIGURE 1.2: *A crossword to complete.*

Across

4. The first step in the research process is to ___ the research question (3)
5. RQs in which groups are compared numerically are called __________ (10)
6. 'Conceptual' and 'Operational' are different types of ___________ (11)
8. RQs may be decision making-type RQs or __________-type RQs (10)
9. The type of research where data are *not* summarised using numerical methods (11)
11. Research is used to provide ________-based answers to RQs (8)
13. The sample size is the number of _____ of analysis (5)
14. What we do with the data to find an answer to the RQ (7)
15. A confounding variable, but is not measured or recorded, is a _______ variable (7)

Down

1. The acronym to remember the four components of a RQ (4)
2. The type of research where data are summarised using numerical methods (and is studied in SCI110) (12)
3. The second step in the research process (6)
7. After designing a study, the next step is to _______ the data (7)
10. Independent collections of units of observations are called unit of ________ (8)
12. The collected information to answer the RQ is called ____ (4)

2

Teaching Week 2: tutorial

You will learn to:

- design quantitative research studies to answer RQs.
- use and understand the language used in research.

The Week 2 content is essential for:

- **Task 2A**, which is largely about the *research design* for answering your own RQ;
- **Quiz 1**, which includes questions about *research design*; and
- the **Exam**, which will contain questions about *research design*.

You will learn and practice the content associated with these chapters of the textbook:

- Chapter 5 (Ethics in research).
- Chapter 6 (External validity: sampling).
- Chapter 7 (Internal validity).
- Chapter 8 (Research design limitations).

You should bring your calculator to tutorials from **next** week onwards.

2.1 Quick revision

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Consider this RQ (Walters et al. 2018):

> What factors are preventing the adoption of household solar technologies in Santiago [Chile]?'

1. For this RQ, what is the *Population*?
2. The study will be *externally valid* if:
   - the sample is representative of all households in the world.
   - the sample is representative of all solar technologies.
   - the sample is representative of all households in Santiago.
   - the sample is representative of all households in Chile.
3. Suppose the researchers mailed surveys to all households in Santiago, and people returned the survey if they wished to. What is the *best* description of this sampling method?
4. Suppose the researchers randomly selected five suburbs in Santiago; then ten streets within each of these suburbs; then ten households within each of these streets. What is the *best* description of this sampling method?
5. Suppose the researchers stood outside a busy supermarket, and surveyed every ninth person who entered. What is the *best* description of this sampling method?

2.2 Class discussion

Discuss: A larger sample is always better than smaller sample.

2.3 Study design

A student group was studying this RQ (which is **not** appropriate for the SCI110 Project; Task 2):

> Among Australians, is the average serum cholesterol concentration different for smokers and non-smokers?

Explain *why*, according to the *Guidelines*, this RQ is *not* appropriate for Task 2.

The students gave the following information about their study. Explain **why** each of these statements is **incorrect**.

1. "The design is observational, as we cannot manipulate each person's serum cholesterol."
2. "The Outcome is *the average serum cholesterol concentration for smokers and non-smokers*."
3. "The study is not externally valid, as the results may not apply to all people in the world."
4. "The response variable is *serum cholesterol*."
5. "In this experiment, the population is *Australians*."
6. "The data file will have two columns: one for smokers, and one for non-smokers."
7. "*Whether or not the person owns a cat* is likely to be a confounding variable."
8. "The *observer effect* is not relevant, as the participants will know they are involved in a study."

2.4 Language

Patients who have suffered a stroke often have restricted limb movement, and rehabilitation therapies are studying robotics to assist these patients. Read the description (line breaks added for clarity) of one such study (Lo et al. 2010), then complete the crossword in Fig. 2.1 regarding this information.

> In this multicenter, randomized, controlled trial involving 127 patients with moderate-to-severe upper-limb impairment 6 months or more after a stroke, we randomly assigned 49 patients to receive intensive robot-assisted therapy, 50 to receive intensive comparison therapy, and 28 to receive usual care.
>
> Therapy consisted of 36 1-hour sessions over […] 12 weeks. The primary outcome was a change in motor function […] at 12 weeks…
>
> We recruited veterans from four participating VA medical centers who were 18 years of age or older and had long-term, moderate-to-severe motor impairment of an upper limb from a stroke that had occurred at least 6 months before enrollment.
>
> Such impairment was defined as a score of 7 to 38 on the Fugl-Meyer Assessment of Sensorimotor Recovery after Stroke, a scale with scores for upper-limb impairment ranging from 0 (no function) to 66 (normal function). All patients provided written informed consent. Patients were randomly assigned to receive robot-assisted therapy, intensive comparison therapy, or usual care…
>
> Trained evaluators who were unaware of study-group assignments assessed patients 6, 12, 24, and 36 weeks after randomization. The primary outcome was a change in the Fugl-Meyer score at 12 weeks, as compared with the baseline value…

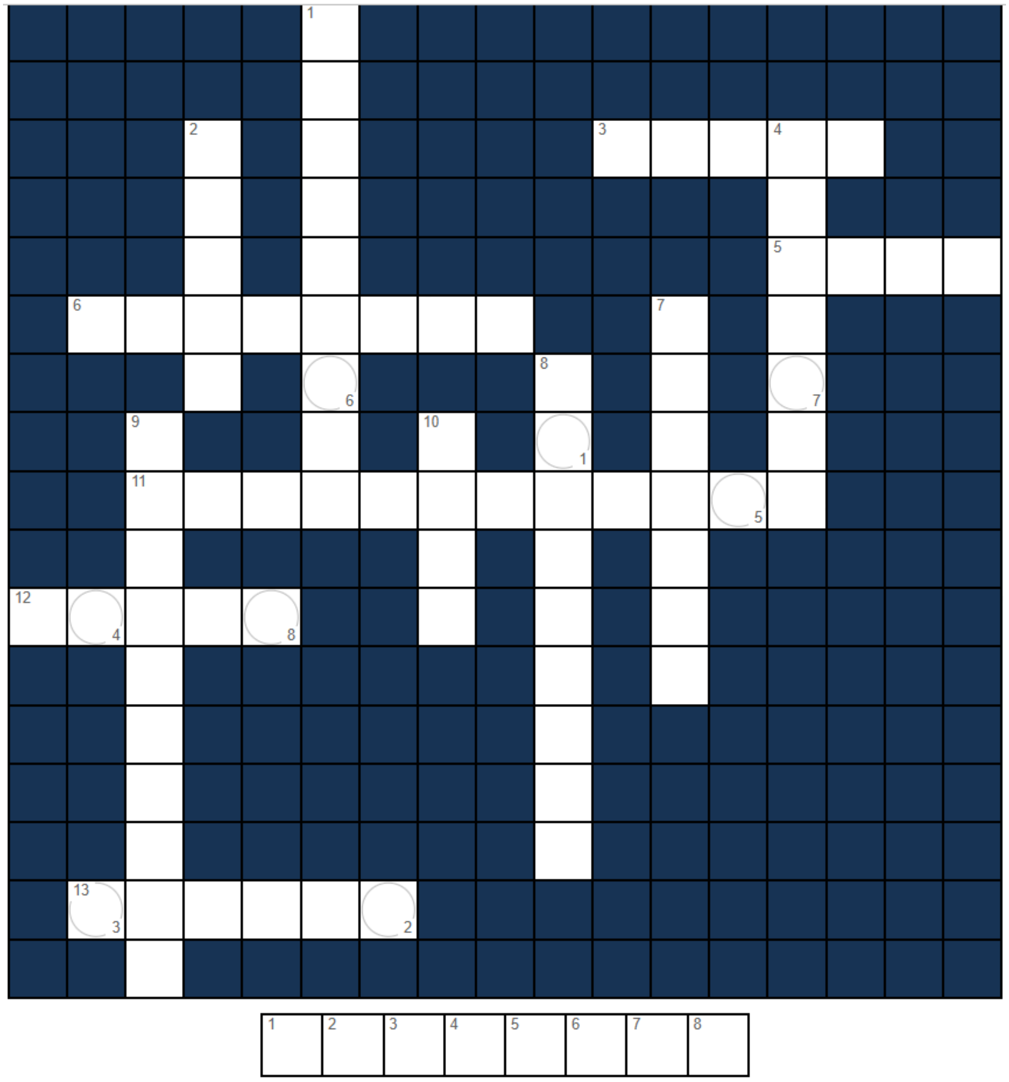

FIGURE 2.1: A crossword to complete.

Across

3. The number of groups being compared (5)
5. Ask if you need ____ (e.g., in tutorial, or on the Canvas *Discussions*) (4)
6. Patients were assigned ________ to the type of therapy received (8)
11. This type of study is called a ____________ study (12)
12. A lovely place to relax, with sand and surf (5)
13. The size of the ______ is 127 (6)

Down

1. When individuals change their behaviour if they think they are being observed: the _________ effect (9)
2. The evaluators were unaware of the group the patient was in; they were _____ to the treatment (5)
4. Informed consent ensures the study is _______ (7)
7. The treatment of 'usual care' would be considered to be the _______ group (7)
8. The type of therapy is called the _________ (9)
9. The research question is of this type (10)
10. Riddle: Feed me, I live; water me, I die (4)

2.5 Assessing reports

A report in *The Sydney Morning Herald* (Berry 2016) was entitled 'Aged cheese could help you age well'. The research was based partly on a study of mice, and partly on a study of people. Based on this headline, answer the following questions.

1. What are reasonable suggestions for the outcome and response variable, given this headline?
2. What are reasonable suggestions for the comparison and explanatory variable, given this headline?
3. Would you expect the study to be observational or experimental? Explain.

Below is an extract from the newspaper article about the mice study that formed part of the research.

> Cheese is not only delicious and nutritious, with its abundance in calcium and vitamin B12, it can also slow down the ageing process, according to a new study… In the study published in the journal *Nature Medicine*, European researchers analysed the effect of a compound found in aged cheese (as well as legumes and whole grains) called spermidine […] the team supplemented the drinking water of mice with spermidine. Control mice drank normal water.

The researchers then found that the addition of spermidine ' "significantly extended" the [average] lifespan of the mice…'. Based on this information, answer the following questions.

4. What are reasonable suggestions for the outcome and response variable for the mice study?
5. What are reasonable suggestions for the comparison and explanatory variable?
6. Is the study observational or experimental? Explain.

7. Does the newspaper headline appear reasonable?

8. Why do the researchers give some mice plain drinking water?

Below is an extract from the newspaper article for the human study that formed part of the research.

> … the researchers surveyed 800 Italians about their diet and found that those who reported a higher intake of spermidine had lower blood pressure…

Based on this information, answer the following questions.

9. What are reasonable suggestions for the outcome and response variable for the human study?

10. What *type* of RQ is suggested: relational, correlational or repeated-measures?

11. Is the study observational or experimental? Explain.

12. Are the following variables likely to be extraneous variables, control variables, confounding variables, or none of these? **Explain.**
    a. The sex of the person.
    b. The number of siblings.
    c. The distance to the nearest hospital.
    d. The amount of exercise per week by the person.
    e. Whether their father had heart problems.

The newspaper article is based on a research paper (Eisenberg et al. 2016), in which the diets of more than 800 Italians were recorded in 1995, 2000 and 2005 for the study of people. From 1995 to 2010, the researchers recorded heart-related events for each subject: incidents of high blood pressure, heart failure, stroke and premature death from heart disease.

Those with the highest spermidine intakes had a 40% lower risk of heart failure (both fatal and non-fatal) compared to those with the lowest spermidine intakes. Those with the highest spermidine intakes also had a significantly lower risk of *any* heart disease, compared to those with the lowest spermidine intakes.

13. Does the newspaper headline appear consistent with the research article?

The study reports that the biggest contributors to spermidine intakes were:

- wholemeal foods (accounting for 13.4% of intake);
- apples and pears (13.3%),
- salad (9.8%),
- vegetable sprouts (7.3%), and
- potatoes (6.4%).
- aged cheese (2.9%).

14. Based on this information, do you agree with, or not agree with, the newspaper article headline? Explain.

15. Is the cause-and-effect relationship, as implied in the article headline, reasonable? Explain.

16. What did you learn from this study?

2.6 Planning a research study

Your tutor will have information.

This activity is very important for helping with your Project.

2.7 Study design features

A study (Truswell et al. 1992) compared the health benefits of crisps ('potato chips') when fried in canola oil or palmolein. The article states (line breaks added):

> Recently canola oil crisps have appeared in the market with the implication that their lower proportion of saturated fatty acids would have beneficial effects on plasma cholesterol.
>
> We decided to examine this experimentally… a major manufacturer agreed to fry batches of potato crisps in canola oil or the conventional palmolein for a human trial. This provided an opportunity for us to run a double blind crossover trial.
> — Truswell et al. (1992)

Part of the protocol for the study is given below:

> Subjects were asked to eat about a third of the day's crisps at the beginning of each meal and then, in the rest of their diets, to eat foods that were low to moderate in fat from a list of such foods…
>
> The crisps were supplied in plain bags, unmarked except for two code characters: 'T4' or 'B9' in 1990 and 'D5' or 'GS' in 1991. Until the end of the experiment, the type of oil was known only by the food scientist who prepared the crisps.
>
> In 1990, 21 subjects (12 men, 9 women) completed the 5 week experiment. They were randomly allocated to take palmolein ('B9') or canola ('T4') crisps for the first 3 weeks, then (without a washout period) changed over to the other type, canola or palmolein for another 2 weeks. Thus half the subjects ate (new) canola crisps for 3 weeks and the (usual) palmolein crisps for the next 2 weeks.
>
> The other half of the subjects ate palmolein crisps for 3 weeks and canola crisps for the next 2 weeks. The first week of the 3 weeks served as an adjustment period. Fasting venous bloods were taken once at the start (usual diet), then for the last 3 mornings of both the first and second experimental periods.
> — Truswell et al. (1992)

1. Explain how random *allocation* is used (and why), if at all.

2. Explain how the impact of the Hawthorne effect has been minimised.

3. Explain how the impact of the observer effect has been minimised.

4. Would the carry-over effect play a role? Explain.

5. Describe the blinding that is used.

3

Teaching Week 3: tutorial

You will learn to:

- collect data.
- classify data.
- produce and understand graphical summaries for quantitative data.
- produce and understand numerical summaries for quantitative data.

The Week 3 content is essential for:

- **Task 2A**, which requires you to *classify variables and suggest appropriate numerical and appropriate graphical summaries*;
- **Task 2B**, which requires you to *collect your own data, classify variables, produce appropriate numerical summaries, and produce appropriate graphs*;
- **Quiz 2**, which includes questions about *classifying variables, numerical summaries and graphs*; and
- the **Exam**, which will contain questions about *classifying variables, numerical summaries and graphs*.

You will learn and practice the content associated with these chapters of the textbook:

- Chapter 9 (Collecting data).
- Chapter 10 (Classifying data and variables).
- Chapter 11 (Summarising quantitative data).

You should bring your calculator to tutorials from this week onwards.

You can search on YouTube for tutorials on using your calculator's **Statistics Mode**. For example:
`"statistics" "mean" "standard deviation" Casio fx82`

3.1 Quick revision

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In a study of obstructive sleep apnoea (OSA) in adults with Down Syndrome (de Carvalho et al. 2020), $n = 60$ adults underwent a sleep study. Part of the data are shown in Table 3.1. (REI is the Respiratory Event Index; REI under 5 refers to no sleep apnoea; REI of 30 or over refers to severe sleep apnoea.)

1. How would you *classify* the variable `Age`?
2. How would you *classify* the variable `Gender`?
3. How would you *classify* the variable `BMI`?
4. How would you *classify* the variable `REI`?
5. What did you learn from this study?

TABLE 3.1: *Part of the obstructive sleep apnoea data set ($n = 60$).*

Age	**Gender**	**BMI**	**Neck circumference (cm)**	**REI**
21	Male	20.3	37	46
24	Male	24.1	40.5	19.3
26	Male	25.2	38	12.4
39	Female	40.8	41	58.6
21	Female	35	37	12.7
29	Male	29.2	41	38.8
20	Male	25.8	42	24.4
21	Male	20.9	37	7
19	Female	20.5	32	37.6
27	Male	22.4	39	21.7
⋮	⋮	⋮	⋮	⋮

3.2 Class discussion

Discuss: Two datasets can have the same numerical summaries but tell very different stories.

3.3 Using the calculator statistics mode for a small data set

Table 3.2 shows the polythene usage, in tonnes, by eight UK cosmetic companies (Gilchrist 2000).

1. Using your calculator's *Statistics Mode*, find the **mean** and **standard deviation** of these numbers.
2. Without using a calculator, find the **median** of the data.
3. Without using a calculator, find the **IQR** of the data.

TABLE 3.2: *The amount of polythene (in tonnes) used by a sample of eight UK cosmetic companies.*

8.001	29.400	266.532	4298.700	94.500	2547.300	676.200	0.000

Most calculators have **two buttons** that compute the standard deviation when in **Statistics Mode**: one computes the standard deviation if the data are a sample, and one if the data are a population. In practice, data are almost *never* a population.

If you are using your calculator correctly, you should get (before rounding) $\bar{x} = 990.0791$ and $s = 1588.514579$. If you get $s = 1485.919327$ for the *standard deviation*, you are using your calculator incorrectly, so please **ask for help**. You are probably pressing the incorrect button to get the standard deviation.

3.4 Understanding centre and variation

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A tutor has recorded the marks (as a percentage) for all students in her four classes for an assignment. All classes have 30 students. The corresponding histograms are shown in Fig. 3.1.

1. In which class would the median be the *largest*? Why?

2. In which class would the median be the *smallest*? Why?

3. In which class would the standard deviation be the *largest*? Why?

4. In which class would the standard deviation be the *smallest*? Why?

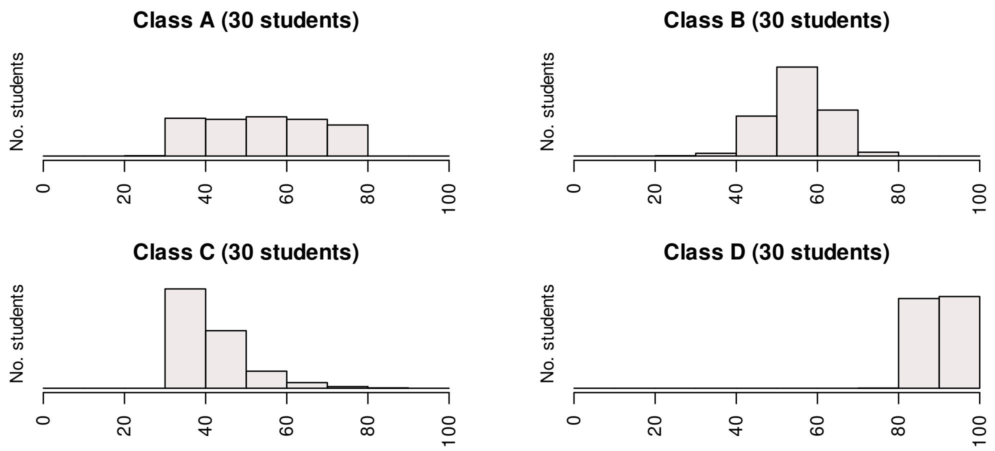

FIGURE 3.1: *Histogram of marks for four classes.*

3.5 Understanding histograms

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⚠ **NOTE**: The first bar of the histogram is not necessarily at zero; it is the **shape** of the histogram that is of interest here: right skewed, left skewed, symmetric, etc.

Consider the four histograms in Fig. 3.2. Which histogram is most likely to describe the *shape* of the following data? Why?

1. The time that students remain in the examination room for a **short** or **easy** two-hour examination.
2. The heights of female students at UniSC.
3. The *starting* salaries of new science graduates employed full-time.
4. The volume of drink in 375 mL cans of soft drink.
5. The time that students remain in the examination room for a **hard** or **long** two-hour examination.

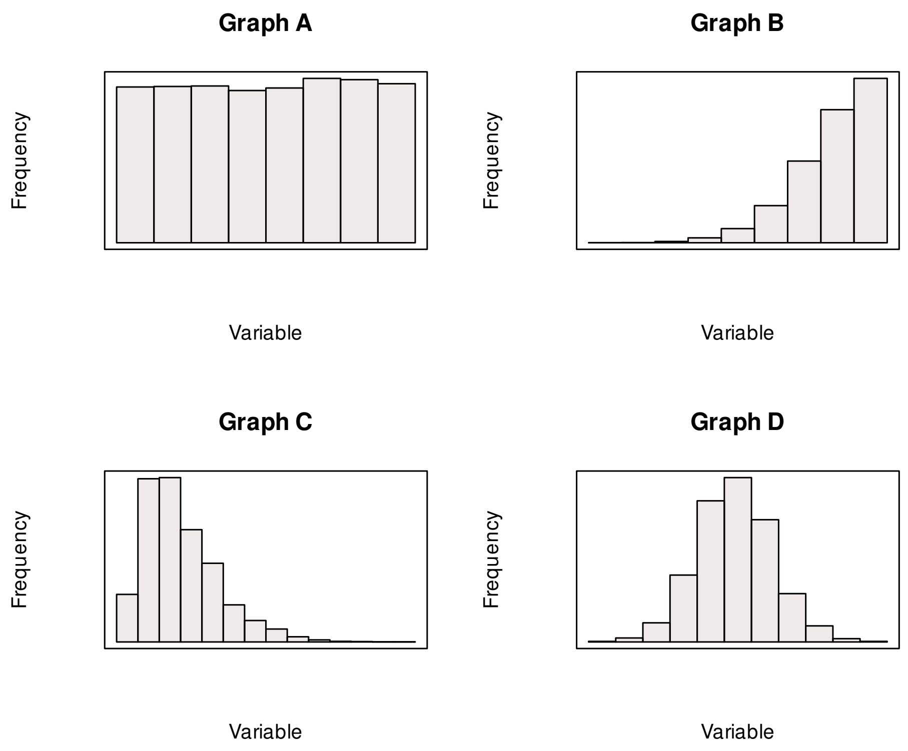

FIGURE 3.2: *Four histograms; what* ***shape*** *is appropriate for which scenario?*

3.6 Collecting data

Your tutor will have information.

3.7 Interpreting graphs

A study of Weddell seals (Bryden et al. 1984) measured, among other things, the seal body length. A histogram of the body length of 90 female seals is shown in Fig. 3.3.

1. Describe this histogram in words (average; variation; shape; outliers).

2. What alternative graphical displays could be used to display these data?

3. Critique the graph.

4. Write a one-sentence summary of the body length of female Weddell seals suitable for a publicity brochure.

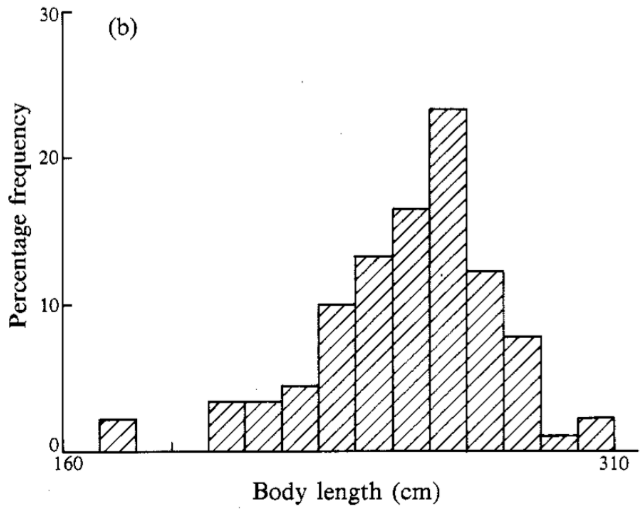

FIGURE 3.3: A histogram of the body length of female Weddell seals

3.8 Optional drills: computational exercises

(Answers appear in Sect. A.3)

These *optional* **Drill** exercises (repeated practice) give you practice at getting computations correct, and using your calculator. These drill questions are more about practising the underlying *mathematics* rather than the *statistics*. If you need help, please **ask**.

1. The data in Table 3.3 give the heights of $n = 7$ female tennis players at the *Australian Institute of Sport* (AIS) in metres (Telford and Cunningham 1991).
   a. Using your calculator's *Statistics Mode*, find the **mean** and **standard deviation** of the data.
   b. Without using a calculator, find the **median** of the data.
   c. Compute the IQR for the data.

TABLE 3.3: *The heights (in metres) of female tennis players at the AIS.*

167.9	177.5	162.5	172.5	166.7	175.0	157.9

2. Bamboo is a fast-growing, strong grass useful for environmentally-friendly building practices. A small research study explored the properties of bamboo when used as flooring material, including the bending strength (the Modulus of Rupture, or MOR, in MPa). Five different bamboo floorboards were provided by Bamboo Flooring Australia Pty Ltd and assessed by the Queensland Department of Primary Industries (Gerber 2004). The five MOR test results (in MPa) are shown in Table 3.4.
   a. Identify the type of RQ being answered: descriptive, relational, cross-sectional or correlational. Justify your answer.
   b. Use your calculator's statistics mode to compute the mean and the standard deviation.
   c. Without using a calculator, find the **median** of the data.

TABLE 3.4: *The MOR (in MPa) for five bamboo boards.*

99.2	111.2	97.6	101.1	104

3. The data in Table 3.3 give the percentage body fat of $n = 9$ female swimmers at the *Australian Institute of Sport* (AIS) (Telford and Cunningham 1991).
   a. Using your calculator's *Statistics Mode*, find the **mean** and **standard deviation** of the data.
   b. Without using a calculator, find the **median** of the data.
   c. Compute the IQR.

TABLE 3.5: *The percentage body fat of female swimmers at the AIS.*

14.52	11.47	17.71	18.48	11.22	13.61	12.78	11.85	13.35

3.9 Optional questions

These questions are **optional**; e.g., if you need more practice, or you are studying for the exam. (Answers appear in Sect. A.3.)

3.9.1 (Optional) Using the calculator Statistics Mode

Batteries are expensive, so comparing the performance of expensive and cheap batteries is helpful. A test on the lifetime of batteries (Lindström 2011) compared the time for two brands of 1.5 volt batteries to reduce their voltage to 1.0 volts under standard testing conditions.

The times (in hours) for nine Energizer Max batteries and nine ALDI brand batteries (Ultracell) are shown in Table 3.6.

TABLE 3.6: *The times taken for batteries to go from* 1.5 *V to* 1.0 *V, in hours.*

Energizer	7.58	7.46	7.46	7.59	7.46	7.52	6.83	6.89	7.45
Ultracell	7.50	7.48	7.47	7.48	7.48	7.41	7.47	6.96	7.48

1. Is the type of research study a true experimental, quasi-experimental, or observational study?

Justify your answer.

2. What are the *units of observation* and *units of analysis*?

3. Use your calculator's *Statistics mode* to compute the following. (Check your answers to ensure you are using your calculator correctly.)
   a. Compute the mean times for the *Energizer* batteries (to two decimal places).
   b. Compute the standard deviation of the *Energizer* battery times (to three decimal places).
   c. Compute the mean times for the *Ultracell* batteries (to two decimal places).
   d. Compute the standard deviation of the *Ultracell* battery times (to three decimal places).

4. Explain what these calculations tell you.

5. Compute the median lifetime for each brand, explain what this tells you. (Many calculators cannot compute medians.)
   a. Compute the median of the *Energizer* batteries (to two decimal places).
   b. Compute the median of the *Ultracell* batteries (to two decimal places).

6. Determine (and justify) if the mean or median would be a more appropriate measure of centre.

7. Do you the average time to reach 1.0 volts is the same for each brand? Explain.

8. Information about the cost of the batteries would also be important information to know (see below). What advice would you give about buying batteries based on these data?

The cost of the batteries, at the time of publication (05 September, 2012), was:

- *Ultracell*: A four-pack cost $2.49 from ALDI online
- *Energizer Max*: A four-pack cost $5.97 from Woolworths online, *on special*; the normal price was $8.01 for the four-pack.

4

Teaching Week 4: tutorial

You will learn to:

- produce and understand graphical summaries for **qualitative** data.
- produce and understand numerical summaries for **qualitative** data (e.g., odds, proportions).
- produce and understand graphical summaries for comparing **qualitative** data.
- produce and understand numerical summaries for comparing **qualitative** data (e.g., odds ratios, difference in proportions).

The Week 4 content is essential for:

- **Task 2A**, which requires you to *suggest appropriate numerical summaries, and suggest appropriate graphs*;
- **Task 2B**, which requires you to *produce and interpret appropriate numerical summaries and produce and interpret appropriate graphs*;
- **Quiz 2**, which includes questions about *numerical summaries and graphs*; and
- the **Exam**, which will contain questions about *numerical summaries and graphs*.

You will learn and practice the content associated with these chapters of the textbook:

- Chapter 12 (Summarising qualitative data).
- Chapter 15 (Comparing qualitative data between individuals.

Take careful note of which chapters are covered in this tutorial!

You should bring your calculator to tutorials from this week onwards.

4.1 Quick revision

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A study of obstructive sleep apnoea (OSA) in adults with Down Syndrome (de Carvalho et al. 2020) had $n = 60$ adults (27 females; 33 males) undergo a sleep study. Part of the data are in Table 4.1. (REI is the Respiratory Event Index; an REI under 5 refers to no sleep apnoea; an REI of 30 or over refers to severe sleep apnoea.)

1. Which of these would be an **inappropriate** numerical summary for the age of the participants? Median, mean, odds, or standard deviation.
2. What might be an **appropriate** way to numerically describe the amount of *variation* in the ages of participants?
3. What might be an **appropriate** way to numerically describe the average Respiratory Event Index (REI) of participants?
4. What might be an **appropriate** way to numerically describe the gender of the participants?
5. In the sample of $n = 60$, what **percentage** of individuals are females?
6. In the sample of $n = 60$, what are the **odds** that an individual is female?
7. Use the jamovi output in Fig. 4.1 to find the mean of *Gender*, and explain what this mean.

4.2 Class discussion

Discuss: The raw data is more useful than a graphical summary.

Descriptives

	Age	Gender	BMI	Neck	REI
N	60	60	60	60	60
Missing	0	0	0	0	0
Mean	27.8	1.45	28.7	40.1	30.4
Median	25.5	1.00	28.3	40.0	23.0
Standard deviation	9.12	0.502	5.89	3.69	19.1
IQR	10.3	1.00	8.55	4.00	24.9
Minimum	18	1	18.7	32.0	7.00
Maximum	56	2	40.8	48.0	97.9

FIGURE 4.1: *A numerical summary of the OSA data: jamovi*

TABLE 4.1: *Part of the Obstructive Sleep Apnoea (OSA) data set.*

Age	**Gender**	**BMI**	**Neck circumference (cm)**	**REI**
21	Male	20.3	37	46
24	Male	24.1	40.5	19.3
26	Male	25.2	38	12.4
39	Female	40.8	41	58.6
21	Female	35	37	12.7
29	Male	29.2	41	38.8
20	Male	25.8	42	24.4
21	Male	20.9	37	7
19	Female	20.5	32	37.6
27	Male	22.4	39	21.7
⋮	⋮	⋮	⋮	⋮

4.3 Percentages and odds

In a study of how well emergency dispatchers recognised signs of stroke (Oostema et al. 2018), the data in Table 4.2 were collected.

TABLE 4.2: *How well emergency dispatchers recognised signs of stroke.*

	Dispatcher suspected stroke	**Dispatcher missed stroke**
Males	67	43
Females	97	39

1. *Why* are the levels 'Male' and 'Female' placed in the rows of the table, rather than the columns?

2. What proportion of patients had their stroke symptoms missed?

3. Sketch a side-by-side or stacked bar chart to display the data.

4. Of the *male* patients, what *proportion* had their stroke symptoms missed by the dispatcher?

5. Of the *female* patients, what *proportion* had their stroke symptoms missed by the dispatcher?

6. For the *male* patients, what are the *odds* that they had their stroke symptoms missed by the dispatcher?

7. For the *female* patients, what are the *odds* that they had their stroke symptoms missed by the dispatcher?

8. What is the *difference between the proportions* of stroke symptoms being missed by the dispatcher, comparing *males* to *females*?

9. What is the *odds ratio* that a patients had their stroke symptoms missed by the dispatcher, comparing *males* to *females*?

10. Construct a numerical summary table by completing Table 4.3.

TABLE 4.3: *Numerical summary table: emergency dispatchers and missing stroke symptoms.*

	Proportion of symptoms missed	**Odds that symptoms missed**	**Sample size**
Males			
Females			
	Difference:	*Odds ratio:*	

4.4 Two-way tables

Soccer (football) is a unique in that one aspect is 'the purposeful use of the unprotected head for controlling and advancing the ball' (Kirkendall et al. 2001). Some researchers suspect that repeatedly 'heading' the ball may impair brain function.

A study (Kirkendall et al. 2001) was conducted to determine (p. 157)

> …whether long-term or chronic neuropsychological dysfunction (i.e. concussion) was present in collegiate soccer players

Data were collected from 240 college students for two variables:

- The student type: one of 'soccer player', 'non-soccer athlete', or 'non-athlete'.
- The number of head concussions: each student was asked about the number of head concussions they had experienced; 'zero' , 'one', or 'two or more' concussions.

Use the study data (Table 4.4) to answer the following questions.

TABLE 4.4: *Data on concussions experienced by college students.*

	Num. concussions			
	0	**1**	**2 or more**	**Total**
Soccer players	45	5	13	63
Non-soccer athletes	68	25	3	96
Non-athletes	45	15	21	81
Total	158	45	37	240

1. *Why* is the number of concussions placed in the columns of the table, rather than the rows?

2. Classify the two variables.

3. Compute the percentage of college students in the *sample* that have received exactly one concussion.

4. Compute the percentage of college students in the population that have received two or more concussions.

5. Many possible graphs exists to display the data; four are shown in Fig. 4.2. What is the main message from each graph? Which graph do you think is best? Why?

6. Among **non-athletes**, compute the odds of receiving two or more concussions. Interpret what this means.

7. Among **soccer players**, compute the odds of receiving two or more concussions. Interpret what this means.

8. Compute the odds ratio comparing the odds of a non-athlete receiving two or more concussions to the odds of a soccer player receiving two or more concussions.

9. Create a table of **column** percentages. What do these tell you?

10. Create a table of **row** percentages. What do these tell you?

11. Which one of these tables is probably more sensible? Why?

12. What did you learn from this study?

4.5 Working with odds

A study (Montalvo et al. 2020) examined 433 bridge collapses. Of these 433 bridges, 36 bridges collapsed due to deterioration, and 82 bridges collapsed due to a collision.

1. The **percentage** of bridges collapsing due to deterioration is _____ divided by _____, or 8.3%.
2. The **odds** of a bridge collapsing due to deterioration is _____ divided by _____, or 0.091.
3. The odds of a bridge collapsing due to deterioration is 0.091. What does this mean?
    a. For every 100 bridges that do collapse due to deterioration, about 9.1 bridges do not collapse.

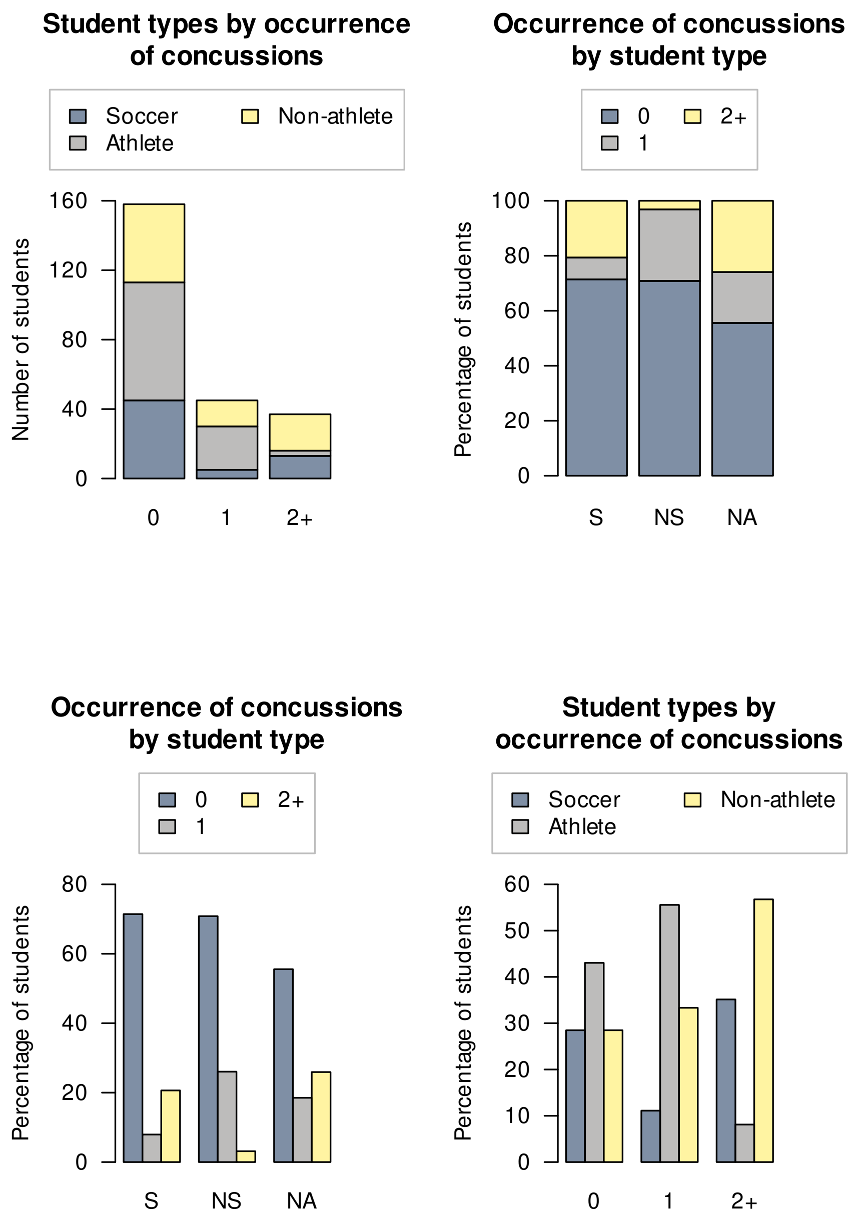

FIGURE 4.2: Four different graphs displaying the soccer-data. 'S' mean a soccer player; 'NS' means a non-soccer athlete; 'NA' means a non-athlete.

b. For every 100 bridges that do not collapse due to deterioration, about 9.1 bridges do collapse.
c. About 9% of bridges collapse due to deterioration.
d. About 90% of bridges collapse due to deterioration.

4. The percentage of bridges **not** collapsing due to deterioration is _____ divided by _____, or 91.7%.
5. The odds of a bridge **not** collapsing due to deterioration is _____ divided by _____, or 11.0.
6. The percentage of bridge collapses due to collisions is _____ divided by _____, or 18.9%.
7. The odds of a bridge collapsing due to collisions is _____ divided by _____, or 0.234.
8. True or false: The odds of an event cannot be larger than one.
9. True or false: The odds of an event cannot be smaller than one.
10. Which **one** of the following statements is true?

    a. Proportions and odds are the same thing.
    b. Percentages must be whole numbers.
    c. Odds cannot be negative

4.6 Optional drills: Computational exercises

(Answers appear in Sect. A.4)

These *optional* **Drill** exercises (repeated practice) give you practice at getting computations correct, and using your calculator. These drill questions are more about practising the underlying *mathematics* rather than the *statistics*. If you need help, please **ask**.

1. A study of drivers (McEvoy et al. 2006) in New South Wales (NSW) and Western Australia (WA) were asked if they use their phone while driving (Table 4.5).

    a. What **percentage** of people use their phone while driving?
    b. What are the **odds** that a person uses their phone while driving?
    c. What are the **odds** that a WA resident uses their phone while driving?
    d. What are the **odds** that a NSW resident uses their phone while driving?
    e. What are the **odds ratio** that a person uses their phone while driving, comparing WA residents to NSW residents?

TABLE 4.5: *The number of drivers who use their phone while driving.*

	Uses phone	Does not use phone
NSW	381	295
WA	345	326

2. The data in Table 4.6 give the number of plum cuttings that were alive or dead, and whether the rootstock was long or short (Bartlett 1935).
   a. Compute the **percentage** of cuttings that were *alive* at the end of the study.
   b. Compute the **percentage** of cuttings that were *dead* at the end of the study.
   c. Compute the **percentage** of cuttings that were *long*.
   d. Compute the **odds** that a cuttings was *alive* at the end of the study.
   e. Compute the **odds** that a cuttings was *alive* at the end of the study, only for the *long* cuttings.
   f. Compute the **odds** that a cuttings was *alive* at the end of the study, only for the *short* cuttings.
   g. Compute the **odds ratio** that a cutting was *alive* at the end of the study, comparing the *short* cuttings to the *long* cuttings.

TABLE 4.6: *The relationship between length and condition of plum rootstocks.*

	Alive	Dead
Long	240	240
Short	138	342

4.7 Optional questions

These questions are **optional**; e.g., if you need more practice, or you are studying for the exam. (Answers appear in Sect. A.4.)

4.7.1 (Optional) Odds ratios

This question has a video solution in the online book, so you can hear and see the solution.

The impact of environmental toxins is not well understood. Pamphlett (2012) examined the association between the environmental exposure of toxins and sporadic motor neuron disease (SMND). A total of 380 SMND cases and 377 controls were studied. Of the 380 SMND cases, 60 had worked with metal in the past; of the 377 controls, 33 had worked with metal in the past.

1. What is the *direction* of this study?
2. Construct a two-way table showing the relationship between disease group, and whether not the specific person had worked with metal, in the sample.
3. Using your table, compute the *odds* that a person *with* SMND had worked with metal. Interpret what this means.
4. Using your table, compute the *odds* that a person *without* SMND had worked with metal. Interpret what this means.
5. How many times greater is the odds that a person *with* SMND having worked with metal, compared to the odds that a person *without* SMND having worked with metal? (This is an *odds ratio*.)

6. Using your table, compute the *percentage* of people *with* SMND that had worked with metal.

7. Using your table, compute the *percentage* of people *without* SMND that had worked with metal.

8. A newspaper report states SMND rates are almost the same between those worked with metal and those who did not. Do you agree or disagree?

9. Sketch a bar chart to display the data.

5

Teaching Week 5: tutorial

You will learn to produce and understand:

- graphical summaries for *between*-individuals comparisons for **quantitative** data.
- numerical summaries for *between*-individuals comparisons for **quantitative** data.
- graphical summaries for *within*-individuals comparisons for **quantitative** data.
- numerical summaries for *within*-individuals comparisons for **quantitative** data.
- graphical summaries for correlational studies.
- numerical summaries for correlational studies.

The Week 5 content is essential for:

- **Task 2B**, which requires you to *produce appropriate numerical summaries and produce appropriate graphs*;
- **Quiz 2**, which includes questions about *numerical summaries and graphs*; and
- the **Exam**, which will contain questions about *numerical summaries and graphs*.

You will learn and practice the content associated with these chapters of the textbook:

- Chapter 13 (Comparing quantitative data within individuals).
- Chapter 14 (Comparing quantitative data between individuals).
- Chapter 16 (Correlations between quantitative variables).
- Chapter 17 (More details about tables and graphs).

Take careful note of which chapters are covered in this tutorial!

5.1 Quick revision

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Lunn and McNeil (1991) compared the dimensions of jellyfish at two sites at Hawkesbury River, NSW (Dangar Island; Salamander Bay) to determine the difference between the jellyfish at each site. A histogram of the breadth of jellyfish at Dangar Island Bay is shown in Fig. 5.1.

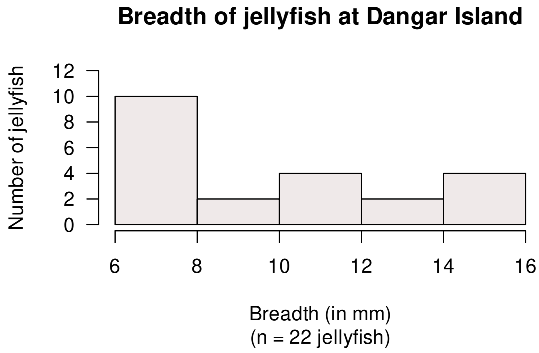

FIGURE 5.1: *A histogram of the breadth of jellyfish at Dangar Island.*

1. Two students are arguing about the median breadth. Who is correct?

Student 1 says:

> The bars in the histogram have heights of 10, 2, 4, 2 and 4. When these numbers are put in order, they are: 2, 2, 4, 4, 10. The median breadth is the median of these numbers, so the median breadth is the middle one: 4 mm is the median.

Student 2 responds:

> You have the correct answer, but for the wrong reason! There are five bars, and the middle bar is the third bar. Since the third bar has a height of 4, the median breadth is 4 mm.

2. Describe the histogram.

3. A boxplot comparing the breadths of jellyfish at Dangar Island and Salamander Bay is shown in Fig. 5.2. Describe and compare the breadths of the jellyfish.

4. Which box in the boxplot represents the Dangar Island jellyfish (shown in Fig. 5.1)?

5. If the difference was defined as $\mu_A - \mu_B$, what does this *measure*?

6. What did you learn from this study?

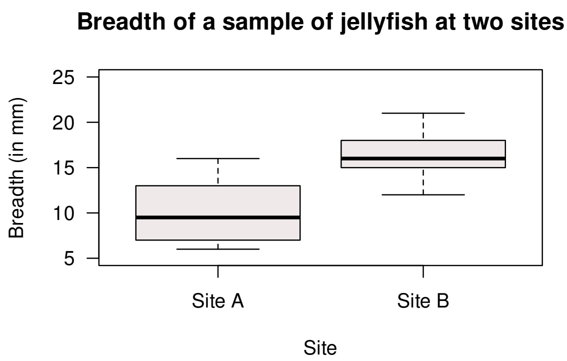

FIGURE 5.2: *A boxplot of the breadth of jellyfish at two sites.*

5.2 Class discussion

Discuss: A badly designed experiment with lots of data is worse than a small, well-designed one.

5.3 Critiquing students' numerical summaries

In this activity, you get to critique actual student Project submissions, so you can critique your own *before* submission.

1. Critique the numerical summary in Fig. 5.3, a screenshot from a student Project Report. Identify ways to improve the numerical summary.

   The RQ is '*What is the mean difference between the kicking distance using dominant and non-dominant legs?*'

⚠ You don't know what 'standard error' means yet, so don't worry about that column.

Table 1: Sample mean, standard dev., standard error and sample size of distances travelled in meters using dominant and non-dominant kicking feet and the difference of dominant value minus non-dominant value of 40 football players on the sunshine coast.

	Mean	Standard Deviation	Standard Error	Sample Size
Dominant	42.6750	6.65789	1.05271	40
Non-dominant	24.1500	6.73129	1.06431	40
Difference	18.525	-0.07343	-0.0116	0

FIGURE 5.3: *A numerical summary from a student Project.*

2. Critique the numerical summary in Fig. 5.4, a screenshot from a student Project Report. Identify ways to improve the numerical summary.

   The RQ is '*Is there a difference between the proportions of drink orders that are for coffee (rather than non-coffee) comparing the morning and the afternoon?*'

Time	Coffee n (%)	Non-Coffee n (%)	Total n	Odds (Coffee / Non-Coffee)
Morning	46 (76.7%)	14 (23.3%)	60	3.285
Afternoon	36 (60.0%)	24 (40.0%)	60	1.50

FIGURE 5.4: *A numerical summary from a student Project.*

Make some notes about important things you learnt from this activity, to remember when creating your numerical summaries for Task 2B.

5.4 Critiquing students' graphs

In this activity, you get to critique actual SCI110 Project submissions, so you can critique your own *before* submission.

1. Critique the graph in Fig. 5.5, a screenshot from a student Project Report comparing the average burn time between two brands of sparklers.

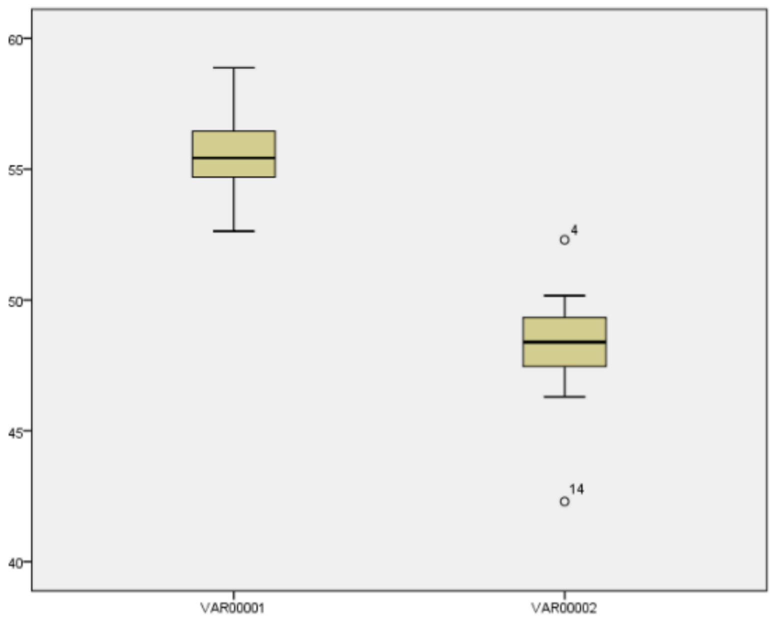

FIGURE 5.5: *A graph from a student Project.*

2. Critique the graph in Fig. 5.6, a screenshot from a student Project Report comparing the percentage of students at UniSC wearing grey-scale clothing between females and males.

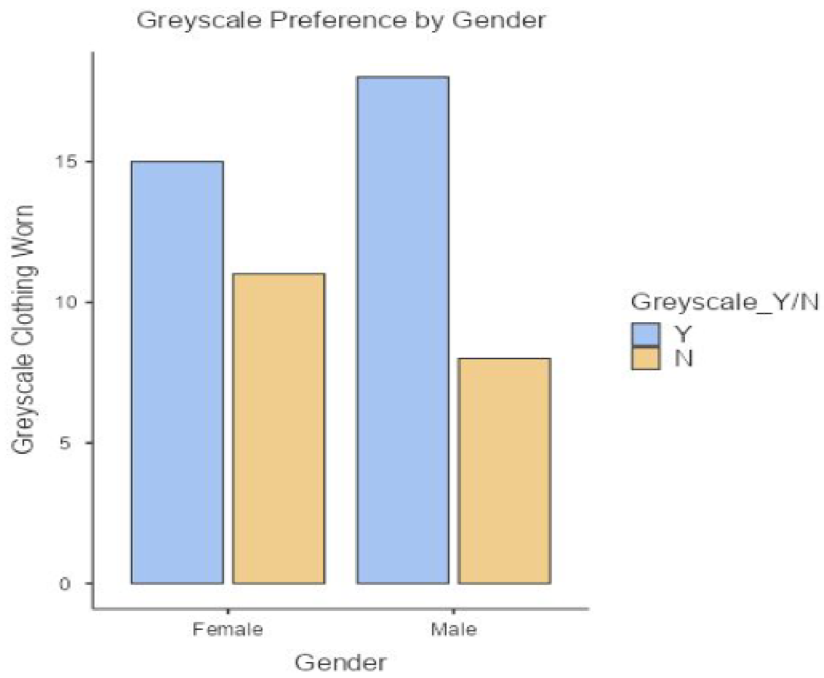

FIGURE 5.6: *A graph from a student Project.*

3. Critique the graph in Fig. 5.7, a screenshot from a student Project Report comparing the 'mean dissolve' time (they probably meant 'mean *dissolving* time') of Panadol in room temperature and cold water. Identify ways to improve the graph.

Make some notes about important things you learnt from this activity, to remember when creating your graphs for Task 2B.

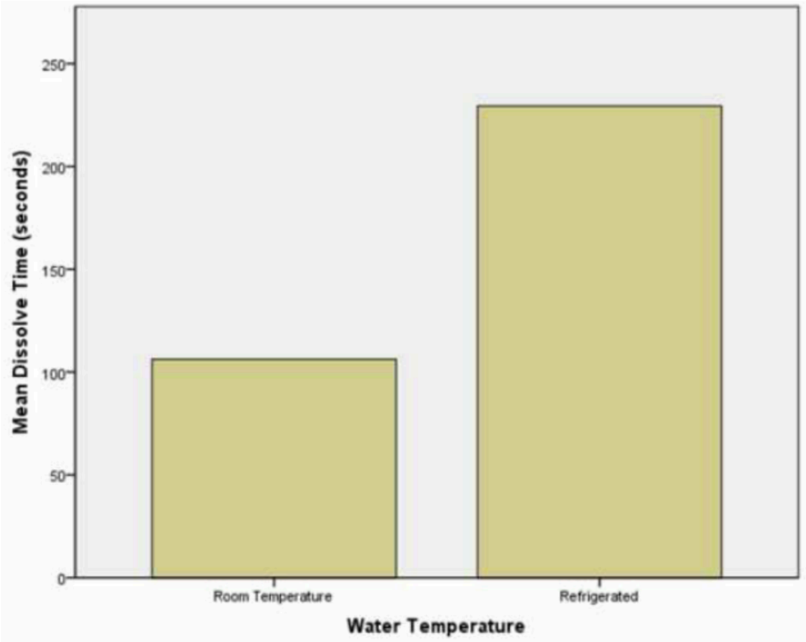

FIGURE 5.7: *A graph from a student Project.*

5.5 Understanding correlations

Answer the following questions, referring to Fig. 5.8. The four correlation coefficients r are:

A: $r = -0.95$	**B**: $r = 0.12$	**C**: $r = 0.75$	**D**: $r = 0.94$

1. For two plots, a correlation is not suitable. Which two? Explain your answer.

2. Determine which plot corresponds to which correlation coefficient (and for which two plots a correlation coefficient is not suitable).

3. Think of an example of two quantitative variables that might produce a plot with a *direction* similar to **Plot 1**.

4. Think of an example of two quantitative variables that might produce a plot with a *direction* similar to **Plot 2**.

5. Compute the values of R^2 for each plot identified in **Part 1** of this question.

5.6 Numerical summary for ruler-drop data

Your tutor will have information.

5.7 Which summary measures to use

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A study of different orthoses (orthotic devices) in children with cerebral palsy (Swinnen et al. 2017)

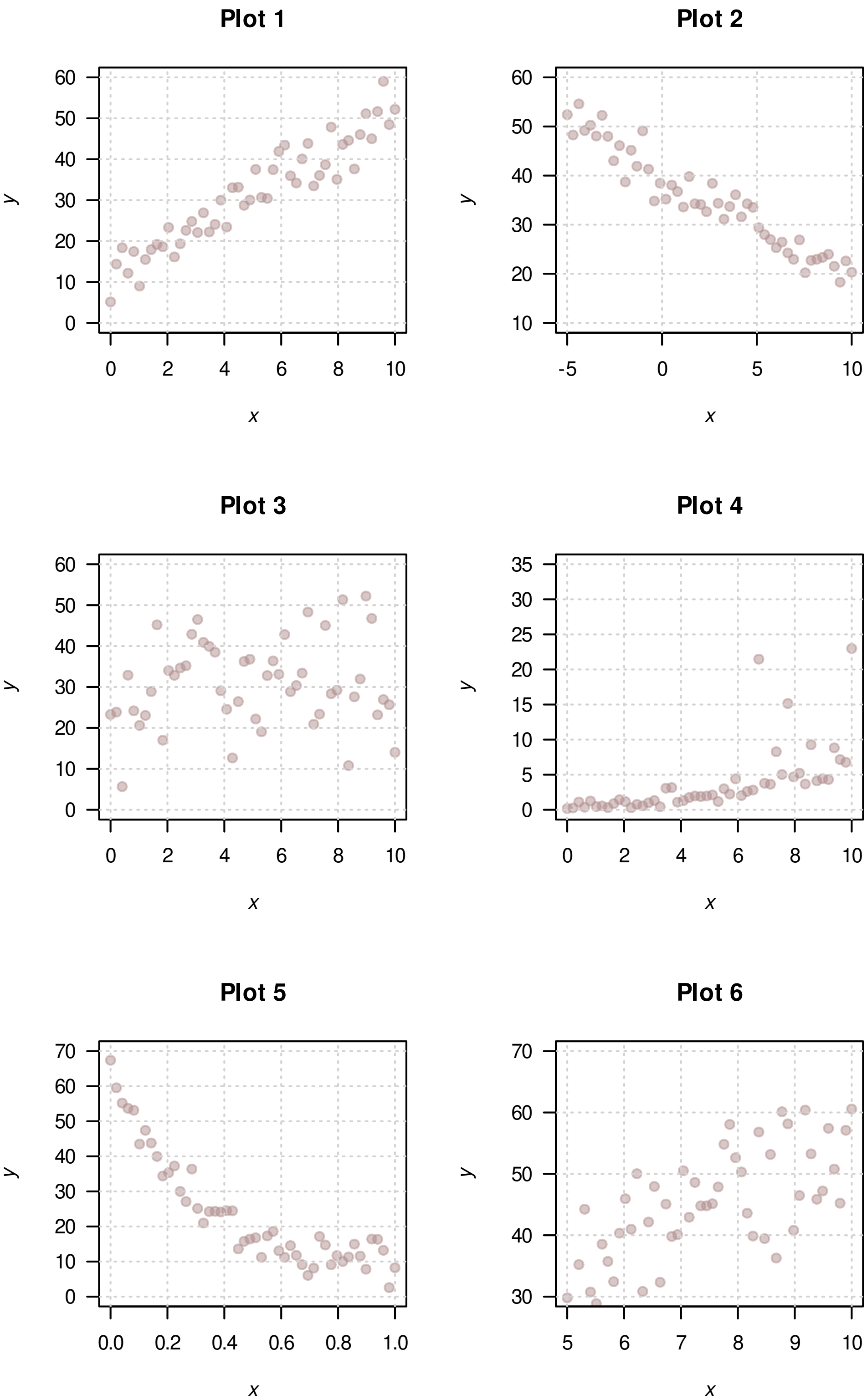

FIGURE 5.8: *Six different scatterplots.*

produced the demographic data of the participants. GMFCS is the 'Gross Motor Function Classification System', which is a five-level classification system describing the gross motor function of people with cerebral palsy, based on self-initiated movement abilities.

The data are shown in Fig. 5.9.

Table 1. Characteristics of the participants.

Participants	Gender	Age (years)	Height (cm)	Weight (kg)	GMFCS
1	M	9	136	34.5	I
2	M	7	106	16.2	II
3	M	7	129	21.1	I
4	M	12	152	40.4	I
5	M	11	146	39.3	II
6	M	5	113	18.1	I
7	M	6	112	16.7	II
8	M	8	112	19.1	I
9	M	8	138	28.6	I
10	M	6	116	19.3	I
11	F	7	113	17.6	I
12	M	11	141	34.9	I
13	M	7	136	34.5	I
14	F	9	128	21.9	I
15	F	8	133	23.0	I

FIGURE 5.9: *The data collected from a study of orthoses (GMFCS: Gross Motor Function Classification System).*

1. How many quantitative *continuous* variable(s) appear in the data set? Which are they?

2. How many qualitative *ordinal* variable(s) appear in the data set? Which are they?

3. How many quantitative *nominal* variable(s) appear in the data set? Which are they?

4. What statistics could be used to *numerically* summarise the variable **Age**?
   a. Odds.
   b. Standard deviation.
   c. Percentage.
   d. Mean.
5. What statistics could be used to *numerically* summarise the variable **Gender**?
   a. Odds.
   b. Standard deviation.
   c. Percentage.
   d. Mean.

6. What statistics could be used to *numerically* summarise the variable **GMFCS**?
   a. Odds.
   b. Standard deviation.
   c. Percentage.
   d. Mean.
7. What graph could be used to *graphically* summarise the variable **Age**?
   a. Pie chart.
   b. Stem-and-leaf plot.
   c. Scatterplot.
   d. Boxplot.
   e. Bar chart.
   f. Histogram.
8. What graph could be used to *graphically* summarise the variable **GMFCS**?
   a. Pie chart.
   b. Stem-and-leaf plot.
   c. Scatterplot.
   d. Boxplot.
   e. Bar chart.
   f. Histogram.
9. What graph could be used to *graphically* summarise the relationship between **Gender** and **Height**?
   a. Pie chart.
   b. Side-by-side bar chart.
   c. Scatterplot.
   d. Boxplot.
   e. Bar chart.
   f. Histogram.
10. What graph could be used to *graphically* summarise the relationship between **Gender** and **GMFCS**?
   a. Pie chart.
   b. Side-by-side bar chart.
   c. Scatterplot.
   d. Boxplot.
   e. Bar chart.
   f. Histogram.

6

Teaching Week 6: tutorial

You will learn to:

- identify different methods for computing probabilities.
- compute probabilities in simple situations.
- understand the role of probability models for describing sampling variation.
- describe sampling variation, and understand the role played by the sample size.
- find the probability associated with a value on a normal distribution.
- find the value associated with a probability in a normal distribution.

The Week 6 content is essential for:

- understanding the content that follows, *all* of which relies on probability, understanding sampling variation, and using normal distributions;
- **Quiz 3**, which includes questions about *probability, sampling variation, and normal distributions*; and
- the **Exam**, which will contain questions about *probability, sampling variation, and normal distributions*.

You will learn and practice the content associated with these chapters of the textbook:

- Chapter 18 (Probability).
- Chapter 19 (Sampling variation).
- Chapter 20 (Models and normal distributions).

You will need to access the normal distribution tables for this tutorial. These Tables can be found in the textbook appendices, and in this document in Appendices B.1 and B.2.

6.1 Quick revision

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A new method for evaluating bridge loads (O'Brien et al. 2018) used a simulation to compare the new method to an existing method. For the simulation, they modelled the gross vehicle mass (GVM) of trucks as having a normal distribution, with a mean of 13 tonnes and a standard deviation of 1.3 tonnes.

The Isuzu F-Series trucks are rated as having a GVM between 10.7 and 26.0 tonnes (depending on the configuration; in 2024).

1. What is the z-score for the lightest truck of 10.7 tonnes?
2. What is the z-score for the heaviest truck of 26.0 tonnes?
3. What does a *negative* z-score mean in this context?
4. **True** or **false**: A standard error measures the size of the difference between the **sample** proportion and the **population** proportion. Explain.

6.2 Class discussion

Discuss: Outliers should always be removed from a dataset before analysis.

6.3 Normal distributions and IQs

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Standard IQs in current use (despite their flaws) are designed so that the IQ scores of the *population* have a normal distribution with a mean of 100, and a standard deviation of 15.

1. Draw a labelled picture of the normal distribution of the IQ scores in the *population*.

2. For a person with an IQ of 130, how many IQ points are they from the mean?

3. For a person with an IQ of 130, how many *standard deviations* is their IQ from the mean?

4. Draw a labelled picture of the normal distribution of the IQ scores, showing the proportion of the population with IQ scores of 85 and greater.

5. For a person with an IQ of 85, how many IQ points are they from the mean?

6. For a person with an IQ of 85, how many *standard deviations* is their IQ from the mean?

7. Approximately, what proportion of the population has an IQ greater than 85? (Use the 68–95–99.7 rule.)

6.4 Normal distributions and heights

In the *National Nutrition Survey* (NNS) (conducted with the 1995 *National Health Survey* (NHS)), trained nutritionists measured the heights of the respondents (Australian Bureau of Statistics 1995).

The values quoted are the population means μ and standard deviations σ of the models used for heights.

1. Take a wild guess: What percentage of your peers (people your age and gender) do you think would be *shorter* than you? ________%.
2. Write down your own height: ________cm.
3. From Fig. 6.1, write down the mean and standard deviation for people in your *age group* and *gender*. Using this information, compute the z-score for your height, relative to people in your age group and gender. Explain what this means.

4. Heights have an approximate normal (bell-shape) distribution, so the 68–95–99.7 rule applies. **Using the** 68**–**95**–**99.7 **rule**, *roughly* estimate the percentage of people of your age and gender that are *shorter* than you. (Hint: Draw a picture!)

5. Using **the normal distribution tables** in the Appendix, together with the information from **Fig. 6.1**, compute the percentage of people of your age and gender that are *shorter* than you.

Age group (years)	Mean cm	Standard deviation cm	Age group (years)	Mean cm	Standard deviation cm
		MALES			FEMALES
18–24			**18–24**		
Measured height	178.4	6.6	Measured height	163.9	6.6
25–44			**25–44**		
Measured height	176.3	6.8	Measured height	162.9	6.4
45–64			**45–64**		
Measured height	174.0	6.7	Measured height	161.2	6.3
65 and over			**65 and over**		
Measured height	170.4	6.5	Measured height	156.9	6.3

FIGURE 6.1: *Part of the ABS report on heights of Australians.*

6. Using the 68–95–99.7 rule, the middle 95% of *females* 18 *and over* are within what height range? (For this group, the mean is $\mu = 161.4$ cm, and the standard deviation is $\sigma = 6.7$ cm, drawn from Fig. 6.1.)

7. For this part, we want to just look at *females* 18 *and over* again. Answer the following questions.

    a. Compute the *percentage* of females 18 and over that are *shorter* than 171 cm.

    b. What are the *odds* that a female 18 and over is *shorter* than 171 cm?

c. Compute the *percentage* of females 18 and over that are *taller* than 171 cm?

d. What are the *odds* that a female 18 and over is *taller* than 171 cm?

e. Compute the *percentage* of females 18 and over that are *between* 170 and 180 cm?

f. What are the *odds* that a female 18 and over is *between* 170 and 180 cm?

8. Approximately 20% of females 18 and over are shorter than what height?

6.5 Probability

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For the following, indicate *how* you could estimate the probability (classical; relative frequency; subjective) in these situations. (**Note:** You are not asked to *compute* the probability.)

1. The probability that a given UniSC student lives in Nambour.

2. The probability that a given Nambour resident is a UniSC student.

3. The probability that Labor wins the next federal election.

4. The probability that you will throw a dart and hit the bullseye.

5. The probability that a roulette wheel spins a 33.

Explain the difference between the meaning of the **first two** statements.

6.6 Random coin tosses

In this activity, we see an example of a sampling distribution that can be described by a normal distribution.

Use the website https://www.random.org/coins/ (you can use the QR code to get there quickly) to flip **ten** Australian one-dollar coins at random (see Fig. 6.2). Click *Games* from the top, then *Coin flipper*, then select the options to toss 10 coins, and use an Australian $1 coin.)

Repeat this process numerous times (if you are in a class, each student can repeat the process numerous times so you get a large number of tosses), and complete Table 6.1.

FIGURE 6.2: Using the online random coin-tosser.

TABLE 6.1: *Tossing coins.*

Proportion of heads in 10 tosses	How many times observed
0.0 (0 heads)	
0.1 (1 head)	
0.2 (2 heads)	
0.3 (3 heads)	
0.4 (4 heads)	
0.5 (5 heads)	
0.6 (6 heads)	
0.7 (7 heads)	
0.8 (8 heads)	
0.9 (9 heads)	
1.0 (10 heads)	
Number of tosses:	

1. Use this data to create a histogram of the proportion of heads. How would you describe the histogram?

2. I also 'tossed' 10 coins, but repeated this random process 400 times. My histogram is shown in Fig. 6.3. How would you describe the histogram?

3. Sketch the theoretical *sampling distribution* of the sampling proportion. What does this sketch show you?

4. How would this *sampling distribution* change if we looked the proportion of heads in 50 tosses of a coin (rather than 10 tosses)?

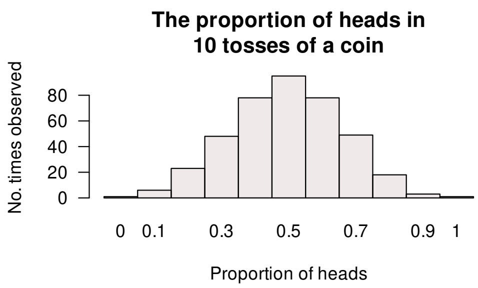

FIGURE 6.3: *The histogram of the proportion of heads in* 10 *tosses, for* 400 *repetitions.*

6.7 Optional questions

These questions are **optional**; e.g., if you need more practice, or you are studying for the exam. (Answers appear in Sect. A.6.)

6.7.1 (Optional) Working with normal distributions

This question has a video solution in the online book, so you can hear and see the solution.

The Southern Oscillation Index (SOI) is a unitless[1] climatological measure that is easily computed, and has been shown to be related to the weather conditions in eastern Australia (Stone and Auliciems 1992; Dunn 2001).

The daily SOI has an approximate normal distribution, and is designed to have a mean of 0 and a standard deviation of 10.

1. Draw a rough, labelled sketch of the theoretical distribution of the SOI values.

[1]That is, it is not measured in kilograms or seconds etc. It is just a number; in fact, it is a bit like a z-score.

2. What is the probability that the daily SOI exceeds 20?

3. What are the *odds* that the daily SOI exceeds 20?

4. What is the probability that the daily SOI is less than -25?

5. What is the probability that the daily SOI is greater than -12?

6. What is the probability that the daily SOI is between -10 and 20?

7. The SOI is *less than* what value about 80% of the time?

8. The SOI is *greater* than what value about 35% of the time?

9. What is the probability that the daily SOI exceeds 55?

10. The SOI is *greater* than what value about 50% of the time?

7

Teaching Week 7: tutorial

You will learn to:

- construct and understand confidence intervals for one proportion.
- construct and understand confidence intervals for one mean.

The Week 7 content is essential for:

- **Task 2B**, which requires you to *produce CIs for your data*;
- **Quiz 3**, which includes questions about *understanding and producing CIs*; and
- the **Exam**, which will contain questions about *understanding and producing CIs*.

You will learn and practice the content associated with these chapters of the textbook:

- Chapter 21 (Introducing inference).
- Chapter 22 (Confidence intervals: one proportions).
- Chapter 23 (Confidence intervals: one mean).
- Chapter 24 (More details about CIs).

7.1 Quick revision

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1. True or false: A *confidence interval* is used to find an interval that probably (but not certainly) contains the *sample* value.
2. True or false: All *confidence interval* are 95% confidence intervals.
3. True or false: A *standard error* is calculated from population data, but a *standard deviation* is calculated from sample data.
4. True or false: CIs answers estimation-type RQs.
5. True or false: A 95% confidence interval will contain approximately 95% of the observations.

7.2 Class discussion

Discuss: If we have *greater* confidence in our answer, we would expect to have a *smaller* range of values to choose from.

7.3 Concepts

Claire and Jake were wondering about the mean number of matches in a box. The boxes contain this statement:

> An average of 45 matches per box.

They purchased a carton containing 25 boxes of matches, and Jake counted the number of matches in *one* of those 25 boxes. There was 44 matches.

"Oh wow. Just wow..." said Jake. "They lie. There's only 44 matches in this box."

1. Identify the two concepts that Jake is confusing. How would you clarify Jake's confusion?

2. Then, Jake and Claire counted the number of matches in *each* of the twenty-five boxes. Claire found that the mean number of matches per box was 44.9 matches, and the standard deviation was 0.124. Jake says, "A mean of 44.9 matches per box? You can't have 0.9 of a match! That's dumb." How would you respond?

3. "The claim is 45 matches per box on average, but the mean really is 44.9!" said Jake. "They're liars! Liar, liar, pants on…"
   Identify the two concepts that Jake is confusing. How would you clarify Jake's confusion?

4. What two broad reasons could explain why the sample mean is *not* 45?

5. Claire says, "The mean won't be *exactly* 45 in every sample of 25 boxes! Let's work out the confidence interval."
   Why does Claire think a CI is needed? What will it tell them?

6. Compute an approximate 95% confidence interval for the mean for Claire's sample.

7. "Aha—I told you so!
   They *are* absolutely lying! Your confidence interval doesn't even include their mean of 45!" said Jake. "The manufacturer *must* be lying! I'm calling my lawyer!"
   Is Jake correct that the manufacturer *must* be lying? Why or why not? What does the CI *mean*?

8. In this scenario, what does $\bar{x}$ represent? What is the *value* of $\bar{x}$?

9. In this scenario, what does μ represent? What is the *value* of μ?

7.4 Standard error for a proportion

The formula for the **standard error for the sample proportion** is:

$$\text{s.e.}(\widehat{p}) = \sqrt{\frac{\widehat{p} \times (1 - \widehat{p})}{n}}.$$

In this formula:

1. What does $\widehat{p}$ represent?

2. What does n represent?

3. What is "s.e." an abbreviation for?

7.5 CIs for one proportion

Endotracheal intubation (ETI) is a method for maintaining an open airway or as a method for administering certain drugs for ill patients. ETI is widely used for airway management of children in the out-of-hospital setting, but for many years little evidence was available on the effect of using ETI. The BVM ('ball valve mask') method is also used.

Gausche et al. (2000) examined the use of ETI for airway management of children in the out-of-hospital setting.

1. For the **BVM sample**, 123 out of the 404 patients survived. For those receiving BVM, compute the sample *odds* of a patient who survived.

Context Endotracheal intubation (ETI) is widely used for airway management of children in the out-of-hospital setting, despite a lack of controlled trials demonstrating a positive effect on survival or neurological outcome.

Objective To compare the survival and neurological outcomes of pediatric patients treated with bag-valve-mask ventilation (BVM) with those of patients treated with BVM followed by ETI.

Design Controlled clinical trial, in which patients were assigned to interventions by calendar day from March 15, 1994, through January 1, 1997.

Setting Two large, urban, rapid-transport emergency medical services (EMS) systems.

Participants A total of 830 consecutive patients aged 12 years or younger or estimated to weigh less than 40 kg who required airway management; 820 were available for follow-up.

Interventions Patients were assigned to receive either BVM (odd days; n = 410) or BVM followed by ETI (even days; n = 420).

Main Outcome Measures Survival to hospital discharge and neurological status at discharge from an acute care hospital compared by treatment group.

FIGURE 7.1: *Part of the Abstract from Gausche et al. (2000).*

2. For the **BVM sample**, 123 out of the 404 patients survived. For those receiving BVM, compute an estimate of the population *proportion* of patients who survive.

3. Explain the difference between the meaning of the two above calculations.

4. In another sample of 404 patients who received the BVM treatment, would we expect to find *exactly* 123 surviving? Why or why not?

5. Compute the *standard error* of $\hat{p}$, which indicates the sampling variability in the estimate of p. Explain what is meant by the term ‘sampling variation’ in this context.

6. Every time a new sample is chosen, the sample will likely yield a different value for $\hat{p}$. Sketch the distribution that shows how much the value of $\hat{p}$ is likely to change from one sample to the next.

7. Compute an approximate 95% confidence interval for the population proportion based on the sample proportion.

8. Write a sentence communicating this confidence interval.

9. If the researchers wished to estimate the true survival proportion to within give-or-take 0.01 with 95% confidence, would the sample size need to be *larger* or *smaller*?

10. Confirm that the conditions necessary for this calculation to be statistically valid are met.

7.6 CIs for one mean

In 2011, *Eagle Boys Pizza* ran a campaign that claimed (among many other claims) that Eagle Boys pizzas were 'Real size 12-inch large pizzas' in an effort to out-market *Dominos Pizza*.

Eagle Boy's made the data behind the campaign publicly available (Dunn 2012). A summary of the diameters of a sample of 125 of Eagle Boys' large pizzas is shown in Fig. 7.2.

Descriptives

	DiameterInches
N	125
Missing	0
Mean	11.486
Median	11.449
Standard deviation	0.247
Minimum	10.465
Maximum	12.228

FIGURE 7.2: *Summary statistics for the diameter of Eagle Boys' large pizzas; jamovi.*

1. What do μ and $\bar{x}$ represent in this context?

2. Write down the *values* of μ and $\bar{x}$.

3. Write down the *values* of σ and s.

4. Compute the value of the standard error of the mean.

5. Explain the difference in *meaning* between s and s.e.$(\bar{x})$ here.

6. If someone else takes a sample of 125 Eagle Boys pizzas, will the sample mean be 11.486 inches again (as it is in this sample)? Why or why not?

7. Draw a picture of the approximate sampling distribution for $\bar{x}$.

8. Compute an approximate 95% confidence interval for the mean pizza diameter.

9. Write a statement that communicates your 95% CI for the mean pizza diameter.

10. What are the **statistical** validity conditions?

11. Which of these conditions must we **assume** are met for this CI to be **statistically** valid? How does Fig. 7.3 help, if at all? Explain.
    - The sample size is greater than about 25.
    - The population has a normal distribution.
    - The population standard deviation is known.
    - The sample has a normal distribution.

12. Do you think that, on average, the pizzas do have a mean diameter of 12 inches in the population, as Eagle Boy's claim? Explain.

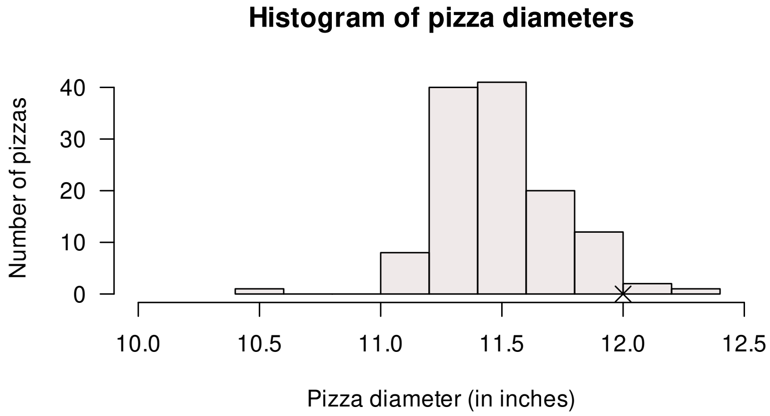

FIGURE 7.3: *Histogram for the diameter of Eagle Boys' large pizzas. The cross is the claimed diameter of 12 inches.*

7.7 Optional drills: Computational exercises

(Answers appear in Sect. A.6)

These *optional* **Drill** exercises (repeated practice) give you practice at getting computations correct, and using your calculator. These drill questions are more about practising the underlying *mathematics* rather than the *statistics*. If you need help, please **ask**.

The *standard error* for a sample proportion is calculated as

$$\text{s.e.}(\hat{p}) = \sqrt{\frac{\hat{p}\times(1-\hat{p})}{n}}.$$

The standard error quantifies how much the sample proportion is likely to vary from sample to sample.

Compute the standard error in these situations:

1. When $n = 25$ and $\hat{p} = 0.70$.

2. When $n = 100$ and $\hat{p} = 0.25$.

3. When $n = 32$ and $\widehat{p} = 0.42$.

4. When $n = 53$ and $\widehat{p} = 0.814$.

Will the CIs be statistically valid in the above situations?

8

Teaching Week 8: tutorial

You will learn to:

- explain and use the decision-making process.
- conduct and understand hypothesis tests for one proportion.
- conduct and understand hypothesis tests for one mean.
- explain and understand P-values.

The Week 8 content is essential for:

- **Task 2B**, which requires you to *conduct a hypothesis test for your data*;
- **Quiz 3**, which includes questions about *understanding and conducting hypothesis tests*; and
- the **Exam**, which will contain questions about *understanding and conducting hypothesis tests*.

You will learn and practice the content associated with these chapters of the textbook:

- Chapter 25 (Making decisions).
- Chapter 26 (Hypothesis tests: one proportion).
- Chapter 27 (Hypothesis tests: one mean).
- Chapter 28 (More details about hypothesis testing).

8.1 Quick revision

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1. True or false: Hypothesis tests are used to *prove* or *disprove* the null hypothesis.
2. True or false: Hypothesis are always written in terms of population parameters.
3. True or false: Alternative hypotheses may be one- or two-tailed, depending on the **data**.
4. True or false: Hypotheses should be constructed without any knowledge of the data.
5. True or false: The sampling distribution shows how the values of the sample statistic are likely to vary from sample to sample (when the null hypothesis is true).

8.2 Class discussion

Discuss: Confidence intervals are more informative than P-values.

8.3 The decision-making process

8.3.1 Part a

Fill in the blanks in this piece of text, using words and phrases from Table 8.1.

- The decision-making process begins with making an assumption about the population __________.
- This means we know what to __________ from the sample __________.
- We never know exactly what value of the statistic we will see in the sample, because of __________.
- But we can have some of idea of what values are reasonable to expect.
- Then we take the __________ (that is, we make the observations).

- Then we __________ what the sample statistic that we observed… to the sample statistic we expected.
- If what we observe is inconsistent with what was expected, then the the assumption is __________ true.
- However, if what we observe is __________ with what was expected, then the the assumption is __________ true.

TABLE 8.1: *Words and phrases to insert.*

compare	probably	expect	sampling variation	consistent
unlikely to be	sample	parameter	statistic	

8.3.2 Part b

Fill in the blanks in this piece of text, using words and phrases from Table 8.2.

- Suppose I wish to see if a six-sided die is fair.
- My __________ would be that the proportion of odd numbers is 0.5. This value is a __________.
- The __________ would be "the proportion of odd numbers in the sample".
- We would __________ that the value of the statistic would be near 0.5 (but maybe not exactly 0.5).
- If I roll the die 100 times and observe **53** odd numbers, this seems __________ with what I would expect if the assumption was true.
- If I roll the die 100 times and observe **zero** odd numbers, this seems __________ with what I would expect if the assumption was true.

TABLE 8.2: *Words and phrases to insert.*

assumption	consistent	inconsistent
statistic	expect	parameter

8.4 Test for one proportion

According to Statistica.com, 30% of individuals in Sweden wore contact lenses in 2020 (the highest percentage in Europe).

Suppose a group of 35 Swedish computer programmers at a large company includes 12 that wear contact lenses. Is there evidence in this sample that a *greater* proportion of Swedish programmers wear contact lenses at this company, compared to the general Swedish population?

1. Write the hypotheses for conducting the appropriate hypothesis test.

2. Compute the value of the appropriate sample statistic.
3. Compute the value of the appropriate standard error of the statistic. What does this value *mean*?
4. Compute the value of the appropriate test statistic.
5. Write a conclusion (including a CI).
6. What assumptions are needed for this test to be statistically valid, and externally valid?

8.5 Tests for one mean

In 2011, *Eagle Boys Pizza* ran a campaign that claimed that Eagle Boys' pizzas were 'Real size 12-inch large pizzas' (Dunn 2012). Eagle Boys made the data from the campaign publicly available.

A summary of the diameters of a sample of 125 of their large pizzas is shown in Fig. 8.1. We would like to test the company's claim, and ask the RQ:

> For Eagle Boys' pizzas, is mean diameter actually 12 inches, or not?

Descriptives

	DiameterInches
N	125
Missing	0
Mean	11.486
Median	11.449
Standard deviation	0.247
Minimum	10.465
Maximum	12.228

FIGURE 8.1: *Summary statistics for the diameter of Eagle Boys' large pizzas; jamovi.*

1. What is the **parameter** of interest?

2. Write down the values of $\bar{x}$ and s.

3. Determine the value of the standard error of the mean.

4. Explain the difference in meaning between s and s.e.$(\bar{x})$ in this context.

5. Write the hypotheses to test if the mean pizza diameter is 12 inches.

6. Is the alternative hypothesis one- or two-tailed? Why?

7. Draw the normal distribution that shows how the *sample mean pizza diameter* would vary by chance, *even if* the population mean diameter was 12 inches.

8. Compute the t-score for testing the hypotheses.

9. What is the approximate P-value using the 68–95–99.7 rule?

10. Write a conclusion. (The CI was found in the Teaching Week 7 tutorial.)

11. Is it reasonable to assume the *statistical* validity conditions are satisfied? How does Fig. 8.2 help, if at all?

12. Do you think that the pizzas do have a mean diameter of 12 inches in the population, as Eagle Boys' claim? Explain.

8.6 Approximate P-values

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Assuming the tests are statistically valid, what is the approximate P-value for a *two*-tailed t-test if:

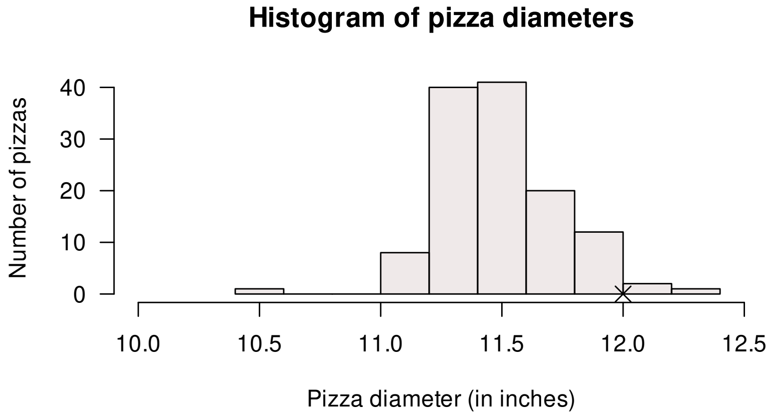

FIGURE 8.2: *Histogram for the diameter of Eagle Boys' large pizzas. The cross represents the claimed diameter of* 12 *inches.*

1. $t = -2.14$.
2. $t = 1.90$.
3. $t = 3.15$.
4. $t = -1.20$.
5. $t = -1.59$.
6. $t = 6.71$.

(**Hint**: Use the 68–95–99.7 rule.)

How would your answers change if these t-scores came from a *one-tailed* test?

9

Teaching Week 9: tutorial

You will learn to:

- construct CIs and conduct hypothesis tests for the mean difference in paired situations.
- construct CIs and conduct hypothesis tests for comparing two independent means.

The Week 9 content is essential for:

- **Task 2B**, which requires you to *produce CIs and conduct a hypothesis test for your data*;
- **Quiz 4**, which includes questions about *understanding and producing CIs, and understanding and conducting hypothesis tests*; and
- the **Exam**, which will contain questions about *understanding and producing CIs, and understanding and conducting hypothesis tests*.

You will learn and practice the content associated with these chapters of the textbook:

- Chapter 29 (CIs and tests: mean differences (paired data)).
- Chapter 30 (CIs and tests: comparing two means)

9.1 Quick revision

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Some of the summary information regarding the number of decayed, missing and filled teeth (DMFT) is shown below. The researchers wanted to compare the mean DMFT for coeliacs and non-coeliacs.

TABLE 9.1: *Coeliacs and dental cavities.*

	Sample size	Mean	Standard deviation	Standard error
Coeliac (C)	23	8.39	4.4	0.92
Non-coeliac (NC)	23	8.17	4.1	0.86
Difference		0.22		1.3

An *exact* 95% CI is given as for the difference is -2.32 to 2.76.

1. Using the 68–95–99.7 rule gives a slightly different CI. Why?
2. True or false: The difference is computed as the number of DMFT for coeliacs minus non-coeliacs.
3. True or false: One of the values for the CI is a negative value, which must be an error, since a negative number of DMFT is impossible.
4. We are 95% confident that the difference between the population means is:
5. True or false: The *null* hypothesis is H_0: $\mu_C - \mu_{NC} = 0$.
6. True or false: The *alternative* hypothesis is H_0: $\mu_C - \mu_{NC} > 0$.
7. True or false: Since the two sample means are different, we reject the null hypothesis.
8. Suppose the difference between the means was computed as $\mu_C - \mu_{NC}$. What does this *measure*?

9. Both samples sizes are less than 25. What does this mean for the statistical validity of the CI?

9.2 Class discussion

Discuss: If the mean of Group A is $\bar{x}_1 = 12.0$ cm and the mean of Group B is $\bar{x}_2 = 14.2$ cm, then the two means are clearly different.

9.3 Matching RQs and hypotheses

In Table 9.2, match the RQs with the appropriate *null hypothesis*. In this question, p represents the proportion of subs shorter than 12-inches one longer than 12-inches as appropriate, and μ represents the mean length of a sub.

Then, match the RQs with the appropriate *alternative hypothesis*.

TABLE 9.2: *Match RQs with the null (and then the alternative) hypotheses.*

Research question	**Hypothesis**
1. At Subway, is the mean length of a 12-inch sub really 12 inches?	A. $p_{\text{white}} - p_{\text{wholemeal}} = 0$
2. At Subway, is the mean length of a 12-inch sub *different* for white	B. $\mu_{\text{white}} - \mu_{\text{wholemeal}} = 0$
and wholemeal subs?	C. $\mu_{\text{white}} = \mu_{\text{wholemeal}} = 12$
3. At Subway, is the proportion of 12-inch subs that are shorter than	D. $\bar{x}_{\text{white}} - \bar{x}_{\text{wholemeal}} = 0$
12 inches *different* for white and wholemeal subs?	E. $\bar{x}_{\text{white}} = \bar{x}_{\text{wholemeal}} = 12$
4. At Subway, is the mean length of a 12-inch sub *longer* for white	F. $\mu = 12$
and wholemeal subs?	G. $\mu_{\text{white}} - \mu_{\text{wholmeal}} \neq 0$
	H. $p_{\text{white}} = p_{\text{wholemeal}} < 12$
	I. $\mu \neq 12$
	J. $\bar{x} \neq 12$
	K. $p_{\text{white}} - p_{\text{wholemeal}} \neq 0$
	L. $\mu_{\text{white}} - \mu_{\text{wholemeal}} > 0$

9.4 CIs and tests for mean differences (paired data)

Guirao et al. (2017) examined the difference between 2-minute walk test score (2MWT) for 10 patients before *and* after receiving a prosthetic implant. (The 2MWT measures how far participants can walk in two minutes, in metres.)

The 2MWT for ten amputees *with* and *without* an implant are shown in Fig. 9.1. The researchers asked:

> For this type of amputee, is the mean 2MWT **improved** (i.e., longer distances walked) after using the implant?

	2MWT (m)	
	Without Imp	With Imp
Patient 1 (TON, female, 44 years)	112	144
Patient 2 (VASC, male, 64 years)	84	96
Patient 3 (VASC, male, 62 years)	96	120
Patient 4 (VASC, male, 87 years)	78	108
Patient 5 (TON, male, 39 years)	132	140
Patient 6 (TON, female, 57 years)	72	86
Patient 7 (TON, male, 63 years)	80	94
Patient 8 (TON, male, 37 years)	106	134
Patient 9 (VASC, female, 46 years)	108	146
Patient 10 (TON, female, 37 years)	116	165

***FIGURE 9.1:** Two-minute Walk Times (2MWT) for 10 patients with implants (With Imp) and without implants (Without Imp), in metres.*

1. Suppose the difference were computed as **With Imp** values *minus* the **Without Imp** values, what would this measure?

2. What type of RQ is implied: descriptive, relational, repeated-measures or correlational?

3. What do μ_d and $\bar{d}$ represent in this context?

4. Explain why these data should be analysed as *paired* data.

5. Compute the *changes* in 2MWT for each patient.

6. Although it doesn't really matter, *why* does it probably makes more sense to compute the **With Imp** values minus the **Without Imp** values?

7. Using the statistics mode on your calculator, compute the sample mean difference $\bar{d}$ and the sample standard deviation of the differences s_d.

8. Compute the standard error of the mean difference s.e.$(\bar{d})$.

9. Explain the *meaning* of the standard error of the mean difference in this context.

10. If another sample of ten subjects were studied, would the same sample mean difference $\bar{d}$ be computed? How much variation would be expected in the sample mean differences found from different samples?

11. Draw the approximate sampling distribution of $\bar{d}$ that shows how the value of $\bar{d}$ varies.

12. Compute an approximate 95% confidence interval for the population mean difference in 2MWT.

13. Do you think the population 2MWT changes because of the prosthetic, on average?

14. Which of the following is the correct null hypothesis for this RQ? **Why** are the others incorrect? Is the test one- or two-tailed? What is the alternative hypothesis?
    - $\mu_{\text{Without Implant}} - \mu_{\text{With implant}} = 0$
    - $\mu_{\text{Difference}} = 0$
    - $\mu_{\text{Difference}} > 0$
    - $\mu_{\text{Without Implant}} = 0$
    - $\mu_{\text{With implant}} = 0$
    - $\mu_{\text{Without Implant}} - \mu_{\text{With implant}} < 0$
    - $\mu_{\text{Difference}} < 0$
    - $\mu_{\text{Without Implant}} < 0$
    - $\mu_{\text{With implant}} > 0$

15. Write down the value of the t-statistic, then estimate the P-value for testing the hypotheses (using Fig. 9.2).

16. Explain why the t-score from jamovi is a negative value, but the t-score from SPSS (Fig. 9.3) is a positive value.

17. Write a statement that communicates the result of the test.

18. What conditions must be met for this test to be valid?

19. Is it reasonable to assume the assumptions are satisfied? How does Fig. 9.4 help, if at all?

20. What did you learn from this study?

Paired Samples T-Test

			statistic	df	p	Mean difference	SE difference
Without Implant	With Implant	Student's t	−6.04	9.00		−24.90	4.12

Descriptives

	N	Mean	Median	SD	SE
Without Implant	10	98.40	101.00	19.55	6.18
With Implant	10	123.30	127.00	26.49	8.38

FIGURE 9.2: *Output from jamovi for the 2MWT example, partially edited.*

9.5 CI and test for the ruler-drop

Your tutor *may* decide to do this activity.

Paired Samples Test

		Paired Differences							
					95% Confidence Interval of the Difference				
		Mean	Std. Deviation	Std. Error Mean	Lower	Upper	t	df	Sig. (2-tailed)
Pair 1	With implant – Without implant	24.900	13.034	4.122	15.58	34.224	6.041	9	

FIGURE 9.3: *Output from SPSS for the 2MWT example, partially edited.*

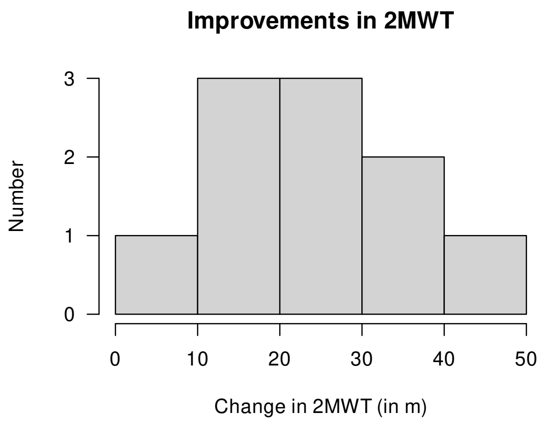

FIGURE 9.4: *Histogram of differences for the increases in 2MWT with implant.*

9.6 CIs and test for difference between two sample means

Researchers were interested in the impact of diet on the lifetime of rats:

> For rats, is the mean lifetime *shorter* for rats on a *free-choice* diet compared to rats on a *healthy, restricted* diet?

Berger et al. (1988) compared the lifetime of rats on a healthy, restricted diet (106 rats) and on a free-eating diet (89 rats). The data set is large, so only an extract of the data is shown (Fig. 9.5).

1. Explain why this study compares two *independent* groups.

2. Carefully define **parameter** of interest, including the *direction*. What does this definition of the difference measure?

3. Use the jamovi output (Fig. 9.7) to prepare a numerical summary table.

4. Use the jamovi output to write down the appropriate 95% CI, to estimate the *difference* between the population means.

5. Which of these short statements best communicates this CI? **Why** are the other statements incorrect?
   a. The sample mean lifetime is between 223.34 and 346.13 days.
   b. The difference between the population mean lifetimes is between 223.34 and 346.13 days.
   c. We are 95% sure that the difference between the sample mean lifetimes is between 223.34 and 346.13 days.
   d. We are 95% sure that the difference between the population mean lifetimes is between 223.34 and 346.13 days.
   e. If we repeated everything many times, 95% of the CIs constructed would contain the difference between the population means.
6. The best statement above for communicating the CI is missing an important piece of information. Write an improved statement communicating the CI.

7. Explain the difference between the *meaning* of what is displayed in the two graphs in Fig. 9.6.

8. Write down the hypotheses being tested. Is this a one- or a two-tailed test? Explain.

9. What are *two* possible reasons why the sample mean lifetimes of rats on the two diets are different?

10. Write down the t-score and the appropriate P-value, using the output in Fig. 9.7 .

11. Calculate the t-score (using the standard error as given in the output), and show it is the same value as given in the output.

12. Make a conclusion, in context.

13. What conditions are necessary for the CI and test to be statistically valid?

14. Is it reasonable to assume these conditions are satisfied? How does Fig. 9.6. help, if at all?

15. What if all these rats only came from only 20 litters?

	Lifetime	Diet
1	105	Restricted
2	193	Restricted
3	211	Restricted
4	236	Restricted
5	302	Restricted
6	363	Restricted
7	389	Restricted
8	390	Restricted
9	391	Restricted
10	403	Restricted
11	530	Restricted
12	604	Restricted

Lifetime	Diet
796	Free-eating
799	Free-eating
801	Free-eating
806	Free-eating
807	Free-eating
815	Free-eating
836	Free-eating
838	Free-eating
850	Free-eating
859	Free-eating
894	Free-eating
963	Free-eating

FIGURE 9.5: *Part of the data for the rat lifetime example (left: the start of the data; right: the end of the data).*

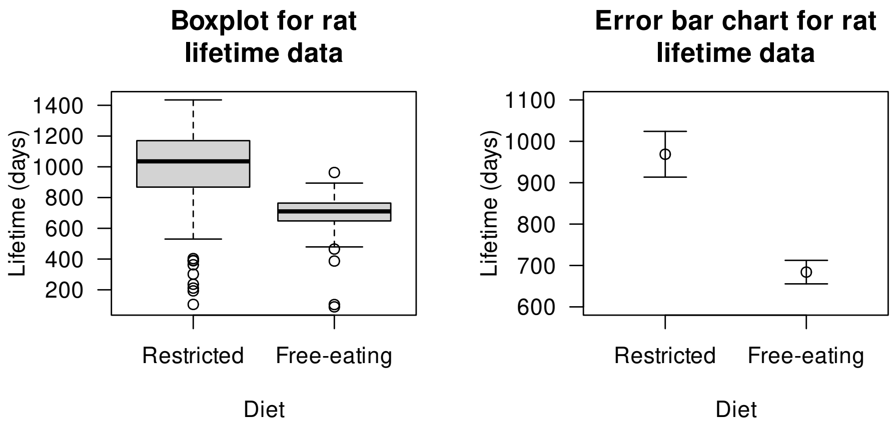

FIGURE 9.6: *Boxplot (left panel) and error-bar chart (right panel) for the rat lifetime data.*

Independent Samples T-Test

							95% Confidence Interval	
		statistic	df	p	Mean difference	SE difference	Lower	Upper
Lifetime	Student's t	8.66 [a]	193.00	< .001	284.73	32.87	219.91	349.56
	Welch's t	9.16	154.94	< .001	284.73	31.08	223.34	346.13

[a] Levene's test is significant (p < .05), suggesting a violation of the assumption of equal variances

Group Descriptives

	Group	N	Mean	Median	SD	SE
Lifetime	Restricted	106	968.75	1035.50	284.58	27.64
	Free-eating	89	684.01	710.00	134.08	14.21

FIGURE 9.7: *The jamovi output summarising the rat lifetimes data.*

9.7 Optional questions

These questions are **optional**; e.g., if you need more practice, or you are studying for the exam. (Answers appear in Sect. A.9.)

9.7.1 (Optional) Paired t-test

Hébert-Losier et al. (2023) recorded double-legged jumping distance for 80 healthy people, when they wore shoes and were barefoot. The data are too large to show here, but use the jamovi output (Fig. 9.8) to answer the following questions.

1. What do the differences *mean* (as given in the output)?
2. Create a numerical summary table for the data.
3. Determine 95% CI for the difference in height of the two jumps.
4. Conduct a hypothesis test to determine if a mean difference exists in the population.

Paired Samples T-Test

			statistic	df	p	Mean difference	SE difference
Shoes	Barefoot	Student's t	−2.231	79.000	0.029	−0.636	0.285

Note. H_a $\mu_{\text{Measure 1 - Measure 2}} \neq 0$

Descriptives

	N	Mean	Median	SD	SE
Shoes	80	32.115	32.450	9.206	1.029
Barefoot	80	32.752	33.733	8.933	0.999

FIGURE 9.8: *The jamovi output summarising the jumping data.*

9.7.2 (Optional) Two-sample t-test

Strayer and Johnston (2001) examined the reaction times, while driving, for students from the University of Utah (Agresti and Franklin 2007). In one study, students were randomly allocated to one of two groups:

- one group *used* a mobile phone while driving in a driving simulator; and
- one group *did not use* a mobile phone while driving in a driving simulator.

The reaction time for each student was measured. The data are shown below. Use the information in Table 9.4 to answer this RQ:

> For students, is there a difference between the mean reaction time while driving when using a mobile phone and when *not* using a mobile phone?

Be sure to include an approximate 95% CI as well as the hypothesis test results.

TABLE 9.3: *Reaction times (in milliseconds) for students using, and not using, mobile phones while driving.*

Reaction time: using phone						
636	600	609	554	578	688	527
623	542	559	626	560	679	536
615	554	595	501	525	960	
672	543	565	574	647	558	
601	520	573	468	456	482	

Reaction time: not using phone						
557	506	626	436	617	539	512
572	648	626	642	528	523	449
457	485	426	476	578	479	
489	610	585	586	472	535	
532	444	487	565	485	603	

TABLE 9.4: *Summary information for the reaction-time data.*

	Mean	**Sample size**	**Std dev.**	**Std error**
Using phone	533.59	32	65.36	11.554
Not using phone	585.19	32	89.65	15.847
Differences	51.59			19.612

10

Teaching Week 10: tutorial

You will learn to:

- construct CIs and conduct hypothesis tests for ORs.
- construct CIs and conduct hypothesis tests for the difference between two proportions.
- estimate sample sizes for finding CIs.
- selecting an appropriate analysis for a given situation.

The Week 10 content is essential for:

- **Task 2B**, which requires you to *produce CIs and conduct a hypothesis test for your data*;
- **Quiz 4**, which includes questions about *understanding and producing CIs, understanding and conducting hypothesis tests, and estimating sample sizes*; and
- the **Exam**, which will contain questions about *understanding and producing CIs, understanding and conducting hypothesis tests, and estimating sample sizes*.

You will learn and practice the content associated with these chapters of the textbook:

- Chapter 31 (CIs and tests: comparing two odds or proportions)
- Chapter 32 (Finding sample sizes for CIs)
- Chapter 34 (Selecting an analysis)

Take careful note of which chapters are covered in this tutorial!

10.1 Quick revision

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An ecologist is studying two different grasses to help combat soil salinity, by comparing to a new grass (Grass A) to a native grass (Grass B). She uses 50 different sites, allocating the two grasses at random to the sites (25 sites for each grass).

After 12 months, the ecologist records whether the soil salinity at each site has improved, and hence computes the *odds* that each grass will improve the salinity. She finds a statistically significant difference between the odds in the two groups.

Which of these statements is *consistent* with this conclusion?

1. The OR $= 4.1$ and $P = 0.36$.
2. The OR $= 4.1$ and $P = 0.0001$.
3. The OR $= 0.91$ and $P = 0.36$.
4. The OR $= 0.91$ and $P = 0.0001$.

How would the other statements be interpreted then?

10.2 Class discussion

Discuss: When comparing two groups, is it better to focus on differences in *proportions*, *odds*, or *percentages*?

10.3 Consistency

Suppose a researcher asked the RQ:

For Australians, are the *odds* of people with mosquito bites the same for people sitting near a citronella candle as for people sitting near an ordinary wax candle?

1. Which would be the appropriate null hypothesis?
   a. The odds of a person being bitten by a mosquito is the same for people near citronella candles and wax candles.
   b. The proportion of people being bitten by a mosquito is the same for people near citronella candles and wax candles.
   c. Either of the above.
2. Which would be the appropriate CI to produce?
   a. A CI for the odds ratio.
   b. A CI for the difference in proportions.
   c. Either of the above.
3. Which would be the appropriate hypothesis test?
   a. A hypothesis tests for the difference between proportions.
   b. A hypothesis tests for the odds ratio.
   c. Either of the above.

10.4 CI and test for odds ratios

The timing of pubertal maturation can vary, which can have impacts upon behaviour. Duncan et al. (1985) studied 'the relationships between maturational timing and body image, school behavior, and deviance.' Sample data were collected from

…children and youth of the entire United States drawn by the National Center for Health Statistics… known as the National Health Examination Survey (1966–1970). Data were collected on… adolescents' physical and psychological status…

The researchers asked the RQ;

For US children, is the odds of a boy maturing late the *same* as the odds of a girl maturing late?

Part of the data are shown entered into jamovi (Fig. 10.1)

1. For the 2 864 males in the sample, 352 were classified as maturing late.
   For the girls, 336 of the 2 664 matured late. Use this information to construct a two-way table of sex against maturation time (Table 10.1).

TABLE 10.1: *Maturation and gender.*

	Matured late	**Did not mature late**	**Total**
Males			
Females			
Total			**5528**

	Sex	Maturation
2859	Male	Not late
2860	Male	Not late
2861	Male	Not late
2862	Male	Not late
2863	Male	Not late
2864	Male	Not late
2865	Female	Late
2866	Female	Late
2867	Female	Late
2868	Female	Late
2869	Female	Late
2870	Female	Late
2871	Female	Late

FIGURE 10.1: *Some of the maturation-data entered into jamovi.*

2. What graphical summary could be used to display the information? Sketch this display.

3. Among the boys, compute the *odds* of maturing late. Interpret this value.

4. Among the girls, compute the *odds* of maturing late. Interpret this value.

5. From the table, compute the odds ratio of a boy maturing late compared to a girl maturing late. Interpret this value, using the software output (Fig. 10.2).

6. What is the **parameter** of interest (make sure you specify the direction)?

7. Write down the OR and the 95% CI for the OR from the jamovi output. What does it mean?

8. Compile the numerical summary table for these data (Table 10.2).
9. Are boys generally more likely to mature later than girls? Perform a hypothesis test.

χ^2 Tests

	Value	df	p
χ^2	0.131	1	0.717
z test difference in 2 proportions	−0.363		0.717
N	5528		

Comparative Measures

		95% Confidence Intervals	
	Value	Lower	Upper
Difference in 2 proportions	−0.003 [a]	−0.021	0.014
Odds ratio	0.971	0.828	1.139

[a] Rows compared

FIGURE 10.2: *The jamovi output from the maturation study*

TABLE 10.2: *Maturation and gender: numerical summary*

	Proportion maturing late	Odds maturing late	Sample size
Males			
Females			
	Diff.:	Odds ratio:	

10.5 CIs and tests for ORs

Singh and Prasad (2016) examined the mortality rates of lower-limb amputation, and factors that may be associated with mortality. As part of the study, the researchers recorded the five-year mortality rate (the number of amputees who died within five years of amputation) and whether or not the person used an artificial limb.

The RQ was:

> For this type of amputees, is the odds of being alive after five years the same for those who used an artificial limb, and those who did not?

A total of 105 subjects were used: 35 died within five years, and 70 were still alive after five years. In addition, 65 subjects used an artificial limb, and 40 did not.

1. After five years, 49 people using an artificial limb were alive. Construct the 2×2 table (Table 10.3) displaying the number of people alive or dead after five years (in columns, say) and whether or not they used an artificial limb or not (in rows).

TABLE 10.3: *Five-year mortality for artifical limb users.*

	Alive	**Dead**	**Total**
Used artificial limb			
Did not use artificial limb			
Total			**105**

2. In Table 10.3, *why* is the use (or not) of artificial limbs listed in the *rows* (rather than the *columns*)?

3. For a person using an artificial limb, compute the odds of being alive after five years. For a person *not* using an artificial limb, compute the odds of being alive after five years. Then compute the odds ratio, complete Table 10.4, and carefully explain what the OR means.

4. Write down the 95% CI for the population OR using the software output (Fig. 10.3).

5. Perform a hypothesis test to compare the odds of being alive after five years for both groups (Fig. 10.3):

a. Write the hypotheses.

b. Write down the value of χ^2.

c. What z-score is this equivalent to?

d. What is the approximate P-value using the 68–95–99.7 rule?

e. What is the exact P-value as reported by software?

f. Write a conclusion.

TABLE 10.4: *Five-year mortality and use of an artificial limb: numerical summary.*

	Proportion alive after 5 years	**Odds of being alive after 5 years**	**Sample size**
Use artificial limb			
Did not use artifical limb			
	Diff.:	Odds ratio:	

10.6 Identify the correct analysis

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χ^2 Tests

	Value	df	p
χ^2	5.84	1	0.016
N	105		

Comparative Measures

		95% Confidence Intervals	
	Value	Lower	Upper
Odds ratio	2.77	1.20	6.41

FIGURE 10.3: *The jamovi output for the question on lower-limb amputees.*

For each of the following scenarios:

a. determine the appropriate method of analysis (if none are appropriate, say so then identify the correct analysis):

   i. A χ^2 test;
   ii. A z-test for comparing two proportions;
   iii. A paired t-test for the mean difference;
   iv. A two-sample t-test
   v. None of the above.

b. carefully define the **parameter** of interest.

1. Populations of the scaled quail have been decreasing. An observational research study (Pleasant et al. 2006) examined the nests of scaled quails to determine some of the reasons for this decline.

In each nest, the researchers recorded the maximum temperature during the first 21 days posthatch. Two groups of nests were compared: when the hen was present at 21 days posthatch ($n = 17$ nests), and when the hen was absent at 21 days posthatch ($n = 37$ nests). The aim was to see if the *mean* maximum temperature was different for both groups (hen present; hen absent).

What is the appropriate method of analysis? Carefully define the parameter of interest.

2. Many studies have observed an association between the presence of airborne allergens and people reporting asthma (e.g., Targonski et al., 1995). To better understand the risks of airborne allergens, a study examined the records for numerous people who died of asthma.

   For each person, they recorded the sex of the person (female or male) who died of asthma, and whether they died in pollen season or non-pollen season.

   The aim was to see if the **odds** of dying of of asthma in pollen season was the same for females and males.

   What is the appropriate method of analysis? Carefully define the parameter of interest.

3. An experimental study (VanLeit and Crowe 2002) "evaluated the impact of an 8-week psychosocial occupational therapy intervention program for mothers who have children with disabilities" (p. 402).

   Each mother was evaluated on their satisfaction in how they are spending their time, using the (quantitative) TUA scale. Mothers have their TUA score measured *before* and *after* the intervention to see if the intervention made a difference to TUA scores (on average).

   What is the appropriate method of analysis? Carefully define the parameter of interest.

4. A experimental study (Ingenbleek and McCully 2012) compared vegetarians to non-vegetarians on many criteria.

   In one example, each person in the study was described as vegetarian or not, and each person's serum lipids concentration (in mmol/L) was assessed.

   What is the appropriate method of analysis? Carefully define the parameter of interest.

5. A study compared the proportion of females in retail working overtime, to the proportion of males on retail working overtime.

   What is the appropriate method of analysis? Carefully define the parameter of interest.

6. A study compared the relationship between the VO2 max (a measure of an individual of an individual physical fitness) and 3 km running velocity, for a set of male long-distance runners.

   What is the appropriate method of analysis? Carefully define the parameter of interest.

10.7 Sample size calculations

This question has a video solution in the online book, so you can hear and see the solution.

Shortly after metric units were introduced to Australia in 1977, a lecturer wondered how accurately students could estimate lengths using the metric measurements (Hand et al. 1996).

The aim of the study was to determine if, on average, students could correctly guess the width of the hall (which was 13.1 m).

The 44 individual students' guesses had a standard deviation of 7.145 m. Suppose the Professor wanted to try the study again with another groups of students, and wanted to estimate the population mean to within 0.5 m.

What size sample would Prof. Lewis need?

11

Teaching Week 11: Tutorial

You will learn to:

- read and understand research.
- write about research.

The Week 11 content is essential for:

- **Task 2B**, which requires you to *write about your own research*;
- **Quiz 4**, which includes questions about *understanding the language of research, and understanding extracts from published research*; and
- the **Exam**, which will contain questions about *understanding the language of research, and understanding extracts from published research*.

You will learn and practice the content associated with these chapters of the textbook:

- Chapter 35 (Reporting and writing research).
- Chapter 36 (Reading and critiquing research).

11.1 Quick revision

In a study of high strength–high ductile concrete, Ranade et al. (2015) measured (among other things) the load required to create the first visible cracks in concrete samples under various strains rates.

Under a strain rate of $10\,\text{s}^{-1}$, the $n = 6$ concrete samples produced a **mean** first-crack strength of 12.4 MPa with a **standard deviation** of 2.8 MPa.

One student computes the approximate 95% CI for the mean first-crack strength as 12.4 ± 2.8 MPa. Another student computes the approximate 95% CI for the mean first-crack strength as $12.4 \pm (2 \times 2.8)$, or from 6.8 to 18.0 MPa.

Both students are incorrect.

1. Explain *why* Student 1 is incorrect, and explain what the student actually calculated.
2. Explain *why* Student 2 is incorrect, and explain what the student actually calculated.
3. Compute the correct (approximate) 95% CI.

11.2 Class discussion

Discuss: A 'statistically significant' result means a real discrepancy or effect.

11.3 Critiquing past students' reports 1

The extracts below are from an actual student Project Report. Critique these extracts, and suggest better ways of communicating the information.

From a study comparing the mean distance an iron golf club and a wooden golf club can hit the ball:

- **Research question**: For casual golfers, will using a metal club rather than a wooden club result in the ball travelling a further distance?
- **Summaries**: The paired means are different with the iron golf club being moderately higher than the wooden club
- **Results (1)**: The 95% CI for the difference in distance in metres (wooden verse iron) differs from 34.74 – 43.25 further for shots with the Iron

- **Results (2)**: Very strong evidence exists in the paired sample (paired $t = 18.406$; $n = 50$; two tailed; $P = 0.000$) of a population mean difference between the wooden and iron golf clubs…

11.4 Critiquing past students' reports 2

The extracts below are from an actual student Project Report. Critique these extracts, and suggest better ways of communicating the information.

From a study comparing resting heart rates between male and female SCI110 students:

- **Null hypothesis**: Among current UniSC students studying SCI110, male and female resting heart rates are the same.

- **Results**: The sample presents evidence ($\chi^2 = 30.0$; two tailed $P = 0.517$) that the mean resting heart rate does not have a significant difference between males and females (males mean: 72.6 bpm; females mean: 77.2 bpm).

11.5 Tests for ORs

This question has a video solution in the online book, so you can hear and see the solution.

A study (Tanagawa and Shigematsu 1998) examined the choice of airway devices used for nontraumatic, out-of-hospital cardiac arrest patients, to evaluate the success and failure of insertion for various devices.

The data for success and failure at insertion is given in Table 11.1, taken from available patient documentation.

TABLE 11.1: *Data from the airways study.*

	Succeed	**Fail**
Esophageal gastric tube airway (EGTA)	545	49
Laryngeal mask (LM)	2701	315

1. Define p as the population proportion of successful attempts at insertion. Write the hypotheses to be tested.

2. Use the software output in Fig. 11.1 to test the hypotheses. Write a proper conclusion to communicate the results.

3. Are the results likely to be statistically valid? Explain.

4. The study was designed to enable medics to choose the best airway device.
   Would you recommend EGTA or LM? Why? Would you like any more information before making your choice? Explain.

χ^2 Tests

	Value	df	p
χ^2	2.64	1	0.104
N	3610		

Comparative Measures

		95% Confidence Intervals	
	Value	Lower	Upper
Odds ratio	1.30	0.95	1.78

FIGURE 11.1: *The jamovi output for the table from Tanagawa and Shigematsu (1998).*

11.6 Critiquing past students' reports 3

The extracts below are from an actual student Project Report. Critique these extracts, and suggest better ways of communicating the information.

From a study comparing the size of lilly pilly leaves on different sides of the tree:

- **Research question**: Among Weeping Lily Pily trees located at the University of the Sunshine Coast, do leaves that grow on the sunrise (east) side become longer than those growing on the sunset (west) side of the tree?

- **Null hypothesis**: The leaves on the east and west sides of Weeping Lily Pily trees are same, therefore no relationship exists between the sunset (east) and sunrise (west) side of the tree and leaf length.

- **Results**: The table of results in Fig. 11.2.

- **Methods**: Chose the 5 trees at the university that receive both easterly and westerly sunlight conditions The 4th leaf on the branch will be chosen 5 times on each side of the tree

	N	Mean	Standard Deviation	Standard Error mean
East and west	50	1.50	0.505	0.071
Length of leaves (mm)	50	80.94	18.710	2.646

FIGURE 11.2: A table from a student project.

11.7 Critiquing past students' reports 4

The extracts below (including typos) are from an actual student Project Report. Critique these extracts, and suggest better ways of communicating the information.

From a study comparing how well sports-playing students can estimate distance compared to non-sports-playing students:

- **Research question**: Among UniSC Sippy Downs students, are individuals who regularly play sports have closer estimations to the actual measurement when identifying the width of a regular path compared to individuals who don't play sports regularly?

- **Null hypothesis**: The Null Hypophysis for our research question is that students who play sports regularly at the UniSC Sippy down campus have an increased level of perception when it comes to estimating distances.

- **Results**: This sample presents strong evidence in support of the Null Hypothesis that sport status has an affect on distance estimation (regular played sport: mean 176.36 and non regular played sport: mean 172.34).

- **Discussion**: From our original research question we have concluded that it supports the Null Hypothesis as it has a large P-value (0.36: one-tailed). Therefore we have proven that playing sports regularly has an effect on the accuracy of distance estimation.

12

Teaching Week 12: Tutorial

You will learn to:

- construct CIs and conduct hypothesis tests for correlation coefficients.
- understand linear equations.
- estimate the parameters in simple linear regression models.
- find the parameters in simple linear regression models from software.
- construct CIs and conduct hypothesis tests for regression coefficients.

The Week 12 content is essential for:

- the **Exam**, which will contain *many* questions about *understanding correlation and regression*.

You will learn and practice the content associated with these chapters of the textbook:

- Chapter 33 (Correlation and regression)

12.1 Quick revision

iconmonstr-computer-6-240.png

A study of how weeds spread (Khan et al. 2018) studied various factors about vehicles that might carry seeds. The researchers found a correlation between the number of grams of mud on a vehicle and the number of seeds carried by the vehicle.

In autumn, the correlation coefficient was given as $r = 0.931$.

1. In this study, what variable would be the x-variable?
2. In this study, what variable would be the y-variable?
3. What is the value of R^2?
4. The regression equation is given as $\hat{y} = 137.4 + 0.3459x$. If 700 gs of mud is found on the car, how many seeds are predicted to be carried by the vehicle?
5. In this regression equation, the slope means:
6. The P-value for the regression slope was 0.002. What does this mean?

12.2 Class discussion

Discuss: The values of y (the data) and $\hat{y}$ (the model predictions) should be as close as possible.

12.3 Understanding correlations and regression

On *each* scatterplot in Fig. 12.1 for which a linear relationship is appropriate (there are *four*!):

1. Draw or estimate the 'best' straight regression line (where appropriate).
2. From the line in **Part 1.** above, estimate the **slope** for each line.
3. From the line in **Part 1.** above, estimate the **intercept** for each line.
4. Then write down an estimate of the regression line equation.

12.4 Interpreting regressions

I was wondering about how the age of second-hand cars impact their price. On 25 June, 2014, I searched Gum Tree, for `Toyota Corolla` in the 'Cars, Vans & Utes' category. The age and the price of each (second-hand) car was recorded from the first two pages of results that were returned.

I then restricted the data to cars 15 years old or younger (and removed one 13-year-old Corolla advertised for sale for $390,000, assuming this was an error). I then produced the scatterplot in Fig. 12.2.

1. Describe the relationship displayed in the graph in words.

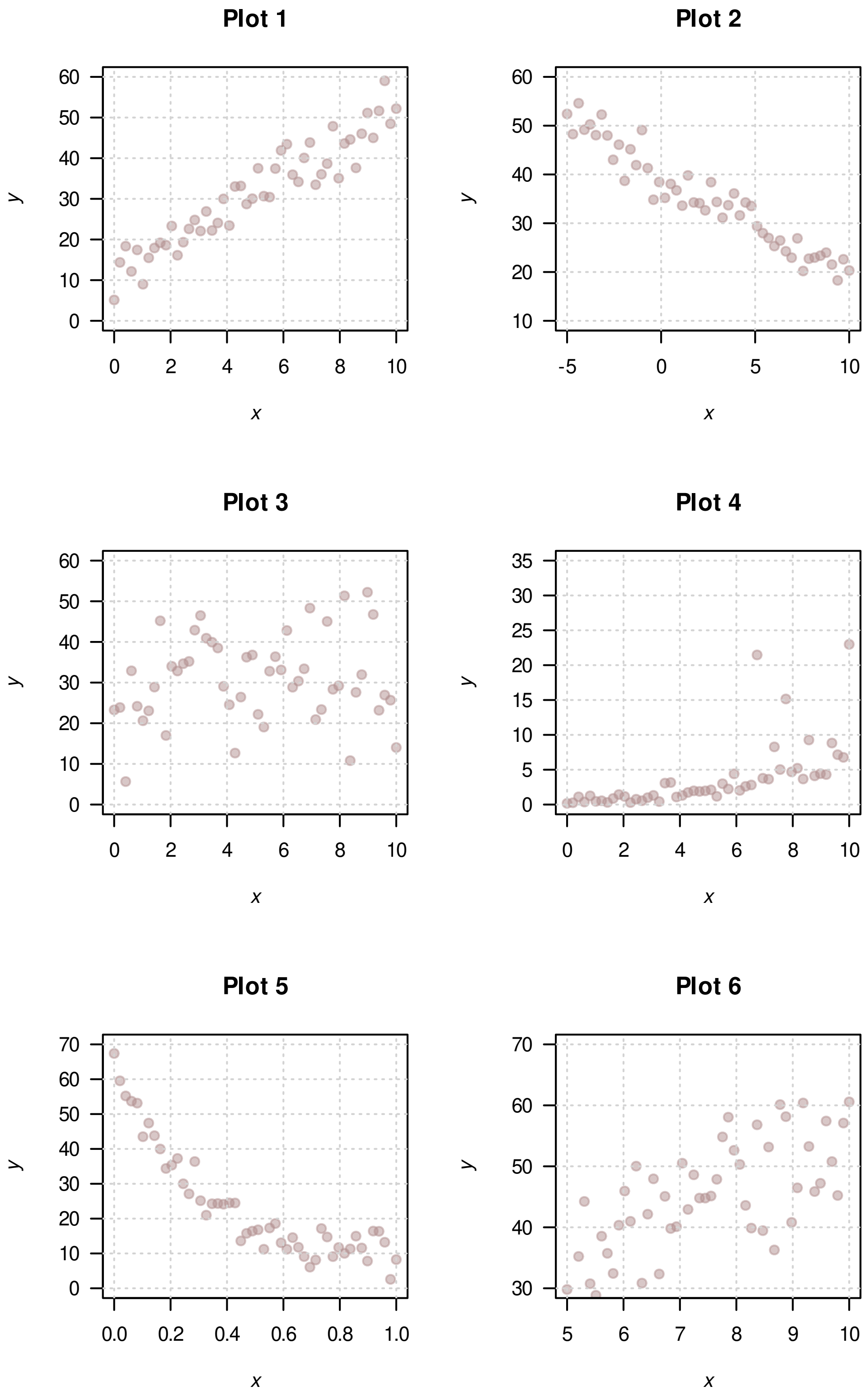


FIGURE 12.1: *Six different scatterplots.*

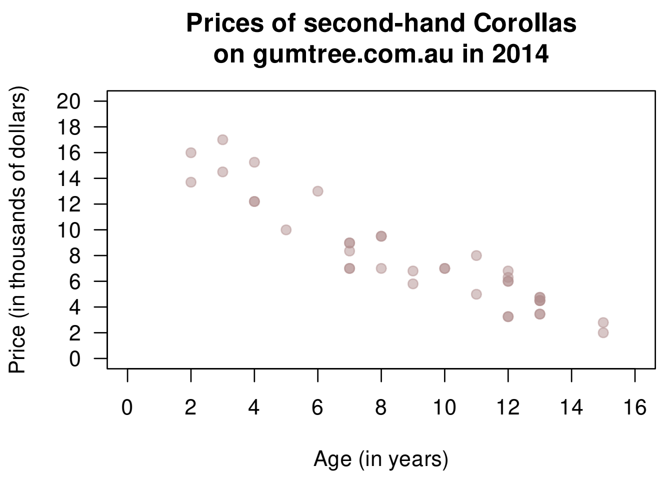


FIGURE 12.2: *The price of second-hand Toyota Corollas as advertised on Gum Tree on 25 June 2014 plotted against age ($n = 38$).*

2. What else could influence the price of a second-hand Corollas?
3. With a ruler or another straight edge (such as a book), draw an estimate of the regression line on the scatterplot.
4. On the scatterplot, locate a seven-year-old Corolla selling for $15,000. Would this be cheap or expensive?
5. As stated, I removed one observation: a 13-year-old Corolla for sale at $390,000. What do you think the price was meant to be listed as, by looking at the scatterplot? Explain.
6. *Estimate* the value of b_0 (the intercept) from the line you drew. What does this mean? Do you think this value is meaningful?

7. *Estimate* the value of b_1 (the slope) from the line you drew. What does this mean? Do you think this value is meaningful?

8. From the line you drew above, write down an *estimate* of the regression equation.

9. Use the software output (Fig. 12.3 and Fig. 12.4 (jamovi)) relating the price (in thousands of dollars) to age to write down the regression equation.

10. Using the software output, write down the value of r. Using this value of r, compute the value of R^2. What does this mean?

11. Use the regression equation from the software output to estimate the sale price of a Corolla that is 20 years old, and explain your answer.

12. Using the software output, perform a suitable hypothesis test to determine if there is evidence that lower prices are associated with older Corollas.

13. Compute an approximate 95% confidence interval for the population slope (use the software output in **Fig. 12.4**).

14. I could have drawn a scatterplot with Price on the vertical (up-and-down) axis and Year of manufacture on the horizontal (left-to-right) axis (**Fig. 12.5**). For this graph:
    a. What is the value of the correlation coefficient?

    b. How would the value of R^2 change?

    c. How would the value of the slope change?

    d. How would the value of the intercept change?

Correlation Matrix

		Age	PriceThous
Age	Pearson's r	—	
	p-value	—	
PriceThous	Pearson's r	−0.929	—
	p-value	< .001	—

FIGURE 12.3: *The jamovi correlation output, analysing the Corolla data.*

Model Coefficients - PriceThous

Predictor	Estimate	SE	t	p
Intercept	16.54	0.63	26.29	< .001
Age	−0.96	0.06	−15.06	< .001

FIGURE 12.4: *The jamovi regression output, analysing the Corolla data.*

12.5 Regression and correlation

Draw the regression line corresponding to $\hat{y} = 5 + 2x$ for values of x between 0 and 10.

1. Add some points to the scatterplot such that the correlation is approximately $r = 0.9$.

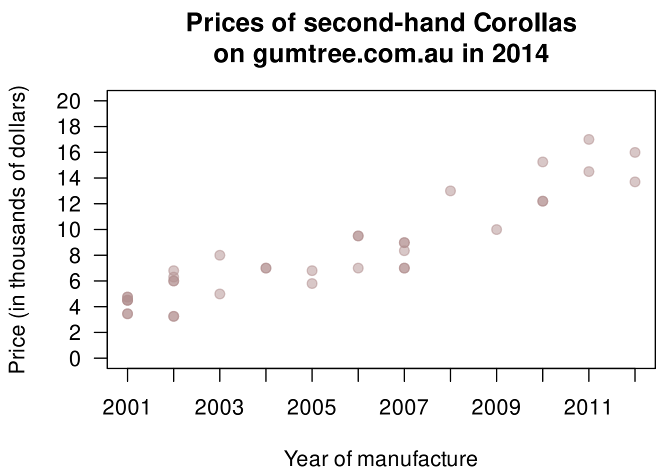


FIGURE 12.5: *The price of second-hand Toyota Corollas as advertised on Gum Tree on 25 June 2014 plotted against the year of manufacture ($n = 38$).*

2. Add some more points to the scatterplot such that the correlation is approximately $r = 0.3$.

A

Answers for exercises

This Appendix contains answers to *most* (not all) exercises. Some are fully worked, and some are only brief solutions.

- Answers: Tutorial 1 (Teaching Week 1): Sect. A.1.
- Answers: Tutorial 2 (Teaching Week 2): Sect. A.2.
- Answers: Tutorial 3 (Teaching Week 3): Sect. A.3.
- Answers: Tutorial 4 (Teaching Week 4): Sect. A.4.
- Answers: Tutorial 5 (Teaching Week 5): Sect. A.5.
- Answers: Tutorial 6 (Teaching Week 6): Sect. A.6.
- Answers: Tutorial 7 (Teaching Week 7): Sect. A.7.
- Answers: Tutorial 8 (Teaching Week 8): Sect. A.8.
- Answers: Tutorial 9 (Teaching Week 9): Sect. A.9.
- Answers: Tutorial 10 (Teaching Week 10): Sect. A.10.
- Answers: Tutorial 11 (Teaching Week 11): Sect. A.11.
- Answers: Tutorial 12 (Teaching Week 12): Sect. A.12.

A.1 Answer: TW 1 tutorial

Answer for Sect. 1.3

Ask the question. **Design** the study. **Collect** the data. **Classify** and **summarize** the data. **Analyse** the data. **Report** the results.

Answers for Sect. 1.4

1. *Neither RQ is actually helping answer the issue of the café owner!* For example, I may be able to tell the difference between diet and regular cola… but I may like them both equally (i.e., the taste rating is the same).
2. One option: For my customers, what proportion can distinguish diet and regular colas? P: The customers. O: The proportion that can distinguish regular ad diet colas. C: None. I: NO.
3. Many options: give a randomly-select cola (some regular; some diet), and record if customers can correctly identify it.

Answers for Sect. 1.5

You can use the **Glossary** in the textbook.

Answers for Sect. 1.6

Answers implied by H5P.

P: People at an American college. **O**: The *average* time taken to walk 50 m. **C**: Between the way subjects were using their phone. **I**: None: an observational study.

Explanatory variable: The way the phone is being used. **Response variable**: The time taken to walk 50 m.

Unit of observation: *Individual* subjects. **Unit of analysis**: *Individual* subjects.

Answers for Sect. 1.7

1. Many populations possible. The “best” population: the easiest, or cheapest to get? All of the options are possibilities.
2. “Relaxing”: Many options possible for quantifying this (heart rate? breathing rate? survey? *change* in heart rate before and after drinking the tea?).
3. Again, many options possible. Some options for comparing to Earl Grey tea: coffee; hot water; black tea; nothing at all; anything else but Early Grey; etc.
4. “Between before and after drinking Earl Grey tea” is *not* a between-individuals comparison because everyone in the population (sample) is treated the same way: there are not two groups in the population being compared.
5. An example RQ might be:

   “Among UniSC students (P), is the average heart rate 30 mins after drinking (O) different between those who drink a cup of Earl Grey tea and those who drinking a cup of hot coffee (C) when the drink is provided to them (I)?”

   I’m not saying this is the best; it is just an example.
6. Get volunteers; place into one of the groups (Earl Grey and coffee); give the beverage; measure heart rate after 30 mins. Note that many design issues are not covered until the next lecture.
7. An example RQ might be:

   “Among UniSC students (P), is the average heart rate 30 mins after drinking (O) is different between those who drink a cup of Earl Grey tea and those who drinking a cup of hot coffee (C)?” I’m not saying this is the best; it is just an example.
8. Based on the above RQ: Find UniSC students who drink Earl Grey tea, and others who drink coffee, and measure their heart rate 30 mins after they drink.

   It would have a lot of confounding effects to account for.
9. For my RQ above: The volume of tea or coffee; brand of tea or coffee; how made; when beverage taken; how long steeped for; how heart rate is measured; student (full-time only?); etc. You don’t need to *define* these, but if you have time you can have a go at one or two definitions.
10. Many options, but for the RQ above: type of beverage; heart rate after 30 mins (or better, the *change* in heart rate).

Answers for Sect. 1.8

1. The unit of analysis is the *floorboard*: because one board is compared to another; because the hardness is a feature of each board, not each test; because the two measurements on each board are not independent of each other (from the same board).
2. There are 5 units of analysis.
3. There are 10 units of observation.
4. Now, only one unit of analysis.
5. *Within* boards, the variation is small, apart from Board 1. *Between* boards it is larger.

Answers for Sect. 1.9.2

The answers, in alphabetical order, are: ask; analysis; analyse; collect; data; definitions; design; estimation; evidence; lurking; POCI; qualitative; quantitative; relational; units.

Answers for Sect. 1.9.1

Answers implied by H5P. Observational. Quasi-experimental (the classes are determined by the *students*, but the lecturer decides which class gets which treatment).

A.2 Answer: TW 2 tutorial

Answers for Sect. 2.3

1. Read the *Guidelines* carefully to find out!
2. The study is *observational*, but because the researchers cannot determine the C (whether the person is a smoker or not). The critical element here is C, *not* O.
3. This is a mix of both C (‘smokers and non-smokers’) and O (‘the median serum cholesterol’).
4. *External validity* only refers to whether the *sample* represents the given target *population*, which is *Australians*. Whether the results apply for the entire world is irrelevant.

5. "Serum cholesterol" is not a variable; nothing here is varying. "Serum cholesterol" is just a type of cholesterol. What actually varies–and so is the variable–is "the serum cholesterol concentration", or the "value of serum cholesterol".
6. This is *not an experiment*, since the individuals cannot be directed into the comparison groups (between smokers and non-smokers) by the researchers.
7. In the data file, each row is a unit of analysis and each column is a variable. So there will be two variables but not those listed: one column will record the smoking status (Yes/No) and one column will record the serum cholesterol concentration.
8. A confounding variable has to be related to *both* the response and explanatory variables.
9. The *observer effect* is about how the *researchers* might respond, *not* the individuals under study.

Answers for Sect. 2.4

The answers are, in alphabetical order: beach; blind; control; ethical; experimental; fire; Hawthorne; help; nominal; randomly; relational; sample; three; treatment.

Answers for Sect. 2.5

1. *Outcome*: average lifespan? Not sure how one measures 'ageing well'. *Response variable*: lifespan?
2. *Comparison*: Between eating and not eating aged cheese. *Explanatory variable*: Whether someone eats aged cheese (or a certain amount of aged cheese) or not.
3. Observational; but a guess.
4. *Outcome*: average lifespan; *response*: lifespan.
5. *Comparison*: Between mice drinking and not drinking spermidine; *explanatory*: the type of water the mice drank.
6. Experimental; researchers treated the water.
7. Possibly; 'aging well' is not the same as having an increased lifespan. Study is for *mice*, not *people* anyway.
8. To act as the control group, to measure the effect of the spermidine.
9. *Outcome*: average blood pressure; *response*: blood pressure.
10. Probably correlational, but possibly relational.
11. Observational.
12. Some answers are debatable.
    a. Probably confounding (often is).
    b. Probably neither.
    c. Probably neither.
    d. Probably confounding.
    e. Probably extraneous.
13. Not really; but 'ageing well' is very vague.
14. No; spermidine may have an effect, but probably not due to aged cheese intake.
15. No; observational.

Answers for Sect. 2.7

1. Yes: 'They were randomly allocated to take palmolein ("B9") or canola ("T4") crisps for the first 3 weeks, then (without a washout period) changed over to the other type, canola or palmolein for another 2 weeks'.
2. It has: 'the type of oil was known only by the food scientist…'
3. 'the type of oil was known only by the food scientist…'
4. Probably.
5. Subjects and researchers blinded.

A.3 Answer: TW 3 tutorial

Answers for Sect. 3.3

1. Some of the significant figures look ridiculous. Mean: 990.0791, or 990.1 tonnes. Standard deviation: 1588.514579, or 1588.5 tonnes. (Specifically, **not** 1485.919327.)
2. Median: 180.5 tonnes.
3. Splitting the data into two parts of four observations each: Q_1 is half-way between 8.0001 and 29.4 tonnes: $Q_1 = 18.7005$. Q_3 is half-way between 676.2 and 2547.3 tonnes: $Q_1 = 1611.75$. The IQR is $1611.75 - 18.7005 = 1593.05$, or about 1593 tonnes.

Answers for Sect. 3.4

Answers implied by H5P. Median largest: **Class D**. Median smallest: **Class C**. Standard deviation largest: **Class A**. Standard deviation smallest: **Class D**.

Answers for Sect. 3.5

Answers implied by H5P.

1. Volume of drinks in $375\,$mL: **Graph D**. (Students sometimes say Graph A, thinking that most cans have very similar amounts.)
2. Time in exam for **short** or **easy** exam: **Graph C**.
3. Time in exam for **long** or **hard** exam: **Graph B**.
4. The heights of females UniSC students: **Graph D**.
5. The *starting* salaries: **Graph C**. (Students sometimes say Graph B, thinking that salaries rise over time.)

Answers for Sect. 3.7

1. Female Weddell: about 260–$270\,$cm long, vary from about 200 to $310\,$cm. Slight negative skewness; possible small outlier at about 170–$180\,$cm.
2. Dotchart.
   (Too much data for a useful stem-and-leaf plot probably.)
3. Better with a title; some more labelled tick marks would be better.
4. Part of this discussion can be about what information is needed on a publicity brochure.

Answers for Sect. 3.8

1. Mean: 168.5714, or $168.57\,$m. Standard deviation: 6.966279, or $6.97\,$m. (Specifically, **not** 6.44952.) Median: $167.9\,$m. IQR: $175.0 - 162.5 = 12.5\,$m (software may give a different value).
2. Descriptive. $\bar{x} = 102.62\,$MPa; $s = 5.356\,$MPa. The median is $101.1\,$MPa.
3. Mean: 13.88778, or 13.888%. Standard deviation: 2.617402, or 2.617%. Median: 13.35%. IQR: $16.115-11.66 = 4.455\%$ (software may give a different value).

Answers for Sect. 3.9.1

1. Observational: the 'treatment' (brand of battery) is not assigned; we simply take measurements from the batteries that exist.
2. Units of observation and units of analysis: The individual batteries.
3. *Energizer*: mean: $7.36\,$h; std dev: $0.289\,$h. *Ultracell*: mean: $7.41\,$h; std dev: $0.172\,$h.
4. Ultracell batteries are *slightly* better (last longer) on average, and more consistent performers.
5. *Energizer*: median: $7.46\,$h. *Ultracell*: median: $7.48\,$h. So Ultracell batteries are *slightly* better (last longer) on average.
6. Probably median (outliers?), but mean and median are similar in any case.
7. Values are close.
   Energizer batteries take, on average, *less* time to reach 1.0 volts, so are 'worse' in this regard. They also have a lot more variation. (Based on means, the difference is $0.05\,$h, or $3\,$mins in over $7\,$h use! Based on medians, the difference is $0.02\,$h, or $1.2\,$mins!) **The *practical* difference is negligible.**
8. At the time of the study, the Ultracell batteries were substantially cheaper, and hence much better value.

A.4 Answer: TW 4 tutorial

Answers for Sect. 4.3

1. Because the gender of the person is the explanatory variable.
2. (No answer.)
3. $43 \div (43 + 67) \times 100 = 39.1\%$.
4. $39 \div (97 + 39) \times 100 = 28.7\%$.
5. $43 \div 67 = 0.642$.
6. $39 \div 97 = 0.402$.

7. $0.642 \div 0.402 = 1.6$.
8. (No answer; see online.)

Answers for Sect. 4.4

1. Because 'Number of concussions' is the response variable.
2. Type of student:nominal (with three levels). Number of concussions: *ordinal* (with three levels).
3. $45 \div 240 = 18.75\%$ of all college students in the sample have received one concussion.
4. We cannot calculate this as we **do not** have a population; we only have a sample.
5. Not sure which is best; they do different things. I'd probably prefer (c) (bottom left graph).
6. Odds (non-athlete receiving 2+ concussions) is $21 \div (81 - 21) = 21 \div 60 = 0.35$. Non-athletes are 0.35 times as likely to receive two or more concussions than fewer than two. Or: For every 100 students with fewer than two concussions, there are 35 with two or more.
7. Odds (soccer player receiving 2+ concussions) is $13 \div (63 - 13) = 13 \div 50 = 0.26$. Soccer players are 0.26 times more likely to receive two or more concussions than fewer than two concussions.
8. OR: $0.35 \div 0.26$, or about 1.35. The odds of a non-athlete receiving two or more concussions is about 1.35 times the odds of a soccer player receiving two or more concussions (that is, smaller odds for soccer players).
9. Not given.
10. Not given.
11. The better table is probably row percentages: the percentages of each type of students with given number of concussions.

Answers for Sect. 4.5

1. The *percentage* of bridges collapsing due to deterioration is *36* divided by *433*, or 8.3%.
2. The *odds* of a bridge collapsing due to deterioration is *36* divided by *397*, or 0.091.
3. The odds of a bridge collapsing due to deterioration is 0.091; this means: *b. For every* 100 *bridges that do not collapse due to deterioration, about* 9.1 *bridges do collapse*.
4. The percentage of bridges *not* collapsing due to deterioration is *397* divided by *433*, or 91.7%.
5. The odds of a bridge *not* collapsing due to deterioration is *397* divided by *36*, or 11.0.
6. The percentage of bridge collapses due to collisions is *82* divided by *433*, or 18.9%.
7. The odds of a bridge collapsing due to collisions is *83* divided by *351*, or 0.234.
8. The odds of an event cannot be larger than one: *FALSE*.
9. The odds of an event cannot be smaller than one. *FALSE*.
10. The following statement is true: *c. Odds cannot be negative*

Answers for Sect. 4.6

1. a. $(381 + 345)/1347 = 0.53898$, or about 53.9%.
   b. $(381 + 345)/(295 + 326) = 1.1691$, or about 1.17.
   c. $345/326 = 1.05828$, or about 1.06.
   d. $381/295 = 1.29153$, or about 1.29.
   e. $1.05828/1.29153 = 0.81940$, or about 0.819.
2. a. $(240 + 138)/960 = 0.39375$, or 39.4%.
   b. $(240 + 342)/960 = 0.60625$, or 60.6%.
   c. $(240 + 240)/960 = 0.5000$, or 50%.
   d. $(240 + 138)/(240 + 342) = 0.64948$, or about 0.649.
   e. $240/240 = 1.000$.
   f. $138/342 = 0.40351$ or about 0.404.
   g. $0.40351/1.000 = 0.40351$ (i.e., the answer to f divided by the answer to e).

Answers for Sect. 4.7.1

1. Backwards.
2. See Table A.1.
3. $60/320 = 0.1875$. A worker with SMND is 0.1875 times as likely to have worked with metal than not.
4. $33/344 = 0.096$. A worker with SMND is 0.096 times as likely to have worked with metal than not.

TABLE A.1: *SMND cases, and whether they had worked with metals.*

	Worked with metals	Did not work with metals	Total
SMND cases	60	320	380
Controls	33	344	377
Total	93	664	757

A.5 Answer: TW 5 tutorial

Answers for Sect. 5.3

1. A few issues: Five decimal places is to the nearest 0.01 of a mm! The standard deviation of the difference is *not* the difference between the individual standard deviations. A standard deviation cannot be negative. (Same applies to standard errors, but we aren't there yet.) Note that there is a sample size of 0 for the difference!
2. The RQ is about a *discrepancy* in proportions, or *odds*, but these are not present in the table.

Answers for Sect. 5.4

1. A few issues: vertical axis is not labelled (presumably burn time in seconds); horizontal axis is not labelled (we have no way of knowing what is happening there.) <!–
2. A few issues: this is not a *summary*: this shows the burn-time of every individual candle, and makes it hard to compare *means* (which is the RQ); why is every bar labelled, which just adds unnecessary clutter; fonts are hard too read (small); a boxplot (or dotchart) would be the appropriate graph. In addition: the largest value is over 70 mins! Do a *quick* project! –>
3. Reasonably good! The label on the vertical axis is a bit misleading: it is the *number* of students; some wear grey-scale clothing and some do not.
4. A few issues: Compares just two numbers (means) using lots of ink; no indication of *variation* in the data; a boxplot (or dotchart) would be appropriate.

Answers for Sect. 5.5

1. Correlation is inappropriate for **Plot 4** (non-linear) and **Plot 5** (non-linear).
2. **Plot 1**: 0.94 (correlation **D**); **Plot 2**: -0.95 (correlation **A**); **Plot 3**: 0.12 (correlation **B**); **Plot 6**: 0.75 (correlation **C**).
3. Examples of the direction in **Plot 1**: any two variables moderately positively correlated, such as height and weight, distance lived from university and travel time, etc.
4. Examples of direction in **Plot 2**: any two variables moderately negatively correlated, such as hours of weekly exercise and body weight, number of SCI110 tutorial missed and final mark, etc.
5. **Plot 1**: 88.4%; **Plot 2**: 90.3%; **Plot 3**: 1.4%; **Plot 6**: 56.3%.

Answers for Sect. 5.7

Answers implied by H5P.

1. Five variables. ('Participants' would not be summarised, it is technically an *identifier* and *not* a *variable*, as each person has a unique value).
2. Age; Height; Weight and quantitative **continuous**.
3. Gender (**nominal**; two levels); GMFCS (**ordinal**; three levels)
4. As follows:
   - 'Gender': Percentages (or number) F and M
   - 'Age': Mean/median; standard deviation/IQR
   - 'Height': Mean/median; standard deviation/IQR
   - 'Weight': Mean/median; standard deviation/IQR
   - 'GMFCS': Percentages (or numbers) in each group
5. As follows:
   - 'Gender': Barchart (not really needed)
   - 'Age': Histogram/stemplot
   - 'Height': Histogram/stemplot

- 'Weight': Histogram/stemplot
- 'GMFCS': Barchart/piechart

6. As follows:
    - Between Gender and Height: Boxplot
    - Between Gender and GMFCS: Side-by-side or stacked bar chart.

A.6 Answer: TW 6 tutorial

Answers for Sect. 6.3

Answers embedded. Some answers:

1. In order from left to right: 70; 85; 100; 115; 130. **3.** 30 points above the mean. **4.** Two standard deviations above the mean. **5.** About 2.5%.

A person with an IQ of 85 has an IQ that is **15** points **below** the mean, equivalent to **1** standard deviation(s) **below** the mean IQ. Using the **68**–95–99.7 rule, the proportion of the population with an IQ less than 85 is about **32**%. In addition, the proportion of the population with an IQ **above** 85 is about **68**%.

Answers for Sect. 6.4

1. Answers vary.
2. Answers vary.
3. Answers vary.
4. Answers vary. You probably cannot be very accurate using the 68–95–99.7 rule.
5. Answers vary.
6. Two standard deviations from the mean is $2 \times 6.7 = 13.4$, so 95% of females aged 18 and over have a measured height between 161.4 ± 13.4 approximately, or from 148.0cm to 174.8cm.
7. As follows:
    - $z = (171 - 161.4)/6.7 = 1.43$, so the probability is 0.9236 or about 92%.
    - So the odds are $92.36/(100 - 92.36) = 12.1$.
    - The answer is just $1 - 0.9236 = 0.0764$ or about 7.6%.
    - The probability of **over 171cm** is 7.64%.
    - So the odds are $7.64/(100 - 7.64) = 0.083$.
    - z-scores are 1.28 and 2.78, so the answer is $0.9973 - 0.8997 = 0.0976$, or about 9.8%.
    - So the odds are $9.8/(100 - 9.8) = 0.11$.
8. Use the Tables: $z = -0.84$; then using unstandardising formula, the height is $x = \mu + (z \times \sigma) = 161.4 + (-0.84 \times 6.7) = 155.772$, or about 156 cm.

Answers for Sect. 6.5

Answers embedded.

Relative frequency. Relative frequency. Subjective. Relative frequency. Classical.

Answers for Sect. 6.6

1. Depends on data.
2. Looking close to normal, centred around 0.5-ish.
3. We'd have an approximate normal distribution with mean $\mu_{\hat{p}} = 0.5$ and standard deviation $\text{s.e.}(\hat{p}) = \sqrt{0.5 \times (1 - 0.5)/10} \approx 0.158$.
4. Still normal; same mean; but standard deviation would be smaller: $\text{s.e.}(\hat{p}) = \sqrt{0.5 \times (1 - 0.5)/50} \approx 0.071$.

Answers for Sect. 6.7.1

2. $z = (20 - 0)/10 = 2$. The area or probability to the *right* is 0.0228, or about 2.3%.
3. The probability the SOI exceeds 20: 2.3%. So odds: $2.3 \div (100 - 2.3)$, or about 0.024.
4. $z = (-25 - 0)/10 = -2.5$. The area to the *left* is 0.0062, or about 0.6%.
5. $z = (-12 - 0)/10 = -1.2$. The area to the *right* is 0.8849, or about 88.5%.

6. The two z-scores: $(-10-0)/10 = -1$ and $(20-0)/10 = 2$. The area *between* these is $0.9772-0.1587 = 0.8185$, or about 81.9%.
7. The z score corresponds to an area of 0.80 to the *left*; from the tables, about $z = 0.84$. This corresponds to an SOI of $0 + (0.84 \times 10)$, or an SOI of about 8.4.
8. The z-score is 0.385 from Appendix 3 in the textbook, remembering that the area to the *left* would be 0.650 (draw a diagram!). So, the z-score is 0.385, so that the SOI values is $x = \mu + (z \times \sigma) = 0 + (0.385 \times 10) = 3.85$.

A.7 Answer: TW 7 tutorial

Answers for Sect. 7.3

1. This is just one box of matches (one *observation*), but the claim is about the *population mean*. Some boxes will have more than 45, and some fewer.
2. Jake is correct in one sense: You can't have 0.9 of a match. But the value is the *mean* number, and that *can* be a decimal. Suppose 10 boxes had 49 matches, and 10 boxes had 50 matches… is the mean 49, or is it 50? Neither are correct; the mean is 49.5.
3. Jake is confusing the *sample* and *population* mean. The claim is that the *population* mean is 45. The sample produced a mean of 44.9.

   Why should the mean of two different things be the same? It's like expecting your height and your Mum's height to be the same: they are both heights, but of different things. Why should they be the same?

   Of course, every sample will produce a different sample mean. This sample may just have an unusually low number of matches.
4. Either (1) The manufacturer is lying; or (2) the manufacturer is *not* lying, and this sample just happens to have a smaller number of matches: (bad) luck.
5. A CI gives some indication of the variation implied by the sample.
6. The standard error for the mean is $0.124 \div \sqrt{25} = 0.0248$. So the approximate 95% CI is: $44.9 \pm (2 \times 0.0248)$, or 44.9 ± 0.0496, or from 44.85 to 44.95.
7. No. A 95% CI may or may not contain the population mean. Of course, the manufacturer *may* indeed be lying… but we'd need to be cautious about making such a bold claim on just this evidence. Ideally, we would repeat this study a few times or take a larger sample. But it *is* looking suspicious…

   If we had many, many sets of 25 matches boxes, 95% of these sets of 25 would have a mean between 44.85 and 44.95.
8. $\bar{x}$ is the mean of the sample, so $\bar{x} = 44.9$.
9. μ is the mean of the population; the true mean if you like. μ is *claimed* to be 45, but the the value of $\bar{x}$ will, of course, vary.

Answers for Sect. 7.4

Answers implied by H5P.

s.e.$(\hat{p}) = \sqrt{(\hat{p} \times (1-\hat{p}))/n}$, where $\hat{p}$ is the *sample proportion*; n the *sample size*; "s.e." the "standard error".

Answers for Sect. 7.5

1. $123/(404-123) = 123/281 = 0.44$.
2. $\hat{p} = 123/404 = 0.304455$.
3. The **odds**: The likelihood of surviving is about 0.44 times the probability of dying (i.e., it is lower). Or: For every 100 that die, about 44 survive.
4. No: sampling variation!
5. s.e.$(\hat{p}) = \sqrt{0.304455 \times (1-0.304455)/404} = \sqrt{0.00052416} = 0.022894$, or about 0.023.

   A definition can be found in the textbook *Glossary*. Essentially, each sample is likely to produce a different value for the sample proportion, $\hat{p}$ (the estimate of the population proportion, p), and that is what we mean by "sampling variation".
6. (Not provided.)
7. The values of $\hat{p}$ will have an approximate normal distribution, with a standard deviation equal to standard error (0.023) and centred around the true proportion p. A 95% CI: $0.304455 \pm (2 \times 0.022894)$, or 0.30446 ± 0.04579, or 0.26 to 0.35.

8. One way of writing communicating: "The population proportion of patients surviving after BVM treatment has a 95% chance of lying between 26% and 35%." This is not strictly correct, but acceptable and very, very commonly used (as explained in the textbook.
9. Larger, to get a tighter (more precise) CI than the one calculated.
10. The number of surviving and non surviving both exceed 5.

Answers for Sect. 7.6

1. μ is the population mean diameter size of *all* EB pizzas; $\bar{x}$ is the mean diameter of the pizzas in the sample.
2. $\bar{x} = 11.486$ inches; it's not sensible to quote the diameter to 0.001 of an inch; what *would* be is sensible? We don't know the value of μ, and we never will. Our best *estimate* is the value of $\bar{x}$.
3. $s = 0.24658$ inches. It's not sensible to quote the diameter to 0.001 of a cm though. σ is the standard deviation of the population. We don't know the value of σ, and we never will.
4. s.e.$(\bar{x}) = s/\sqrt{n} = 0.24658/\sqrt{125} = 0.02205$.
5. The first measures the variation in the diameters of individual pizzas; the second measures the precision of the sample mean when used to estimate the population mean.
6. Almost certainly not the same. Probably close to $\bar{x} = 11.486$ inches. More precisely, probably within three standard errors (3×0.022) of $\bar{x}$.
7. Normal; mean μ; std. dev is the standard error of 0.02205.
8. The approximate 95% CI is $11.486 \pm (2 \times 0.02205)$ or 11.486 ± 0.044, which is from 11.44 to 11.53 inches.
9. Based on the sample, a 95% confidence interval for the population mean for the pizza diameter is between 11.44 and 11.53 inches.
10. $n > 25$ **or** $n \leq 25$ and population has normal distribution.
11. We do not need to **assume** that $n > 25$ because we know that it is. We do *not* require that the sample or the population has a normal distribution. We require that the *sample means* have an approximate normal distribution, which they will if $n > 25$. So the CI is statistically valid, and the histogram is not needed.
12. Population mean diameter probably not 12 inches based on the CI.

Answers for Sect. 7.7

1. $\sqrt{0.70 \times (1 - 0.70)/25} = \sqrt{0.0084} = 0.09165151$, or about 0.09165.
2. $\sqrt{0.25 \times (1 - 0.25)/100} = \sqrt{0.001875} = 0.04330127$, or about 0.04330.
3. 0.08724964, or about 0.08725.
4. 0.0534479, or about 0.05345.

All statistically valid.

Note: Students commonly forget to take the square root.

Note: If you calculator gives an answer something like `1.875 E-03` or similar, it is using **scientific notation**. It means 1.875×10^{-3}, or 0.001875.

A.8 Answer: TW 8 tutorial

Answers for Sect. 8.3

Answers implied by H5P.

1. The decision-making process begins with making an assumption about the population **parameter**. This means we know what to **expect** from the sample **statistic**. We never know exactly what value of the statistic we will see in the sample, because of **sampling variation**. But we can have some of idea of what values are reasonable to expect. Then we take the **sample** (that is, we make the observations). Then we **compare** the sample statistic that we observed… to the sample statistic we expected. If what we observe is inconsistent with what was expected, then the the assumption is **unlikely to be** true. However, if what we observe is **consistent** with what was expected, then the the assumption is **probably** true.

Answers for Sect. 8.4

1. H_0: $p = 0.3$ and H_1: $p > 0.3$.
2. $\hat{p} = 12/35 = 0.3428571$.

3. s.e.$(\hat{p}) = \sqrt{0.3 \times 0.7/35} = 0.07745967$. (Make sure to use p not $\hat{p}$ in the standard error calculation for the test!)
4. $z = (0.3428571 - 0.3)/0.07745967 = 0.553$.
5. The z-value is very small—less than one standard deviation from the mean—so the P-value will be quite large: 'The data present no evidence ($z = 0.55$; $P > 0.05$) that computer programmers are more likely to wear contact lenses ($\hat{p} = 0.342$; approx. 95% CI from 0.182 to 0.503).' (For the CI, remember to use $\hat{p}$ in the standard error calculation: s.e.$(\hat{p}) = 0.0802329$.)
6. For *statistical validity*, we require that the number of people *with* and *without* contact lenses to be at least 5, which is true (12 and 23 respectively). For *external validity*, the sample must be a random sample of Swedish programmers.

Answers for Sect. 8.5

1. The **parameter** of interest is the *population mean pizza diameter*, say μ.
2. From the output: $\bar{x} = 11.486$ inches and $s = 0.247$ inches.
3. Use s.e.$(\bar{x}) = s/\sqrt{n} = 0.247/\sqrt{125} = 0.02205479$ inches.
4. s tells us the variation in diameter from pizza to pizza. s.e.$(\bar{x})$ tells us how much the sample mean is likely to vary from sample to sample, in sample of size 125.
5. The hypotheses are:
   H_0: The mean diameter is 12 inches; or $\mu = 12$.
   H_1: The mean diameter is not 12 inches; or $\mu \neq 12$
6. Two-tailed, because the RQ asks if the diameter is 12 inches, or not.
7. The normal distribution has a mean of 12, and a standard deviation of s.e.$(\bar{x}) = 0.022092$.
8. $t = (11.486 - 12)/0.022092 = -23.3$.
9. The t-value is *huge* (and negative), so the P-value is *very* small. (Two-tailed $P < 0.001$ from the table or from software.)
10. Very strong evidence exists in the sample ($t = -23.3$\$; two-tailed P less than 0.001) that the population mean pizza diameter of pizzas from Eagle Boys' is less than 12 inches (sample mean diameter: 11.49 inches; std. dev.: 0.246; 95% CI from 11.44 to 11.53 inches; approximate 95% CI is 11.486 ± 0.0441).
11. Since n is much larger than 25, we do not require that the population has a normal distribution, just that the population is not grossly non-normal; the sample means will still have an approximate normal distribution.
12. *Very* unlikely!

Answers for Sect. 8.6

Use the 68–95–99.7 rule to *approximate* the P-values:

Less than 0.05; Greater than 0.05; *Less than* 0.003; Greater than 0.05; Greater than 0.05; Very small.

A.9 Answer: TW 9 tutorial

Answers for Sect. 9.3

Answers implied by H5P.

Null hypotheses:

1. Is the mean length of a 12-inch sub really 12 inches? **F**: $\mu = 12$.
2. Is the mean length of a 12-inch sub different for white and wholemeal subs? **B**: $\mu_{\text{white}} - \mu_{\text{wholemeal}} = 0$.
3. Is the proportion of 12-inch subs shorter than 12 inches different for white, wholemeal subs? **A**: $p_{\text{white}} - p_{\text{wholemeal}} = 0$.
4. Is the mean length of a 12-inch sub longer for white (compared to wholemeal) subs? **B**: $\mu_{\text{white}} - \mu_{\text{wholemeal}} = 0$.

Alternative hypotheses:

1. Is the mean length of a 12-inch sub really 12 inches? **I**: $\mu \neq 12$.
2. Is the mean length of a 12-inch sub different for white and wholemeal subs? **G**: $\mu_{\text{white}} - \mu_{\text{wholemeal}} \neq 0$.
3. Is the proportion of 12-inch subs shorter than 12 inches different for white, wholemeal subs? **K**: $p_{\text{white}} - p_{\text{wholemeal}} \neq 0$.
4. Is the mean length of a 12-inch sub longer for white (compared to wholemeal) subs? **L**: $\mu_{\text{white}} - \mu_{\text{wholemeal}} > 0$.

Answers for Sect. 9.4

1. How much further patients walk *with* the implant.

2. **Repeated-measures**: Every subject has two TWMT recorded.
3. μ_d is the mean difference in the target population; $\bar{d}$ is the mean difference in this sample.
4. Each measurement is measured after and before on the same subject.
5. 32; 12; 24; 30; 8; 14; 14; 28; 38; 49. (Differences in the other direction are also acceptable; it just changes the signs of these differences and so on. **Importantly, the direction should be stated somewhere.**)
6. It makes more sense to define directions this way, so that the difference is the *increase* in 2MWT.
7. $\bar{d} = 24.9$; $s_d = 13.03372\,\text{m}$.
8. s.e.$(\bar{d}) = s_d/\sqrt{n} = 13.03372/\sqrt{10} = 4.121623\,\text{m}$.
9. This is the standard deviation of the sample mean difference, a measurement of how precisely the sample mean difference measures the population mean difference.
10. Almost impossible. Sample means vary every time we take a sample around the true mean difference, with a normal distribution with standard error 4.12. Since we don't know μ, the best we can say is that the sample mean will vary about our best guess of the population mean; in other words, the sample means vary around 24.9 with a standard deviation of about 4.12.
11. Normal; mean μ_d, std deviation is the standard error of 4.121.
12. $24.9 \pm (2 \times 4.121623)$, or 24.9 ± 8.243246, or from 16.65675 to $33.143245\,\text{m}$.
13. Only *one sample of people* have been studied, so it is (as always) hard to be sure.
14. H_0: $\mu_d = 0$, differences defined as 'with' minus 'without'. H_1: $\mu_d > 0$ the way I defined differences.

15. $t = 6.041$ and since t is very large, expect P to be very small.
16. The differences have just been defined in the opposite directions. (Notice the P-values are the same.)
17. Very strong evidence exists in the sample (paired $t = 6.041$; one-tailed P less than 0.0005) that the population mean 2MWT are higher after receiving the implant compared to without the implant (mean difference: $24.9\,\text{m}$ higher after receiving the implant; standard deviation: $13.034\,\text{m}$; 95% CI from 16.66 to $33.14\,\text{m}$).
18. The population of differences has a normal distribution, and/or $n > 25$ or so.
19. Since $n < 25$, must assume the population of differences has a normal distribution. The histogram suggests this is not unreasonable.

Answers for Sect. 9.6

1. The two groups are different rats.
2. The **parameter** of interest is the *difference between the population mean lifetimes*, say $\mu_R - \mu_F$. Measure how much longer the lifetimes are for rats on a restricted diet.
3. Table not shown.
4. The 95% CI is the *bottom* one: from 223.34 to 346.13 days.
5. The best of these is Option (e)… but in practice, we usually think about CIs in terms of Option (d) so Option (d) is fine.
6. The CI explanation can be improved by (i) indicating *which* diet leads to larger average lifetimes; and (ii) providing sample summary info. Here is a better answer:
   "The 95% confidence interval for the difference between the populations mean lifetimes of rats on the restricted diet (sample mean: 968.8 days; std dev: 284.6 days) and on the free-eating diet (684.0 days; std dev: 134.1 days) is that rats on a restricted diet live between 223.34 and 346.13 days longer."
7. Boxplots: shows the variation in the lifetimes of individual rats. Error bar chart: displays the variation that the sample means would be expected to show from sample to sample.
8. The null hypothesis is that there is no difference in the mean lifetimes of the two groups of rats. In symbols (where μ represents the mean lifetime in the population):
   H_0: $\mu_R = \mu_{FE}$; and H_1: $\mu_R > \mu_{FE}$; **one**-tailed, because of the RQ.
9. Either sampling variation explain the difference, or the diets really are different.
10. Use the not-equal variance (Welch's test) row: $t = 9.161$; one-tailed $P < 0.0005$ (test is one-tailed).
11. $t = (284.73 - 0)/31.08 = 9.16$, as per output.
12. The evidence contradicts H_0. Very strong evidence exists in the sample ($t = 9.161$ for two independent samples; df $= 154.94$; one-tailed $P < 0.0005$) that the population mean lifetime of rats on a restricted diet (mean lifetime: 968.75 days; std. dev.: 284.6 days) is greater than rats on a free-eating diet (684.01 days; 134.1) (95% CI for the difference from 223.3 to 346.1 days).
13. The *sample means* have a normal distribution. Since both sample sizes are greater than 25, this is true. The figure suggests not very severe non-normality.
14. Rats from the same litter are likely to be similar to each other. The litter would probably be the unit of analysis then, not the individual rat.

Answers for Sect. 9.7.1

H_0: $\mu_d = 0$ and H_1: $\mu_d \neq 0$, where μ_d is the mean difference (shoes minus barefoot, from the second table). $t = -2.231$; $P = 0.029$. Moderate evidence of a mean difference (approx. 95% CI: 0.066 to $1.206\,\text{cm}$ greater barefoot).

Answers for Sect. 9.7.2

H_0: $\mu_P = \mu_N$ and H_1: $\mu_P \neq \mu_N$, where μ_P is the mean reaction time *with* a phone and μ_N is the mean reaction time *with no phone*. Using summary table: $t = (51.59 - 0)/19.612 = 2.63$, so P-value will be smaller than 0.05 (based on 68–95–99.7 rule). Strong evidence of a difference between the mean reaction times; approx. 95% CI: 12.37 to 90.81 s slower (i.e., greater) with phone.

A.10 Answer: TW 10 tutorial

Answers for Sect. 10.3

If the RQ is about odds (and hence odds ratios), then the hypothesis, CI and hypothesis tests should also be about odds.

(Similarly, if the RQ is about proportions, then the hypothesis, CI and hypothesis tests should also be about proportions.)

Answers for Sect. 10.4

1. See Table A.2.
2. Side-by-side or stacked bar chart.
3. $352/2512 = 0.1401$.
4. $336/2328 = 0.1443$.
5. $0.1401/0.1443 = 0.971$.
6. The odds ratio, of a boy maturing late compared to a girl maturing late.
7. OR: 0.971, with 95% CI from 0.828 to 1.139.
8. See Table A.3.
9. $P = 0.717$; no evidence of a difference between boys and girls.

***TABLE A.2:** Maturation data table.*

	Late	Not late	Total
Males	352	2512	2864
Females	336	2328	2664
Total	688	4840	5528

***TABLE A.3:** Numerical summary table for maturation data.*

	Prop late	Odds late	Sample size
Males	0.123	0.1401	2864
Females	0.126	0.1443	2664
	Diff: -0.003	OR: 0.971	

Answers for Sect. 10.5

Some answers embedded.

1. See Table A.4.
2. Use of artificial limb is the explanatory variable.
3. Using artificial limb: $49/16 = 3.0625$. Not using artificial limb: $21/19 = 1.105263$. The OR is $3.0625/1.105263 = 2.771$; that is, the odds of being alive after five years is almost three times higher for those using an artificial limb compared to those who do not. See Table A.5.
4. The CI for the OR is from 1.20 to 6.41.
5. Chi-squared: 5.84; like $z = \sqrt{5.84} = 2.42$: large; P about 0.016.
   The *sample* provides evidence to suggest that the odds of dying within five years is not the same between having a wearing an artificial limb and the five-year mortality rate *in the population* (chi-square $= 5.84$; $P = 0.016$; OR: 2.77 and 95% CI from 1.20 to 6.41).

TABLE A.4: *Five-year mortality for artifical limb users.*

	Alive	**Dead**	**Total**
Used art. limb	49	16	65
Did not use art. limb	21	19	40
Total	70	35	105

TABLE A.5: *Five-year mortality and use of an artificial limb: Numerical summary.*

	Proportion alive after 5 years	**Odds alive after 5 years**	**Sample size**
Use artificial limb	0.754	3.06	65
Did not use artifical limb	0.525	1.11	40
	Diff in percentages: 22.9	OR: 2.771	626

Answers for Sect. 10.6

1. Two-sample t-test. **2.** A χ^2 test. **3.** A paired t-test. **4.** A two-sample t-test. **5.** A z-test comparing two proportions. **6.** None of the other options are correct: requires regression or correlation.

Answers for Sect. 10.7

$n = (2 \times 7.145 \div 0.5)^2 = 816.8$, so use guesses from 817 students.

A.11 Answer: TW 11 tutorial

Answers for Sect. 11.3

- **Research question:** use *mean* distance. So more directly:

> For casual golfers, is the *mean* distance travelled by a golf ball the same when hit by a wooden or metal club?

- **Summaries:** There is no mention of the distance travelled by the ball. It really doesn't make sense to say 'the iron golf club' is 'moderately higher than the wooden golf club'. That sounds like the height of the clubs are different!
- **Results (1):** Spelling/grammar errors ('verse'; 'differes'). Is two decimal places suitable? Certainly no more. The CI is for the difference between the *means* of hitting distances.
- **Results (2):** The results just say there is a difference between metal and wooden clubs (of course!); there is no mention of hitting distance.

Answers for Sect. 11.4

Null hypothesis: The *mean* heart rates. **Results:** A χ^2 test is inappropriate for comparing means.

Answers for Sect. 11.5

1. The null hypothesis is that the proportion (or the odds) of successful attempts is the same using EGTA and LM, in the two populations.
2. Using symbols, based on using proportions: H_0: $p_{\text{EGTA}} = p_{\text{LM}}$; H_1: $p_{\text{EGTA}} \neq p_{\text{LM}}$ (two-tailed).
3. The sample presents insufficient evidence (chi-square $= 2.63$; df $= 1$; $P = 0.104$) that the success rates between EGTA and LM are different in the population.
4. Yes: All expected counts are greater than 5 (using the textbook criteria (smallest row total, times smallest column total, divided by grand total).
5. No evidence of a difference, so perhaps consider other issues such as cost, longevity, pain associated with insertion etc.

Answers for Sect. 11.6

- **Research question**: Spelling error ('lily pily'); *mean* size. It sounds odd, too, to ask if they 'become' longer. We are just comparing the mean length of leaves of eastern and western sides.
- **Null hypothesis**: Spelling again; and the hypothesis should be explicitly comparing the *means*: "For weeping lilly pilly trees at UniSC, is the mean leaf length the same for leaves on the western and eastern sides of the tree?"
- **Results**: Cannot find the mean of a qualitative variable ('Side of tree'); the table should give the mean, standard deviation and standard error for leaves taken from the west side (one row) and the east side (second row), plus information about the *difference*.
- **Methods**: There are only five units of analysis. Leaves taken from the same tree are likely to share a lot in common: genetics; sunlight, watering regime, soil condition, etc. The students should have used 50 trees.

  Alternatively, I suppose they could have used one tree (so the RQ was about what happens on one tree only), and compared 50 leaves on either side. While this is kinda OK, I do **not** recommend it.

Answers for Sect. 11.7

- **Research question**: Cumbersome wording…; *mean* estimates.
- **Null hypothesis**: Spelling/grammar errors; *mean* again; where does 'increased level of perception' come from? We are just comparing estimation distances; perception is much more than just that.
- **Results**: this doesn't answer the question. This is comparing the estimates from both groups—and yes, there may be a difference. But the RQ was about which group could estimate closer (on average) to the actual true width of the path. We still don't know: the RQ was not answered.
- **Discussion**: you don't *prove* anything like this… Again, what was tested doesn't help answer the RQ.

A.12 Answer: TW 12 tutorial

Answers for Sect. 12.3

1. No answer. But a line is fine for **Plots 1, 2, 6** and **3** (but very weak!), but not for **Plots 4 and 5**.
2. My **very rough** slope estimates are:
   Plot 1: $(50 - 10)/10 \approx 4$; **Plot 2**: $(20 - 50)/15 \approx -2$; **Plot 3**: 0; **Plot 6**: $(55 - 35)/5 = 4$.
3. My **very rough** intercept estimates are: **Plot 1**: 8; **Plot 2**: 40; **Plot 3**: 32; **Plot 6**: 10.
4. My very rough estimates are: **Plot 1**; $\hat{y} = 8 + 4x$; **Plot 2**: $\hat{y} = 40 - 2x$; **Plot 3**: $\hat{y} = 32$; **Plot 6**: $\hat{y} = 10 + 4x$.

Answers for Sect. 12.4

1. Approximately linear, negative, reasonably strong.
2. Condition, extras (air con, etc.), sedan/hatch, colour, when rego due, new/old tyres, location, etc.
3. No answer.
4. Looks to be expensive, as $15,000 would be above the line (at least for the line I'd draw).
5. Probably $3900.
6. My guess is about $b_0 \approx 17$ or $17,000. This would mean the average price of a 2014 second-hand Corolla can be expected to be about $17,000.
7. $b_1 = (1 - 17)/(16 - 0) \approx -1$. That is, the price reduces by about $1000 each year older the Corolla gets.
8. Using the above, we have $\hat{y} = 17 - x$ approximately. Guessing the regression line won't, of course, produce this level of precision, so anything close-ish to this is fine.
9. $\hat{y} = 16.54 - 0.96x$ (jamovi) or $\hat{y} = 16.536 - 0.963x$ (SPSS), where y is the price in thousands of dollars, and x is the age in years.
10. $r = -0.929$, and so $R^2 = (-0.929)^2 = 0.863$, or about 86%, so about 86% of the variation in prices can be explained by age alone. The rest can be explained by the car's condition, odometer reading, accessories, service history, etc.
11. jamovi: $\hat{y} = 16.54 - (0.96 \times 20) = -2.66$, or -$2660.
    SPSS: $\hat{y} = 16.536 - (0.963 \times 20) = -2.72$, or -$2720.
    This is clearly silly, as we are extrapolating.
12. Hardly need a test… H_0: $\beta_1 = 0$ vs H_1: $\beta_1 < 0$. From output, $t = -15.059$, and $P = 0.000/2 = 0.000$, so $P < 0.0001$: very strong evidence that older cars fetch lower second-hand prices, on average.
13. $-0.963 \pm (2 \times 0.064)$, or -0.963 ± 0.128, or from -1.091 to -0.835.
14. Looks the same really, just reflected left-to-right.

- *Size* of r won't change, sign from negative to positive; i.e., $r = 0.93$.
- Value of R^2 will be the same.
  - Slope will be the same except sign will change (in both cases, the values on the horizontal axis are one year apart).
- Intercept will change a lot… it is the predicted value of the price if the line is extended to year 0 (which is, of course, meaningless).

Answers for Sect. 12.5

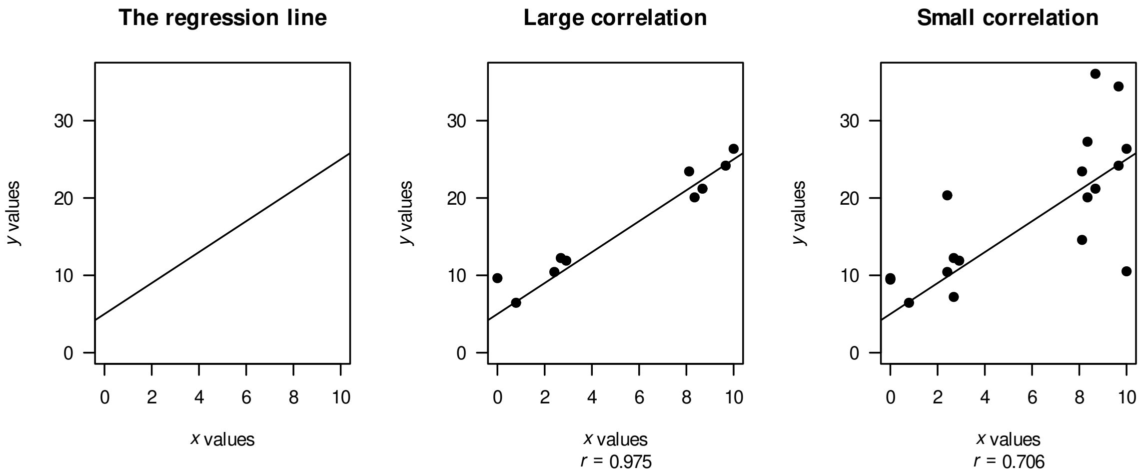


FIGURE A.1: *Plots giving different correlations.*

B

Appendix: Tables

This Appendix contains tables that may be useful:

- z-tables: Find the area associated with a normal distribution *given* the z-score (Appendices B.1 and B.2).
- Random numbers (Appendix B.3).

The z-score tables provide the area *to the left* of a given z-score associated with a normal distribution.

Using the z-score tables

Details for using these tables are provided in Sect. 20.6 of teh **textbook**. Here we give a short summary.

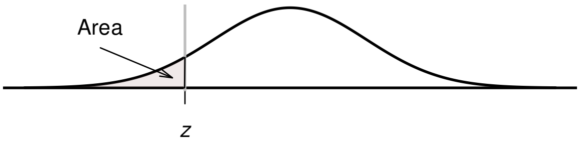
To find the probability that z is less than -2.43, follow these steps:

1. Use Appendix B.1 to find -2.4 in the *left* margin of the table (see image below).
2. Then, find the second decimal place (in this case, 3) in the *top* margin of the table.
3. Where these intersect is the area (or probability) *less than* the z-score of -2.43.

The probability of finding a z-score less than $z = -2.43$ is 0.0075, or about 0.75%.

	0.00	0.01	0.02	$\boxed{0.03}$	0.04
−3.5	0.0002	0.0002	0.0002	0.0002	0.0002
−3.4	0.0003	0.0003	0.0003	0.0003	0.0003
−3.3	0.0005	0.0005	0.0005	0.0004	0.0004
−3.2	0.0007	0.0007	0.0006	0.0006	0.0006
−3.1	0.0010	0.0009	0.0009	0.0009	0.0008
−3.0	0.0013	0.0013	0.0013	0.0012	0.0012
−2.9	0.0019	0.0018	0.0018	0.0017	0.0016
−2.8	0.0026	0.0025	0.0024	0.0023	0.0023
−2.7	0.0035	0.0034	0.0033	0.0032	0.0031
−2.6	0.0047	0.0045	0.0044	0.0043	0.0041
−2.5	0.0062	0.0060	0.0059	[illegible]	0.0055
$\boxed{-2.4}$	0.0082	0.0080	0.0078	0.0075	0.0073
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096

B.1 Normal distribution: negative z-values probabilities

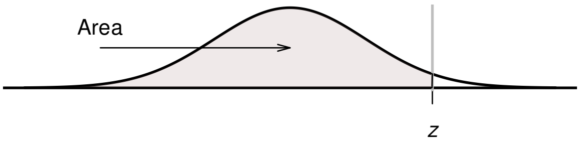


The table gives the probability (area) that a z-score is *less* than a given value. For example: the area *less than* $z = -1.38$ is 0.0838, or 8.38%.

	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
−3.5	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002
−3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
−3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
−3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
−3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
−3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
−2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
−2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
−2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
−2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
−2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
−2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
−2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
−2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
−1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
−1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
−1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
−1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
−1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
−1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
−1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
−1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
−1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
−1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
−0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
−0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
−0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
−0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
−0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
−0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
−0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
−0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
−0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
−0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

For $z = -4$, the probability is 0.00003. For $z = -5$, the probability is 0.0000003.

B.2 Normal distribution: positive z-values probabilities



The table gives the probability (area) that a z-score is *less* than a given value. For example: the area *less than* $z = 1.87$ is 0.9693, or 96.93%.

	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998
3.5	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998

For $z = 4$, the probability is 0.99997. For $z = 5$, the probability is 0.9999997.

B.3 Random numbers

TABLE B.1: *Some random numbers*

	Column I	Column II	Column III	Column IV	Column V
Row A	1 1 3 1 8	2 8 6 1 4	5 0 9 4 4	0 8 3 4 3	2 7 3 7 8
Row B	9 5 9 6 0	2 6 4 2 1	7 2 3 8 3	4 4 0 2 8	6 5 8 2 6
Row C	2 1 8 2 1	0 4 0 3 4	8 7 2 5 4	4 2 3 2 7	1 8 2 7 5
Row D	8 5 2 8 4	0 2 0 1 8	9 7 7 3 3	3 2 2 9 5	6 8 2 2 7
Row E	5 6 2 8 0	0 5 6 6 1	5 0 5 0 1	3 2 5 7 9	5 3 3 6 6
Row F	4 9 5 2 1	9 7 6 1 8	9 6 8 7 5	6 2 7 3 7	2 5 1 5 9
Row G	5 7 2 1 4	5 6 7 5 3	9 1 2 8 3	1 6 8 2 1	3 2 8 2 8
Row H	5 0 0 3 1	1 6 4 1 7	3 5 8 8 6	5 3 5 2 1	2 8 9 0 8
Row I	7 6 0 4 1	7 0 0 0 3	2 9 3 7 3	5 6 2 4 0	8 7 8 4 3
Row J	4 0 5 3 0	1 9 6 6 7	7 4 9 9 3	3 3 1 6 6	1 9 3 9 6
Row K	3 5 0 9 7	9 7 9 8 3	8 9 8 4 1	9 4 5 4 3	6 3 5 0 9
Row L	7 8 0 4 6	1 7 8 2 5	3 8 4 2 9	8 4 0 6 5	0 4 2 1 5
Row M	6 9 9 9 3	2 7 5 6 8	2 6 8 2 1	9 1 7 4 8	4 6 8 8 1
Row N	1 4 0 4 9	8 0 8 2 6	8 0 7 0 9	7 7 1 7 7	3 9 8 1 7
Row O	9 2 1 5 6	2 1 5 5 1	3 4 9 0 4	7 5 5 9 1	5 3 4 0 8
Row P	7 6 9 4 0	1 8 6 6 1	6 0 7 2 2	8 3 4 1 4	8 2 9 9 8
Row Q	7 5 7 5 6	5 1 9 0 9	0 5 2 9 5	3 9 5 2 3	9 8 2 5 2
Row R	1 5 0 5 9	3 4 6 9 0	9 9 1 6 5	8 7 9 0 9	6 5 1 4 8
Row S	3 2 7 5 4	0 1 3 9 8	6 4 4 5 0	8 6 5 5 4	5 0 7 3 4
Row T	6 9 9 1 6	3 4 0 5 4	0 6 9 0 8	6 8 1 8 5	1 4 4 3 3
Row U	2 6 1 9 1	2 6 3 7 8	9 1 2 8 1	4 4 9 6 6	5 9 0 8 9
Row V	3 0 8 2 1	8 2 1 5 7	6 0 4 2 6	8 0 8 8 6	8 0 2 8 9
Row W	2 7 6 2 7	2 0 7 1 4	9 3 0 2 2	1 4 7 2 8	9 4 0 0 5
Row X	6 2 7 0 9	5 8 4 6 5	2 9 9 6 0	5 5 3 3 4	5 0 9 4 9
Row Y	6 5 4 4 5	3 7 9 6 2	0 5 1 3 0	8 8 7 8 8	0 3 0 5 5
Row Z	6 6 4 4 9	7 6 0 7 4	6 2 8 3 1	5 5 9 5 8	8 5 0 2 2

C

Symbols, formulas, statistics and parameters

C.1 Symbols and standard errors

- The following table lists the statistics used to estimate unknown population parameters.
- When the sampling distribution is approximately normally distributed, under appropriate statistical validity conditions, this is indicated by ✔.
- The value of the mean of the sampling distribution (the *sampling mean*) is:
  - unknown, for *confidence intervals*.
  - assumed to be the value given in the null hypothesis, for *hypothesis tests*.

TABLE C.1: *Sample statistics used to estimate population parameters. Some statistics have appproximately normally-distributed sampling distributions under appropriate (statistical validity) conditions, as indicated using a ✔.*

		Sampling distribution			
	Statistic	**Parameter, and sampling mean**	**Normal distn?**	**Standard error**	**Ref.**
Proportion	$\hat{p}$	p	✔	CI: $\sqrt{\dfrac{\hat{p}\times(1-\hat{p})}{n}}$	Ch. 22
				HT: $\sqrt{\dfrac{p\times(1-p)}{n}}$	Ch. 26
Mean	$\bar{x}$	μ	✔	$\dfrac{s}{\sqrt{n}}$	Chs. 23, 27
Mean difference	$\bar{d}$	μ_d	✔	$\dfrac{s_d}{\sqrt{n}}$	Ch. 29
Difference between means	$\bar{x}_1-\bar{x}_2$	$\mu_1-\mu_2$	✔	$\sqrt{\text{s.e.}(\bar{x}_1)^2+\text{s.e.}(\bar{x}_2)^2}$	Ch. 30
Difference between proportions	$\hat{p}_1-\hat{p}_2$	p_1-p_2	✔	CI: $\sqrt{\text{s.e.}(\hat{p}_1)^2+\text{s.e.}(\hat{p}_2)^2}$ HT: $\sqrt{\text{s.e.}(\hat{p}_1)^2+\text{s.e.}(\hat{p}_2)^2}$ using *common* proportion $\hat{p}$	Ch. 31
Odds ratio (OR)	Sample OR	Pop. OR	✗	(Not given)	Ch. 31
Correlation	r	ρ	✗	(Not given)	Ch. 33
Regression: slope	b_1	β_1	✔	s.e.(b_1) (value from software)	Ch. 33
Regression: intercept	b_0	β_0	✔	s.e.(b_0) (value from software)	Ch. 33

C.2 Confidence intervals

For statistics whose sampling distribution has an approximate normal distribution, *confidence intervals (CIs)* have the form

$$\text{statistic} \pm (\text{multiplier} \times \text{s.e.}(\text{statistic})).$$

Notes:

- The multiplier is *approximately* 2 to create an *approximate* 95% CI (based on the 68–95–99.7 rule).
- The quantity 'multiplier × s.e.(statistic)' is called the *margin of error*.
- Software uses *exact* multipliers to form *exact* confidence intervals.
- When the sampling distribution for the statistic does *not* have an approximate normal distribution (e.g., for ORs and correlation coefficients), *this formula does not apply* and the CIs are taken directly from software output when available.

C.3 Hypothesis testing

For statistics whose sampling distribution has an approximate normal distribution, the *test statistic* has the form:

$$\text{test statistic} = \frac{\text{statistic} - \text{parameter}}{\text{s.e.}(\text{statistic})},$$

where s.e.(statistic) is the standard error of the statistic. The test-statistic is a t-score for most hypothesis tests in this book when the sampling distribution is described by a normal distribution, but is a z-score for a hypothesis test involving one or two *proportions*.

Notes:

- If the test-statistic is a z-score, the P-value can be found using tables (Appendices B.1 and B.2), or *approximated* using the 68–95–99.7 rule.
- If the test-statistic is a t-score, the P-value can be *approximated* using tables (Appendices B.1 and B.2), or *approximated* using the 68–95–99.7 rule (since t-scores are similar to z-scores; Sect. 28.4.
- When the sampling distribution for the statistic does not have an approximate normal distribution (e.g., for ORs and correlation coefficients), *this formula does not apply* and P-values are taken from software when available.
- A hypothesis test about ORs uses a χ^2 test statistic. For 2×2 tables only, the χ^2-value is equivalent to a z-score with a value of $\sqrt{\chi^2}$.

C.4 Sample size estimation

The following formulas compute the *approximate* minimum (i.e., conservative) sample size needed to produce a 95% CI with a specified margin of error (i.e., the 'give-or-take' amount).

- To estimate the sample size needed for *estimating a proportion* (Sect. 32.3), use:

$$n = \frac{1}{(\text{Margin of error})^2}.$$

- To estimate the sample size needed for *estimating a mean* (Sect. 32.4) use:

$$n = \left(\frac{2 \times s}{\text{Margin of error}}\right)^2$$

  for some estimate s of the standard deviation of the data.

- To estimate the sample size needed for *estimating a mean difference* (Sect. 32.5) use:

$$n = \left(\frac{2 \times s_d}{\text{Margin of error}}\right)^2$$

  for some estimate s_d of the standard deviation of the differences.

- To estimate the sample size needed for *estimating the difference between two means* (Sect. 32.6) use:

$$n = 2 \times \left(\frac{2 \times s}{\text{Margin of error}}\right)^2$$

  for *each* group being compared, where s is an estimate of the common standard deviation in the population for both groups. This formula assumes:

  - the sample size for each group will be the same; and
  - the standard deviation in each group is the same.

- To estimate the sample size needed for *estimating the difference between two proportions* (Sect. 32.7) use:

$$n = \frac{2}{(\text{Margin of error})^2}$$

  for *each* group being compared. This formula assumes the sample size in each group will be the same.

Notes:

- In *sample size* calculations, *round up* the sample size found from the above formulas.

C.5 Other formulas

- To *calculate z-scores* (Sect. 20.4), use

$$z = \frac{\text{value of variable} - \text{mean of the distribution of the variable}}{\text{standard deviation of the distribution of the variable}}.$$

  t-scores are like z-scores. When the 'variable' is a sample estimate (such as $\bar{x}$), the 'standard deviation of the distribution' is a standard error (such as s.e.$(\bar{x})$).
- The *unstandardising formula* (Sect. 20.8) is $x = \mu + (z \times \sigma)$.
- The *interquartile range* (IQR) is $Q_3 - Q_1$, where Q_1 and Q_3 are the first and third quartiles respectively (or, equivalently, the 25th and 75th percentiles).
- The smallest expected value (for assessing statistical validity when forming CIs and conducting hypothesis tests with proportions or ORs) is

$$\frac{(\text{Smallest row total}) \times (\text{Smallest column total})}{\text{Overall total}}.$$

- The *regression equation* in the *sample* is $\hat{y} = b_0 + b_1 x$, where b_0 is the sample intercept and b_1 is the sample slope.

C.6 Other symbols and abbreviations used

Symbol or abbreviation	Meaning	Textbook reference
RQ	Research question	Chap. 2
s	Sample standard deviation	Sect. 11.7.2
σ	Population standard deviation	Sect. 11.7.2
s_d	Sample standard deviation of differences	Sect. 11.7.2
σ_d	Population standard deviation of differences	Sect. 11.7.2
R^2	R-squared	Sect. 16.4.2
H_0	Null hypothesis	Sect. 28.2
H_1	Alternative hypothesis	Sect. 28.2
CI	Confidence interval	Chap. 24
s.e.	Standard error	Def. 19.4
n	Sample size	Def. 2.21
χ^2	The chi-squared test statistic	Sect. 31.6.3
$\pm$	Plus-or-minus (give-or-take)	Sect. 22.3

D

Thoughts for teaching

Tutor information

This Appendix contains some thoughts on teaching in the tutorial, and answers to the *Quick Revision* questions.

Thoughts on *Class Discussions*

Most teaching weeks include an *optional* class discussion. These discussions hopefully will, across the various discussions, help to:

- Develop students' *statistical reasoning* rather than just technical skills.
- Demonstrate that statistics is more than just calculations.
- Expose *common misconceptions*.
- Encourage students to *think statistically*.
- Normalise the idea that *context matters in statistics*.
- Sometimes, emphasise that there is *not always a single correct answer*.

Here is one framework for how to **facilitate** a discussion:

1. Pose the question (1 minute)

- Display or read the discussion question.
- Emphasise that the discussion is about *reasoning*, not correctness. In fact, there may not *be* a correct answer.

2. Think–Pair–Share (say, 5 to 10 minutes)

- **Think:** Students reflect individually (say, 30–60 seconds).

- **Pair/Group:** Students discuss in small groups or pairs (2–4 minutes).
- **Share**: Whole-class discussion (3–5 minutes).
  - Ask groups to *share* one idea or concern.
  - Ask groups to summarise the discussion in **one sentence**.
  - Invite contrasting views without framing them as "for vs against".
  - Write key points on the board if helpful.

3. Consolidation (2 to 3 minutes)

- Summarise the *range of perspectives*.
- Highlight what the discussion reveals about that week's topic.
- Explicitly connect back to lectures or upcoming material.
- Ask: *"What would you want to know next?"*.
- Revisit the question briefly at the end of class.

General tips:

- Emphasise productive contribution.
- Identifying *assumptions* when appropriate.
- Asking clarifying questions.
- Using *examples*, even informal ones.
- Avoid turning the discussion into a lecture.
- Avoid signalling a "correct" answer (if it exists) too early.
- Avoid letting one person or view dominate.
- If students ask: *"What's the right answer?"*, you can reply something like: 'In statistics, the answer often depends on the context; let's focus on when each idea might apply.'
- If discussion is quiet or stalls, use prompts such as:
  - Can anyone think of a situation where this might not hold?
  - What assumption are we making here?
  - How would this change with a different sample size?
- If discussion becomes opinion-based, gently redirect:
  - What data or evidence would help answer that?
  - What statistical idea from this week relates to that point?

Return to index of tutor information.

Thoughts on Teaching Week 1 tutorial

Return to index of tutor information.

Quick Revision answers

1. **Relational**: there is a comparison (between Chinese and Indian meals).
2. Decision-making.
3. **No**: the researchers did not determine whether a meal was Chinese or Indian.
4. The **Outcome** refers to something quantitatively measured in a **group** (such as an average or percentage) that is the outcome or result of the comparison.
5. A variable is measured on an **individual**; in this case, the amount of salt in the individual meals.
6. The **unit of analysis** depends on how the data were collected.
   Any given restaurant probably uses salt in a similar way, so if many meals were taken from each restaurant, the restaurant would be the unit of analysis.
   Or perhaps restaurant *chains* were used and restaurants in the same chain are likely to operate in a similar way, and the restaurant chain would be the unit of analysis.
   Or it may be the individual meals. *We need more information.*
7. The **unit of observation** is the meal, as the salt content must be assessed from each individual meal.
8. Conceptual: it defines what a word or term means.

Thoughts on Class Discussion

Discuss: In *this* course, in *this* semester, what percentage of students are female?

- Ask students to guess the percentage of males or females in SCI110 *this* semester.
  - Some will make an estimate using the people in their own tutorial: is this a good guesstimate?
  - Why or why not?
- Reveal that about 65% of UniSC students are female.
  - Does this change their guess?
- Then reveal the *Fun Survey* results from the last 10 years or so.
  - Have students comment on the closeness or otherwise of their guesses to this percentage.
- Then reveal the survey results for the *current* semester.
  - Have students comment on the closeness or otherwise of their guesses to this percentage.
- Point out that the survey results are not from everyone enrolled in SCI110 this semester.
  - How close do they think the survey percentages are to the true percentage?
  - Within 1%? 5% 10% of the true value?
  - How many would we need in our sample to get a percentage to within 1% either side of the true percentage?
  - What if (say) males were less likely to answer the survey than females?
  - Does the voluntary-response aspect cause you to doubt the accuracy and precision of the answers?
- How could you ensure that the sample results were as close as possible to the true answers?

Thoughts on Sect. 1.7

One way to manage this: arrange students into groups, and have each group discuss the answers separately. Then have each group share their answer with the class and discuss.

Or: have one person from each of Group 1, 2 and 3 form new groups (all containing someone from each group) to discuss the parts.

The last two parts of this question can be done by groups separately, then have the groups share with other groups.

Encourage and promote student discussion and debate about the answers. The point is that there are many possible options, all of which are acceptable, and all of which lead to different RQs.

1. Note that Jane's actual assertion in the last sentence explicitly refers to "people", though the context really is (UniSC?) students as she asserts earlier. Encourage this discussion!
2. In some ways, the answer is not important. But clearly stating what you are doing is important!
3. Again, many options are possible. Some options for comparing to Earl Grey tea: coffee; hot water; black tea; nothing at all; anything else but Early Grey; etc.
4. "Between before and after drinking Earl Grey tea" is a *within-individuals* comparison, *not a between-individuals comparison* because everyone in the population (sample) is treated the same way. There are not two groups in the population being compared.

Thoughts on Sect. 1.8

Encourage and promote student discussion and debate about the answers, within and then between groups. Then let the groups share their answers with another group(s).

Return to index of tutor information.

Thoughts on Teaching Week 2 tutorial

Return to index of tutor information.

Quick Revision answers

1. The RQ states that only households in Santiago will be studied.
2. External validity means that the sample represents the intended population (i.e., households in Santiago).
3. Since people only return the survey if they *want* to, this is *self-selecting* (or, equivalently, *voluntary response*).
4. There are multiple stages of *random* selection, so this is multi-stage.
5. It is a *little* bit like *systematic*… but is at a supermarket only. So perhaps it could be described as a systematic sample from a non-random sample (i.e., a combination).

Thoughts on Class Discussion

Discuss: A larger sample is always better than smaller sample.

This discussion helps students:

- Distinguish *accuracy* and *precision*.

- Distinguish *internal* and *external* validity.
- Recognise that *more data does not fix bad design*.

You could introduce by asking:

> At first glance this may sound obviously true. Let's unpack when it *is* true, and when it might *not* be true.

Many students initially may **agree** with the statement, saying:

- More data is more accurate.
- Larger samples are more 'reliable'.

Note that larger samples:

- Reduce **random variation**.
- Narrow **confidence intervals** and increase precision.
- Increase **power** to detect effects (though we don't discuss *power* explicitly).
- Produce more stable estimates *if the sample is random*.

However, larger samples

- May still produce *biased samples* (e.g., convenience samples).
- May produce systematic errors (e.g., non-response bias).
- Have ethical, cost, and/or feasibility constraints.

Use these **prompts** to move the discussion forward:

- What if the sample is large but non-random?
- Can increasing the sample size fix a bad survey question?
- Would 10 randomly-chosen observations ever be better than 1 000 non-randomly chosen observations?
- Is there a point where more data stops being useful?

Possible **misconceptions** you may meet:

- Large samples guarantee representativeness.
- Small samples are automatically useless.
- Confusing **precision** with **accuracy**

If these arise, acknowledge them and redirect:

> What assumption are we making about how the data were collected?

Summarise with something like:

> Larger samples reduce randomness (improve external validity), but they don't fix bias or poor design (internal validity). Sample size is important, but it's only one part of good research.

Ideally, explicitly **connect** to:

- Sampling methods.
- Study design.
- Upcoming topics on variability and inference.

Thoughts on Sect. 2.4

Remind students that there is a *Glossary* in the textbook!

Thoughts on Sect. 2.5

For newspaper articles, remember that the *reporting* may not be very good, even though the *research* itself may be good. Sometimes we need to assume answers or just state that we don't know.

Thoughts on Sect. 2.6

The purpose of this question is to get students thinking about *study design* for Assessment Task 2A. It is **very important**!

I have prepared a H5P artefact for you to use, thinking mostly of **online tutorials**:

In online classes, you can **talk** about the issues, rather than **do** the study, or use the above.

There are many correct ways to do this activity. However, for many reasons, we suggest following the suggestions here in the end (though having students suggest other ideas first is a great idea too).

You could show the *Project Proposal* form on the overhead screen, and get the students to work through it one page at a time.

It is easier to measure reaction time using an app or website on a smartphone… but **do not do so**. The *purpose* of this activity is **not** getting the reaction times, but to think about planning a study. Using the ruler drop as the activity allows a *lot* of opportunities to talk about many of the issues in study design and data collection, which **is** the purpose of this activty.

Resources: Rulers.

We *plan* a study that we will continue next week (i.e., collect the data). The whole class will do the same study, and contribute data.

We will compare the 'average ruler reaction time' for students after using their dominant and non-dominant hands. Being a little vague at this stage:

- **P**opulation: Could be SCI110 students, UniSC students, …
- **O**utcome: The average amount that the dominant hand is faster than the non-dominant hand
- **C**: None.
- **I**ntervention: None.

Seeing that there is no C is important! This is a relational RQ, with a within-individuals comparison (there are not two subsets being compared).

As a result, **a Task 2 project like this is not permitted**.

Some points:

- To begin, state that we will compare left-hand reaction 'time' to right-hand reaction 'time'. This is not the best idea, as some people and left-handed and some are right-handed, and it is better is to compare dominant and non-dominant hands. But start like this, and lead a discussion that encourages students to come to this realisation.
- Initially tell them what you want in very general terms ("we'll measure the reaction time in left and right hands"), and tell them to *Go!* They'll probably all start measuring without any planning. So let them go for one or two minutes, then pull them up and start ranting and raving: "*What about*

the planning process that we've been learning about!?" Even observe and then tell the student how different groups were doing things differently to make your point.

- Then have the students discuss planning as a class. Form a class consensus on how to proceed, and **keep a record of what the class decides**. You will actually **collect** this data, either this week or next weeks (probably more likely), so also be practical!
- You may like to prepare a Word document with this info (during class time; ask a student to be the scribe), then over the following weeks as we compile more about the study (e.g., numerical and graphical summaries) add to this Word document.
- Things to decide upon (some are arbitrary; some will have better answers than others):
  - How do we decide which hand goes first?
  - How will we determine the *dominant hand*? (Perhaps easiest: hand we write with.)
  - What if someone is ambidexterous, or only has one arm? (Use exclusion criteria!)
  - How will be grab the ruler? Anyway we like, or with index finger and thumb?
  - How do we measure the reaction time on the ruler? What is grabbing finger/thumb is not horizontal exactly? Top of thumb?
  - Where will we start the ruler relative to fingers?
  - When will we release the ruler? After (say) 3 seconds? After a random time?
  - Will catchers be sitting or standing?
  - What of the catcher drops their hand while catching? (Maybe arm should rest on a table?)
  - How far apart should fingers be?
  - Just one catch per hand, or more? If more, what (single) number to report: the slowest? fastest? median? mean?
  - What if the catchers fails to catch the ruler: have another go? or just their bad luck, and they miss a measurement?
  - Can we manage Hawthorne (no), observer effect (ask which hand is dominant *after* collecting data), etc.
- For example, we may decide to toss a coin to start (*Heads*: start with left; *Tails*: start with right), then alternate to have three tries each hand. Report the median for each hand. *After* the toss, ask which hand is dominant.

Notes:

- The unit of analysis is the *person*, and each person gets two units of observation. Remember: jamovi and SPSS like data arranged with one unit of analysis (i.e. person in this case) per row.
- If you have time, you can collect the data in class. But this activity often takes me about an hour, so we collect data next week.

Save the protocol!

Return to index of tutor information.

Thoughts on Teaching Week 3 tutorial

A note on calculators

At present, exams are online. This means that the need for students to use a calculator is diminished; they can use whatever they want (e.g., spreadsheet or jamovi) I guess. We have no way of knowing anyway.

For that reason, you may wish to reconsider how this is taught. Nonetheless, I think that learning to use a calculator for simple calculations is a useful skill. Using the **Stats** mode can also be useful.

There *will* be a question in the exam where students are expected to compute a mean and/or standard deviation, so finding a quick way (i.e., not by hand) is important (and you **can tell the students this**). Students could use as jamovi or the *Stats* mode on a calculator, for example.

Quick Revision answers

1. *Age* is quantitative continuous. (Values are rounded down to the nearest year for convenience, but the *variable* is continuous. The age of babies, for example, if often quoted in days, weeks or months.)
2. Qualitative nominal.
3. Quantitative continuous.
4. Quantitative continuous.

Thoughts on Class Discussion

Discuss: Two datasets can have the same numerical summaries but tell very different stories.

This discussion helps students:

- Appreciate the purpose and limits of **numerical summaries**.
- Understand why **graphs are essential** alongside data.
- Recognise how distribution matters.
- Avoid over-reliance on a small set of summary statistics.

You could introduce by asking

> If two datasets have the same mean and standard deviation, do they describe the same data?
> What might still be different?

You may find students start by saying:

- If the summaries are the same, the data must be similar.
- The mean captures the centre, so it's enough.
- Graphs just make things look nicer.

Some may assume summaries fully describe data; encourage them to think about **centre, spread, shape, and unusual values**.

Numerical summaries:

- Provide concise descriptions of centre and spread, etc.
- Allow quick comparisons between datasets.
- Assist with further statistical analysis.
- Do not show **distribution shape**
- Can hide **outliers or clusters**

- May mask skewness or multimodality.

Use these prompts to move the discussion forward:

- What features of data can't be seen from the mean and standard deviation?
- How could two datasets with the same summaries look very different?
- When might numerical summaries be misleading?
- How do graphs *complement* numerical summaries?

Common **misconceptions**:

- The mean represents a "typical" value in all situations.
- One summary statistic can describe a dataset.
- Graphs are optional extras.

If these arise, redirect with:

> What does the *distribution* actually look like?

Summarise with something like:

> Numerical summaries give useful but limited information. Graphical summaries reveal structure, patterns, and anomalies. Good data analysis uses both.

Ideally, explicitly **connect** to:

- One-variable quantitative summaries.
- Histograms, boxplots, dot plots, etc.
- Data collection and classification choices

Time permitting, you could:

- Show two datasets with identical summaries but different graphs (see below).
- Ask students to choose the most informative graph for a given dataset.

A **small example**:

- Dataset A: 6, 8, 9, 10, 11, 12, 13, 14, 16
  - Roughly symmetric
  - Values spread fairly evenly
  - No obvious outliers
- Dataset B: 2, 6, 9, 10, 11, 14, 17, 20, 20
  - Strong clustering around the centre
  - Extreme values at both ends

For both datasets:

- $n = 9$;
- $\bar{x} = 11$; and
- $s = 4.47$.

You could ask two groups of students to compute the statistics for the two group, and a third group to draw a 2D dotchart (on the whiteboard in a F2F class).

Thoughts on Sect. 3.3

You may like to suggest that students search on YouTube for some tutorials on using their calculator's **Statistics Mode**.

The Course Outline indicates that a calculator is needed. Help students as much as you can, but you cannot be expected to know how to work every type of calculator that is out there. You can perhaps direct students to work with other students having similar calculators.

If students use the *wrong* standard deviation button, they will get $\sigma = s = 1485.919327$ tonnes in error. **This is one important outcome of this question: that students know what button to press *on their calculator* to get the sample standard deviation.**

Thoughts on Sect. 3.4

You can have groups answer each part separately, then share their explanations with the class. You may wish to quiz the students about means and IQRs too.

For your info only (note the means and medians are very similar), see Table D.1.

TABLE D.1: *Some statistics from the graphs*

	A	**B**	**C**	**D**
Std devs	7.04	5.90	5.10	2.84
IQRs	12.10	7.96	5.82	4.71
Medians	27.13	25.19	20.07	45.03
Means	27.15	25.26	21.49	44.95

Return to index of tutor information.

Thoughts on Sect. 3.6

Following on from the ruler-drop planning activity from Week 2 (Sect. 2.6): use the protocol established then, and have students actually collect data.

I suggest starting jamovi. (Remember: each unit of analysis (student) is a row; the columns will be `Dom` and `NonDom` hand timings), Show students how to set up the variables, and have them come to the computer and enter their own data.

Then compute the differences (automatically; see me if unsure), and then show them how to produce an appropriate graph (e.g., histogram of differences) and some numerical summary information.

Save the data!

Return to index of tutor information.

Thoughts on Teaching Week 4 tutorial

Return to index of tutor information.

Quick Revision answers

1. *Age* is quantitative, so odds are not appropriate (odds are used with qualitative data).
2. Variation is measured using standard deviation or IQR.
3. The average value (of a quantitative variable) can be described using a mean or a median.
4. Percentages or odds can be used to numerically summarise a qualitative variable like *gender*.
5. $27/60 \times 100 = 45\%$.
6. $27/33 = 0.8181$.
7. The *mean* is silly, and meaningless, for qualitative variables!

Thoughts on Class Discussion

Discuss: The raw data is more useful than a graphical summary.

This discussion helps students:

- Think carefully about what 'useful' means in different contexts.
- Understand *why we summarise data*.
- Recognise that *raw data* and *visual summaries* are complementary.
- Avoid the idea that one representation is universally superior

You could introduce by asking

> When we say something is *useful*… useful for what? Think about who is using the data and what they want to do with it.

You may find students start by saying:

- Raw data allows you to see everything.
- Graphs hide information.
- Graphs are quicker to understand.

Some will treat usefulness as absolute; encourage them to think about *purpose* and *audience*.

Raw data is more useful for:

- Checking for *errors, outliers, or unusual values*.
- Reproducibility and transparency.
- Small datasets where individual observations matter.
- Re-analysis, alternative summaries, or new research questions.

Graphical summaries are more useful for:

- Quickly identifying *patterns, trends, and anomalies*.
- Communicating results to others.
- Comparing distributions or groups.
- Reducing cognitive load for large datasets.

Use these prompts to move the discussion forward:

- Useful for *whom*: the analyst, a decision-maker, or the public?
- Could you work effectively with 1 000 raw values and no graph?
- What kinds of information are easier to see in a graph than in raw data?
- When would you need to go back to the raw data after seeing a graph?

Common **misconceptions**:

- Raw data is always more informative.
- Graphs necessarily distort the data.
- Confusing **detail** with **insight**

If these arise, redirect with:

> What task are we trying to do, and which representation supports that task best?

Summarise with something like:

> Raw data and graphical summaries are useful in different ways. Raw data supports checking and flexibility, while graphs support understanding and communication. Good statistical practice uses both.

Ideally, explicitly **connect** to:

- One-variable numerical and graphical summaries.
- Choosing graphs based on purpose.
- Communicating data responsibly.

Time permitting, you could:

- Show the same dataset in raw form and as a plot; ask what each reveals.
- Ask students how their answer changes as the dataset grows larger.

Return to index of tutor information.

Thoughts on Teaching Week 5 tutorial

Return to index of tutor information.

Quick Revision answers

1. Neither is correct; 4\ is the median height of the bars… which is meaningless.
2. The sample is small, so hard to be certain: *Average* about\ 10 mms; data from about\ 6 to\ 16 mms; slightly right skewed; no outliers.
3. Discuss in class.
4. Site\ A.
5. How much larger the mean breath is for Site\ A jellyfish, compared to Site\ B jellyfish.

Thoughts on Class Discussion

Discuss: A badly designed experiment with lots of data is worse than a small, well-designed one.

This discussion helps students:

- Distinguish *research design* (internal validity) from *sample size* (external validity) and other issues.
- Understand the impact of *confounding*.

- Recognise that more data does not fix poor design.
- Appreciate trade-offs between *precision*, *accuracy* and *validity*.

You could introduce by asking

> Imagine two studies answering the same question; one study has lots of data but poor research design, the other study has very little data but is carefully designed.
> Which would you trust more, and why?

You may find students start by saying:

- More data must be better.
- Small studies can't be trusted.
- Bad design can be fixed with enough data.

Some will focus only on sample size; encourage them to think about *what kind of error* each study is vulnerable to.

Badly designed studies are problematic because

- Bias does not disappear with larger samples.
- Confounding can produce precise but *wrong* estimates.
- Measurement error is repeated at scale.
- Results may be misleading yet appear very convincing.

Small, well-designed studies are valuable because

- They can provide **unbiased** estimates.
- Assumptions and procedures are clearer.
- Results are easier to interpret.
- They are often appropriate for pilot or feasibility studies.

Use these prompts to move the discussion forward:

- Can more data fix bias or confounding?
- Which is more dangerous: a noisy estimate or a precise but biased one?
- In what situations would a small study still be useful?
- When might a large, messy dataset still be worth analysing?

Common **misconceptions**:

- Large samples guarantee correctness.
- Small studies are automatically useless.
- Precision and accuracy mean the same thing.

If these arise, redirect with:

> What type of error are we worried about here: random error or systematic error?

Summarise with something like:

> Sample size impacts precision; sampling type impacts acuracy; research design impacts validity. A large amount of bad data can give very confident wrong answers, while a small, well-designed study can still be informative.

Ideally, explicitly **connect** to:

- Sampling and study design.

- Bias, confounding, and measurement error.
- The distinction between precision and validity.

Thoughts on Sect. 5.1

With 22 observations, the median is halfway between the 11th and 12th ordered observations (observation number $(22 + 1)/2 = 11.5$): both be in the second bar, so the median is somewhere between 8 and 10 mm (we cannot be sure based on the histogram, as information is lost). If they understand the histogram, they should also see that the smallest breadth is about 6 mm, so a median of 4 mm makes no sense.

If you have the time: **Ask** *how* to explain why they are wrong, and **ask approximately what the median would be**.

Return to index of tutor information.

Thoughts on Sect. 5.4

Have students work in groups, and then share their answers.

Thoughts on Sect. 5.6

This activity follows on from the ruler-drop planning activity from Week 2 (Sect. 2.6) and the data collection (Sect. 3.6. Use jamovi and the data file to create the numerical summary.

You can save this info to add to later if you wish.

In this activity, have students actually collect the data (quickly), according to the plan prepared in Sect. 2.6, at least as best as you can.

The data themselves aren't actually that important: it is more about *using* the data to show how to

- enter the data into software;
- draw graphs; and
- compute numerical summaries.

They are **not** meant to become experts in the software, but it useful for them to see.

I suggest having students enter their data into jamovi (Fig. D.1), and then you show them how to do some basic stuff.

If you have collect data from each person left and right hand (to study the average difference between left and right hand reaction, for instance), remember that the unit of analysis is the *person*, and each person gets two units of observation: jamovi and SPSS like data arranged with one unit of analysis (i.e. person in this case) per row (Fig. D.1).

	Name	Dom. hand	Non-dom...	Faster (d...
1	Barry	13	16	3
2	Niko	18	17	-1
3	Seng	23	24	1
4	Uri	18	12	-6
5	Sena	19	28	9
6	Scotty	22	23	1
7	Masud	13	13	0

FIGURE D.1: *Data entered in jamovi for the ruler-drop study*

Return to index of tutor information.

Thoughts on Teaching Week 6 tutorial

Return to index of tutor information.

Quick Revision answers

1. $z = (10.7 - 13)$
   $div 1.3 = -1.77$.
2. $z = (26.0 - 13)/1.3 = 10$.
3. A *negative* z-score means that the value is *less than* the **mean**.
4. **FALSE**.

Thoughts on Class Discussion

Discuss: Outliers should always be removed from a dataset before analysis.

This discussion helps students:

- Understand the role of *outliers* in data analysis.
- Recognise that removal is *context-dependent*.
- Learn to distinguish between *errors* and *true extreme values*.
- Avoid blindly cleaning data without justification.

You could introduce by asking

> When you see outliers in your data, should you always remove them? What could happen if you do, or if you don't?

You may find students start by saying:

- Yes, outliers distort or bias results.
- No, all data should be kept.
- Only remove obvious mistakes.

Some may confuse **extremes** with **errors**. Remember: *any* rule for declaring outliers is arbitrary!

Reasons to consider **removing** outliers:

- They result from measurement or data entry errors.
- They disproportionately affect means and some statistical tests.
- They may obscure patterns in the main body of data.

Reasons to **keep** outliers:

- They may reflect true variation or rare but important phenomena.
- Incorrectly removing them can bias results.
- Robust statistical methods can accommodate them.

Use these prompts to move the discussion forward:

- How can you tell if an extreme value is an error or genuine observation?
- What impact do outliers have on different summary statistics?
- Can removing outliers sometimes mislead your conclusions?

If it comes up, make students aware that other (robust; nonparametric) methods can be used instead of removing data.

Common **misconceptions**:

- All outliers are mistakes.
- Removing outliers always improves results.
- Outliers don't carry useful information.

If these arise, redirect with:

> Consider the *reason* for the outlier: is it an error, or does it reflect real variation?

Summarise with something like:

> Outliers should not be removed automatically. Their treatment depends on context, the type of analysis, and whether they represent real variation or errors. Thoughtful consideration and transparency are key.

Ideally, explicitly **connect** to:

- Data cleaning and pre-processing.
- Choice of summary statistics and models.
- Ethical reporting of data handling.

Time permitting, you could show examples where removing outliers changes conclusions.

Section 11.8.4 of the textbook says:

> Managing outliers depends on why they occur (Dunn and Smyth (2018), p. 138):
> - the outlier is clearly a mistake (e.g., an age of 222). If the mistake cannot be fixed (e.g., the completed questionnaire form is lost), the observation can be deleted. Similarly, if the outlier comes from an error or mistake in the data collection (e.g., too much fertiliser was accidentally applied), the observation can be deleted.
> - the outlier represents a different population. Suppose an outlier is identified in a study of students, corresponding to a student aged 65. If the next oldest student in the data is aged 39, the outlier can be removed, since it belongs to a different population ('students aged over 40') than the other observations ('students aged 40 and under'). The remaining observations can be analysed, but the results only apply to students aged under 40 (which should be clearly communicated).
> - the reason for the outlier is unknown. In these cases, discarding outliers routinely is not recommended; the outliers are probably real observations that are just as valid as the others. Perhaps a different analysis is necessary (e.g., using medians rather than means). Furthermore, very large datasets are expected to have a small number of observations identified as outliers using the above arbitrary rules.
>
> In all cases, whenever observations are removed from a dataset, this should be clearly explained and documented.
>
> — *Scientific Research and Methodology*, p. 138.

Thoughts on Sect. 6.4

Encourage drawing diagrams! A common error: plugging numbers into calculators wrongly, and effectively computing $z = x - (\mu/\sigma)$ rather than $z = (x - \mu)/\sigma$.

Notice that students cannot be very accurate using the 68–95–99.7 rule, which is one of the points to be made: only a guess can be made, but using z-scores leads to more accurate answers.

You could have a discussion about rounding. Is it sensible to quote to the nearest millimetre, or tenth of a millimetre?

Thoughts on Sect. 6.6

Since samples are easy to generate, you can have each group/person in class generate a few simulations. On the whiteboard, you can tally these, and produce a histogram. Obviously, the more sets of 10 you can generate, the better… and the webpage is fast… but don't take *too* long doing so.

One purpose is to show students about *sampling variation*: not every set of 10 tosses produces the same value. The main purpose is to demonstrate what we mean by a sampling distribution: the distribution that shows how a sample statistic varies from sample to sample. After drawing the histogram from your classes data, you may wish to point out where certain groups' $\hat{p}$ values are located.

Return to index of tutor information.

Thoughts on Teaching Week 7 tutorial

Return to index of tutor information.

Quick Revision answers

1. **FALSE**. A CI is an interval which probably contains the **population** value. We already know the **sample** value… we don't need an interval for it.
2. **FALSE**. Confidence intervals can be any percentage (though 90%, 95% and 99% CIs are the most common, and 95% easily the most common of all).
3. **FALSE**. A *standard deviation* is a measure of variation in the *individuals*; a *standard error* is a measure of how much a sample summary value (such as the sample mean, or sample proportion) varies between samples.
4. **TRUE**.
5. **FALSE**. A CI is an interval that is likely to contain the unknown value of the *parameter*, not a certain percentage of the observations.

Thoughts on Class Discussion

Discuss: If we have *greater* confidence in our answer, we would expect to have a *smaller* range of values to choose from.

This discussion helps students:

- Clarify what "confidence" means in statistics.

- Connect **confidence level** with **interval width**.
- Distinguish *confidence* from *precision*.
- Avoid common misinterpretations of confidence intervals.

You could introduce by asking

> When we say we are 'more confident', what exactly do we mean? Does it mean we know the answer more precisely, or something else?

You may find students start by saying:

- Greater confidence should mean a narrower interval.
- More certainty means less spread.
- A wider range feels less confident.

Some will confuse *confidence* with *precision*; encourage them to separate these ideas. Reasons students may think this is **true** include:

- Higher certainty intuitively feels like tighter bounds.
- Everyday language uses "confidence" to mean precision.
- Narrow intervals appear more informative.

However:

- Increasing the **confidence level** (e.g. 90% to 95%) increases interval width.
- Increasing **sample size** reduces interval width.
- Reducing variability reduces interval width.
- Confidence level and precision are controlled by different mechanisms.

Use these prompts to move the discussion forward:

- What changes when we go from a 90% CI to a 95% CI?
- Can we be more confident *and* more precise at the same time?
- What would you change to get a narrower interval *without* lowering confidence?
- Why might wider intervals be more honest?

Common **misconceptions**:

- Higher confidence means greater accuracy.
- Confidence level controls precision.
- A narrow interval must be correct.

If these arise, redirect with:

> What do we have to trade off when choosing a confidence level?

Summarise with something like:

> In statistics, confidence and precision are not the same thing. Greater confidence means we are more likely to capture the true value, but this usually requires a wider range of plausible values.

Ideally, explicitly **connect** to:

- Confidence intervals for means and proportions.
- The role of sample size and variability.
- Interpretation of confidence levels.

Time permitting, you could:

- Compare two confidence intervals with different confidence levels (or use the textbook animation).
- Ask how sample size could change the conclusion.

Thoughts on Sect. 7.5

Students commonly forget the square root when computing s.e.$(\hat{p})$, and some just have trouble with the maths anyway. Furthermore, **the calculators may use scientific notation** to display the results from some calculations (e.g. 2.2894×10^{-2} or 2.2894^{-2} as it displays on some calculators); many students will have no idea what's going on.

Return to index of tutor information.

Thoughts on Teaching Week 8 tutorial

Quick Revision answers

1. **FALSE**. Hypothesis testing does not **prove** anything; we cannot prove anything about the *population* by studying just one of the many possible *samples*. In loose terms, a hypothesis test quantifies how much evidence there is in the *sample* to support the alternative hypothesis about the *population*.
2. **TRUE**. Hypothesis are always about the *unknown* vaues of *population* parameters.
3. **FALSE**. Alternative hypotheses may be one- or two-tailed, but it does **not** depend on the *data*; it depends on the RQ.
4. **TRUE**.
5. **TRUE**.

Thoughts on Class Discussion

Discuss: Confidence intervals are more informative than P-values.

This discussion helps students:

- Compare what information confidence intervals and P-values provide.
- Understand the limits of *binary decision-making*.
- Connect statistical results to **practical interpretation**.
- Avoid treating inference as a single number.

You could introduce by asking

> Suppose I tell you only a P-value, and nothing else. What do you still not know about the result?

You may find students start by saying:

- P-values tell you whether something is statistically significant. (You can follow with: What do you mean by 'statistically significant?')
- Confidence intervals show the range of possible values.
- P-values are simpler and easier to interpret.

Some will focus on ease rather than information; encourage them to think about *what questions each tool answers*.

CIs are informative because they:

- Show the **range of plausible values** for a parameter.
- Convey **precision and uncertainty**.
- Allow assessment of **practical significance**.
- Can be used to make comparisons across studies.

P-values are informative because they:

- Provide evidence against a null hypothesis.
- Support formal hypothesis testing.
- Are familiar and widely reported.
- Offer a quick summary for decision rules.

Use these prompts to move the discussion forward:

- What can a confidence interval tell you that a P-value cannot?
- Can a result be statistically significant but practically unimportant?
- Can two studies have the same P-value but very different confidence intervals?
- When might a P-value still be useful?

Common **misconceptions**:

- A small P-value means a *large* effect, or an *important* effect.
- Confidence intervals contain the true value with 95% probability (though we do kind-of do this…).
- One method should completely replace the other.

If these arise, redirect with:

> What question are we trying to answer: "Is there evidence?" or "How large is the effect?"

Summarise with something like:

> P-values answer a narrow question about evidence against a null hypothesis.
> Confidence intervals provide richer information about effect size and uncertainty.
> Used together, they give a more complete picture.

Ideally, explicitly **connect** to:

- Hypothesis testing vs estimation
- Interpretation of confidence intervals
- Reporting statistical results responsibly

Time permitting, you could:

- Show two results with the same P-value but different confidence intervals
- Ask students which result they would trust more, and why

Thoughts on Sect. 8.6

> We only need to use the 68–95–99.7 rule. The one-tailed P-values are *half* of the two-tailed P-values.

Return to index of tutor information.

Thoughts on Teaching Week 9 tutorial

Return to index of tutor information.

Quick Revision answers

1. The researchers used an exact 95%CI, which gives very similar answers to using an approximate 95%\ CI. (The 68–95–99.7 rule gives approximate multipliers only.)
2. **TRUE**. If we take the *coeliac* mean minus the *non-coeliac* mean, we would have $8.39 - 8.17 = 0.22$, which is what is given in the table.
3. **FALSE**. The CI is for the *difference* in the DMFT. The value is **not** about the *number* of teeth; it is about the *difference* in the number of teeth between coeliacs and non-coeliacs. A negative value is fine if we understand what the CI is estimating.
4. The *positive* value of 2.76 means (as we saw in the first Quick Revision question) that coeliacs have a mean of 2.76 more DMFT.
   So the *negative* value of -2.32 means that *non*-coeliacs have a mean of 2.32 more DMFT.
5. **TRUE**. Remember that all hypotheses are about population parameters, and have an 'equals to' in there somewhere (the 'no difference, no change, no relationship' position).
6. **FALSE**. The hypothesis is about population parameters (
   mu), which is good.
   But the actual question was only about studying the difference, which suggests a *two*-tailed alternative hypothesis (not a *one*-tailed alternative hypothesis as given).
7. **FALSE**. The *sample* means could be different for one of two reasons: (a)\ the population means are actually the same, but the *sample means* are just different due to sampling variation; or (b)\ because the *population* means are different.
   We don't know *which* one of these reasons is the reason here.
8. How much larger the mean number of cavities are for coeliacs.
9. When both sample sizes are larger than\ 25, the CI is statistically vaid. Since both sample sizes are only *just* smaller than\ 25, the test is *probably* statistically valid.

Thoughts on Class Discussion

Discuss: If the mean of Group A is $\bar{x}_1 = 12.0$ cm and the mean of Group B is $\bar{x}_2 = 14.2$ cm, then the two means are clearly different.

This discussion helps students:

- Understand that **observed differences** do not automatically imply statistical significance.
- Connect **mean differences** to **variability and sample size**.
- Recognise the importance of **confidence intervals and hypothesis tests**.
- Avoid over-interpreting raw summary statistics.

You could introduce by asking

> Two groups have means 12.0 cm and 14.2 cm. Can we conclude there is a difference? What else do we need to know?"

You may find students start by saying:

- The difference of 2.2 cm seems obvious!
- Means alone tell the story.
- We don't need any more calculations.

Some may ignore **spread and sample size**; encourage them to think about **uncertainty**. Some will confuse *sample* means and *population* means.

The sample means are obviously different… but RQs are about *population* means. The differences between the sample means may due to:

- Large **within-group variability** that can make the difference uninformative.
- Small sample sizes that can produce **unstable estimates**.
- Overlapping **confidence intervals** that indicate uncertainty.
- Chance.

Of course, the difference between the sample means *might* indicate a difference between the population means:

- Low variability and sufficiently large samples.
- Statistical tests confirm significance.
- Effect size is meaningful in context.

Use these prompts to move the discussion forward:

- What else would you need to decide if the difference is real in the population?
- How does variability affect our confidence in this difference?
- Could a small sample exaggerate the observed difference?
- How might confidence intervals or hypothesis tests help?

Common **misconceptions**:

- A difference between sample means implies a difference between population means.
- Sample means alone provide the full story.
- Larger differences are always more important.

If these arise, redirect with:

> What role does variability and sample size play in interpreting these means?

Summarise with something like:

> Observed differences in sample means are suggestive, but not conclusive. We need to consider variability, sample size, and use confidence intervals or statistical tests to decide if the difference is meaningful.

Ideally, explicitly **connect** to:

- Hypothesis testing for two means.
- Confidence intervals for group differences.
- Importance of considering variability and sample size.

Time permitting, you could:

- Compare two datasets with the same mean difference but different variability.
- Ask students to interpret confidence intervals for the difference between means.

Thoughts on Sect. 9.5

> If you wish, you can conduct a hypothesis test for the ruler-drop (using jamovi).
>
> In particular, if you get a negative number for one (or both) of the CI limits, you can discuss what the negative number (i.e., negative *difference*) means.

Return to index of tutor information.

Thoughts on Teaching Week 10 tutorial

Return to index of tutor information.

Quick Revision answers

Statements 2 and 4 are **both** consistent with the information (small P-value).

Thoughts on Class Discussion

Discuss: When comparing two groups, is it better to focus on differences in *proportions*, *odds*, or *percentages*?

This discussion helps students:

- Understand different ways of **quantifying group differences** for comparing qualitative variables.
- Recognise that **choice of measure depends** on context but are ultimately the same.
- Practice **selecting an appropriate and consistent analysis** for a research question.

You could introduce by asking

> Imagine you want to compare two treatments on whether patients improve. Should you look at the difference in proportions, the odds, or percentages? Why?

You may find students start by saying:

- Differences in proportions are easier to understand.
- It depends on which measure is statistically significant.

Differences in **proportions** are effectively the same as differences in **percentages** (percentages are proportions multiplied by 100), and:

- Are often easier to understand.
- Are useful for reporting to a general audience.

However, we teach inference for the difference in **odds** (or using **odds ratios**), since this is useful:

- For adjusting for confounders (in later courses for some students).
- For logistic regression models.
- When outcomes are rare.

In SCI110, the main issue is to **keep consistency** between the RQ, hypothesis and the conclusions.

Use these prompts to move the discussion forward:

- How does sample size affect confidence intervals for each measure?
- How would your choice change if the outcome is rare vs common?
- What measure would be easier to communicate to non-statisticians?
- How do CIs help interpret practical importance, not just significance?

Common **misconceptions**:

- Odds and proportions are much the same. They are different, but conclusions will have the same meaning (apart from a few edge cases we don't need to worry about).

If these arise, redirect with the ideas of **consistency**.

Summarise with something like:

> The choice between differences in proportions, odds and perecentages depends on context, and the research question. Confidence intervals, hypothesis tests, and sample size planning should guide the analysis.

Ideally, explicitly **connect** to:

- Confidence intervals and hypothesis tests for ORs and differences in proportions.
- Sample size estimation.
- Selecting the most appropriate analysis for a given scenario.

Return to index of tutor information.

Thoughts on Teaching Week 11 tutorial

Return to index of tutor information.

Quick Revision answers

Student 1: Not using the standard error; hence, **if** the *data* have a normal distribution, this is an interval that contains about 68% of the *data*.
Student 2: Not using the standard error; hence, **if** the *data* have a normal distribution, this is an interval that contains about 95% of the *data*.
Correct: 95% CI for the *mean* is $12.4 \pm (2 \times 2.8/sqrt6)$, or from 10.11 to 14.69 MPa. :::

Thoughts on Class Discussion

– **Discuss**: A 'statistically significant' result means a real discrepancy or effect.

This discussion helps students:

– Understand what **statistical significance** actually means and measures.
– Distinguish between **evidence against the null** and **practical reality**.
– Recognise the effects of **sample size, variability, and research design**.
– Avoid over-interpreting P-values as 'proof' of truth.

You could introduce by asking

> If a result is statistically significant, does that guarantee there is a discrepancy or effect in the population? What does 'significant' mean in statistical terms?

You may find students start by saying:

– Yes, significant means the effect really exists.
– Maybe, but it could be a fluke.
– Significance implies importance.

Some may confuse **significance** with **magnitude or reality**; encourage them to separate these concepts.
Statistical significance indicates:

– Evidence against a null hypothesis, however small the discrepancy is.
– The observed data is unlikely under the null, assuming all assumptions hold.
– That chance alone is an unlikely explanation, **given the model**.

Statistical significance does **not** necessarily indicate:

– That the effect is practically or scientifically important.
– That the effect is guaranteed to exist in the population.
– That the estimate is accurate or free from bias.

Use these prompts to move the discussion forward:

– Could a very small effect be statistically significant with a large sample?
– Could a large effect fail to reach significance with a small sample?
– How do variability and study design influence significance?
– How could confidence intervals provide a fuller picture?

Common **misconceptions**:

– 'Statistical significance' is the same as 'a discrepancy in the population'.
– 'Non-statistical significant' is the same as 'no discrepancy in the population'.
– P-values tell the probability that the null hypothesis is true.

If these arise, redirect with:

> Statistical significance only quantifies how consistent the data are with the null hypothesis, not whether the effect is real in practice.

Summarise with something like:

> A statistically significant result suggests the data are unlikely under the null hypothesis, but it does not prove a discrepancy in the population. or practically important effect. Always consider effect size, uncertainty, and context alongside P-values.

Return to index of tutor information.

References

Agresti A, Franklin CA. Statistics: The art and science of learning from data. Third. Pearson Education Limited; 2007.

Australian Bureau of Statistics. How Australians measure up [Internet]. Australian Bureau of Statistics; 1995. Report No.: 4359.0. Available from: http://www.ausstats.abs.gov.au/.

Barkley JE, Lepp A. Cellular telephone use during free-living walking significantly reduces average walking speed. BMC Research Notes. 2016;9(195):1–5.

Barr D, DeBruine L. Webexercises: Create interactive web exercises in 'R Markdown' (formerly 'webex') [Internet]. 2021. Available from: https://github.com/psyteachr/webexercises.

Bartlett MS. Contingency table interactions. Supplement to the Journal of the Royal Statistical Society. JSTOR; 1935;2(2):248–52.

Berger RL, Boos DD, Guess FM. Tests and confidence sets for comparing two mean residual life functions. Biometrics. 1988;44(1):103–15.

Berry S. Aged cheese could help you age well [Internet]. The Sydney Morning Herald. 2016. Available from: https://www.smh.com.au/lifestyle/health-and-wellness/aged-cheese-could-help-you-age-well-20161114-gsp5m0.html.

Bryden MM, Smith MSR, Tedman RA, Featherston DW. Growth of the Weddell seal, *Leptonychotes weddelli* (Pinnipedia). Australian Journal of Zoology. 1984;32:33–41.

de Carvalho AA, Amorim FF, Santana LA, de Almeida KJQ, Santana ANC, Neves F de AR. STOP-Bang questionnaire should be used in all adults with Down Syndrome to screen for moderate to severe obstructive sleep apnea. PloS ONE. 2020;15(5):e0232596.

Duncan PD, Ritter PL, Dornbusch SM, Gross RT, Carlsmith JM. The effects of pubertal timing on body image, school behavior, and deviance. Journal of Youth and Adolescence. 1985;14(3):227–35.

Dunn PK. Bootstrap confidence intervals for predicted rainfall quantiles. International Journal of Climatology. 2001;21(1):89–94.

Dunn PK. Assessing claims made by a pizza chain. Journal of Statistical Education [Internet]. 2012;20(1). Available from: https://www.tandfonline.com/doi/abs/10.1080/10691898.2012.11889637.

Eisenberg T, Abdellatif M, Schroeder S, Primessnig U, Stekovic S, Pendl T, et al. Cardioprotection and lifespan extension by the natural polyamine spermidine. Nature Medicine. Nature Publishing Group US New York; 2016;22(12):1428–38.

Garnier S, Ross N, Rudis R, Camargo AP, Sciaini M, Scherer C. viridis(Lite) - colorblind-friendly color maps for R [Internet]. 2024. Available from: https://sjmgarnier.github.io/viridis/.

Gausche M, Lewis RJ, Stratton SJ, Haynes BE, Gunter CS, Goodrich SM, et al. Effect of out-of-hospital pediatric endotracheal intubation on survival and neurological outcome: A controlled clinical trial. Journal of the American Medical Association. 2000;283(6):783–90.

Gerber S. Material properties of bamboo flooring. Indooroopilly, Queensland: Queensland Department of Primary Industries, for Bamboo Flooring Pty Ltd; 2004. Report No.: 2-04.

Gilchrist R. Regression models for data with a non-zero probability of a zero response. Communications in Statistics-Theory and Methods. Taylor & Francis; 2000;29(9-10):1987–2003.

Guirao L, Samitier CB, Costea M, Camos JM, Majo M, Pleguezuelos E. Improvement in walking abilities in transfemoral amputees with a distal weight bearing implant. Prosthetics and Orthotics International. 2017;4(26–32).

Hand DJ, Daly F, Lunn AD, McConway KY, Ostrowski E. A handbook of small data sets. London: Chapman; Hall; 1996.

Hébert-Losier K, Boswell-Smith C, Hanzlíková I. Effect of footwear versus barefoot on double-leg jump-landing and jump height measures: A randomized cross-over study. International Journal of Sports Physical Therapy. 2023;18(4):845.

Ingenbleek Y, McCully KS. Vegetarianism produces subclinical malnutrition, hyperhomocysteinemia and atherogenesis. Nutrition. 2012;28:148–53.

Jaworowska A, Blackham T, Stevenson L, Davies IG. Determination of salt content in hot takeaway meals in the United Kingdom. Appetite. 2012;59(2):517–22.

Khan I, Navie S, George D, O'Donnell C, Adkins SW. Alien and native plant seed dispersal by vehicles. Austral Ecology. Wiley Online Library; 2018;43(1):76–88.

Kirkendall DT, Jordan SE, Garrett WE. Heading and head injuries in soccer. Sports Medicine. Springer; 2001;31(5):369–86.

Lindström P. Test report for primary battery testing for ALDI Stores Australia. Intertek; 2011.

Lo AC, Guarino PD, Richards LG, Haselkorn JK, Wittenberg GF, Federman DG, et al. Robot-assisted therapy for long-term upper-limb impairment after stroke. The New England Journal of Medicine. 2010;362(19):1772–83.

Lunn AD, McNeil DR. Computer-intensive data analysis. Chichester: John Wiley; Sons; 1991.

McEvoy SP, Stevenson MR, Woodward M. Phone use and crashes while driving: A representative survey of drivers in two Australian states. Medical Journal of Australia. 2006;185(11–12):630–4.

Montalvo C, Cook W, Keeney T. Retrospective analysis of hydraulic bridge collapse. Journal of Performance of Constructed Facilities. American Society of Civil Engineers; 2020;34(1):1–8.

O'Brien EJ, Zhang L, Zhao H, Hajializadeh D. Probabilistic bridge weigh-in-motion. Canadian Journal of Civil Engineering. NRC Research Press; 2018;45(8):667–75.

Oostema JA, Chassee T, Reeves M. Emergency dispatcher stroke recognition: Associations with downstream care. Prehospital Emergency Care. 2018;22(4):466–71.

Pamphlett R. Exposure to environmental toxins and the risk of sporadic motor neuron disease: An expanded Australian case–control study. European Journal of Neurology. 2012;19:1343–8.

Pleasant GD, Dabbert CB, Mitchell RB. Nesting ecology and survival of scaled quail in the southern high plains of Texas. The Journal of Wildlife Managament. 2006;70(3):632–40.

R Core Team. R: A language and environment for statistical computing [Internet]. Vienna, Austria: R Foundation for Statistical Computing; 2018. Available from: https://www.R-project.org/.

Ranade R, Li VC, Heard WF. Tensile rate effects in high strength-high ductility concrete. Cement and Concrete Research. 2015;68:94–104.

Singh RK, Prasad G. Long-term mortality after lower-limb amputation. Prosthetics and Orthotics International. 2016;40(5).

Soetaert K. Diagram: Functions for visualising simple graphs (networks), plotting flow diagrams [Internet]. 2017. Available from: https://CRAN.R-project.org/package=diagram.

Stone RC, Auliciems A. SOI phase relationships with rainfall in eastern Australia. International Journal of Climatology. 1992;12:625–36.

Strayer DL, Johnston WA. Driven to distraction: dual-task studies of simulated driving and conversing on a cellular telephone. Psychological Science. 2001;12(6):462–6.

Swinnen E, Baeyens J-P, Mulders B Van, Verspecht J, Degelaen M. The influence of the use of ankle-foot orthoses on thorax, spine, and pelvis kinematics during walking in children with cerebral palsy. Prosthetics and Orthotics International. 2017;2017.

Tanagawa K, Shigematsu A. Choice of airway devices for 12,020 cases of nontraumatic cardiac arrest in Japan. Prehospital Emergency Care. 1998;2:96–100.

Telford RD, Cunningham RB. Sex, sport, and body-size dependency of hematology in highly trained athletes. Medicine and Science in Sports and Exercise. 1991;23(7):788–94.

The jamovi Project. jamovi (version 1.0) [computer software] [Internet]. Available from: https://www.jamovi.org.

Truswell AS, Choudhury N, Roberts DCK. Double blind comparison of plasma lipids in healthy subjects eating potato crisps fried in palmolein or canola oil. Nutrition Research. 1992;12:S43–52.

VanLeit B, Crowe TK. Outcomes of an occupational therapy program for mothers of children with disabilities: Impact on satisfaction with time use and occupational performance. The American of Occupational Therapy. 2002;56(4):402–10.

Walters JP, Kaminsky J, Huepe C. Factors influencing household solar adoption in Santiago, Chile. Journal of Construction Engineering and Management. American Society of Civil Engineers; 2018;144(6):05018004.

Xie Y. Dynamic documents with R and knitr [Internet]. 2nd ed. Boca Raton, Florida: Chapman; Hall/CRC; 2015. Available from: https://yihui.name/knitr/.

Xie Y. Bookdown: Authoring books and technical documents with R markdown [Internet]. Boca Raton, Florida: Chapman; Hall/CRC; 2016. Available from: https://github.com/rstudio/bookdown.

Xie Y, Cheng J, Tan X. DT: A wrapper of the JavaScript library 'DataTables' [Internet]. 2018. Available from: https://CRAN.R-project.org/package=DT.

Zhu H. kableExtra: Construct complex table with 'kable' and pipe syntax [Internet]. 2018. Available from: https://CRAN.R-project.org/package=kableExtra.